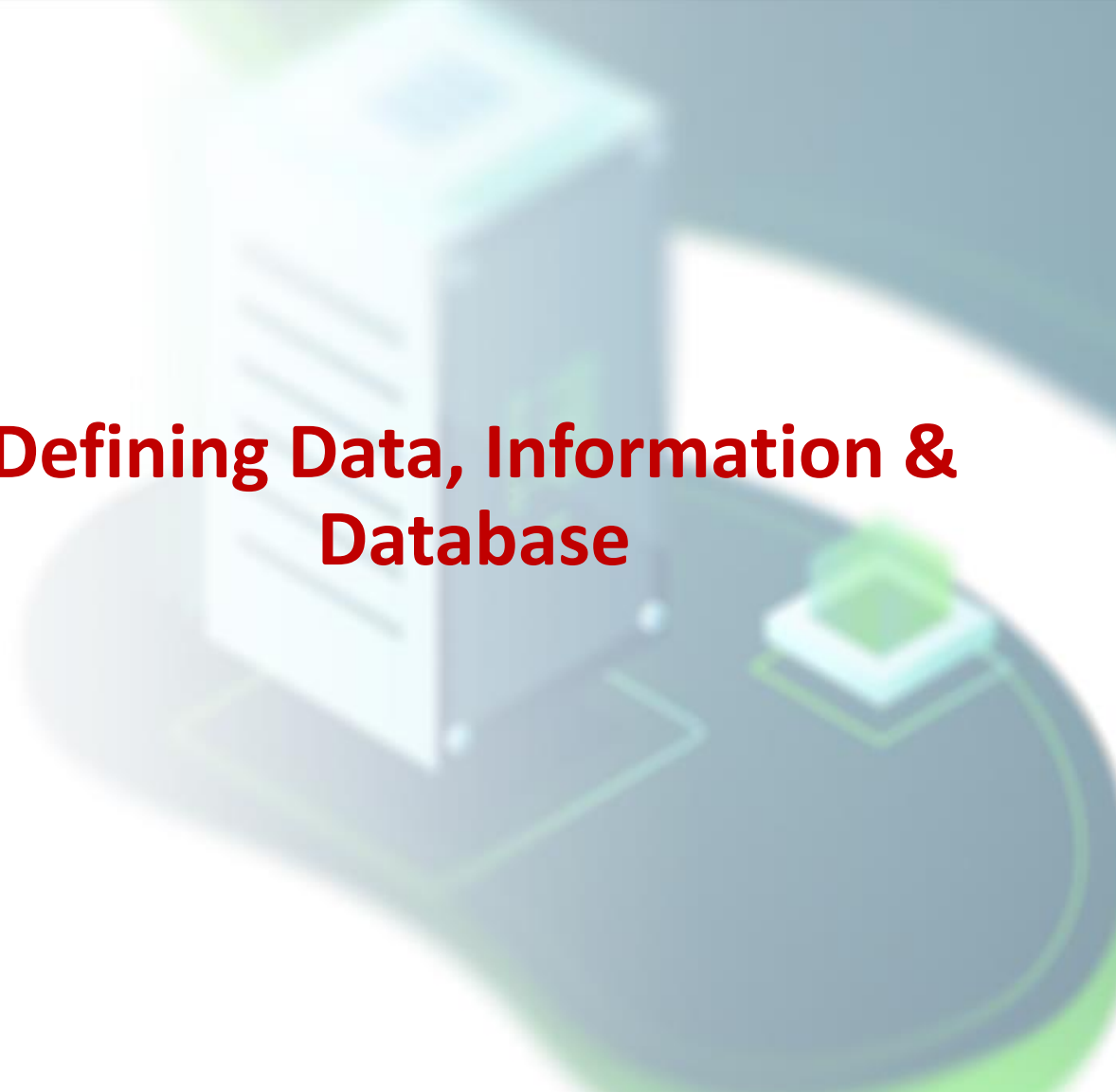


Introduction to Database Administration

Chapter 1

**Defining Data, Information &
Database**

A stylized illustration of a server rack and a database component. The server rack is a tall, white, rectangular unit with a blue square on top. It sits on a dark blue circular base. To the right of the server, on the same base, is a smaller, white, rectangular component with a blue square on top. Green lines connect the base of the server to the base of the smaller component. The background is a light blue gradient.

Defining Data, Information & Database

Data, Information **From Data to Information**

- For Knowledge
- For Business Intelligence



Defining Data, Information & Database

Data

- Anything which is not necessarily meaningful to human being.
- Raw facts/Un-processed information.

Defining Data, Information & Database



Examples of Data

The numbers 10, 22, and 119 are data

Some one can think these are lengths and other may think room numbers

Similarly, 1.5, 2.5, 31.5, ... **MAY** represent lengths of some iron rods

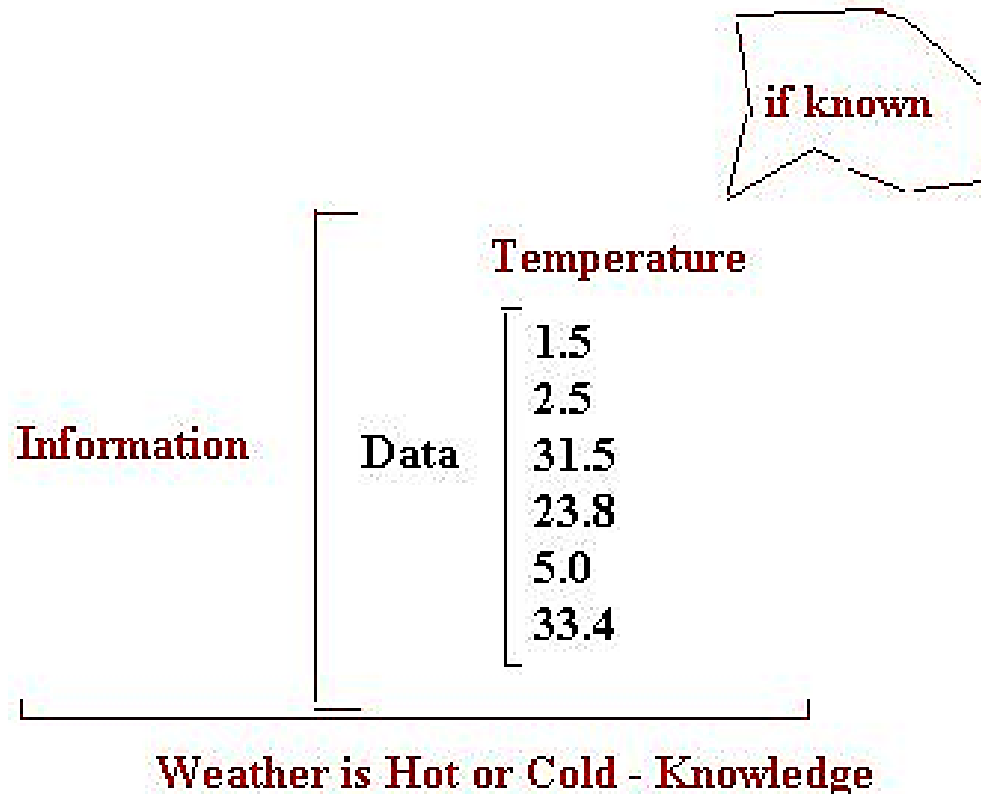
Defining Data, Information & Database

Information

Data Processed to reveal its meaning

- Information is meaningful
- In today's world, accurate, relevant and timely information is the key to good decision making
- Good decision making is key to survival in today's competitive and global environment

Defining Data, Information & Database



Examples of Information

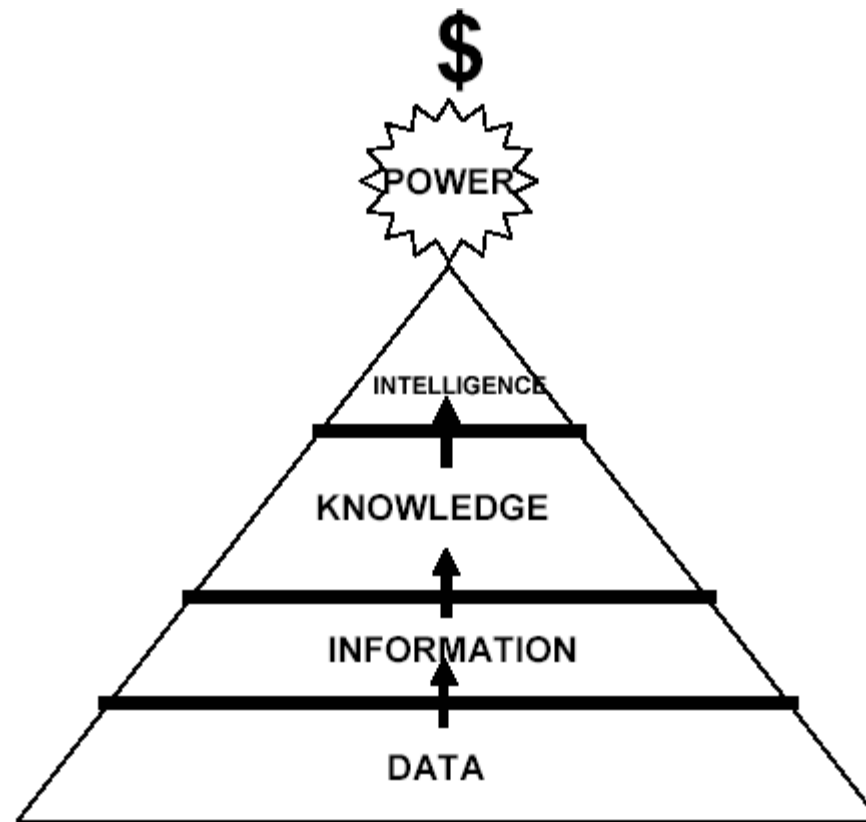
Software Engineer designed some system

- May be monthly reports are not ok
- Reports are giving wrong information, etc.

This means DESIGN is not ok

Defining Data, Information & Database

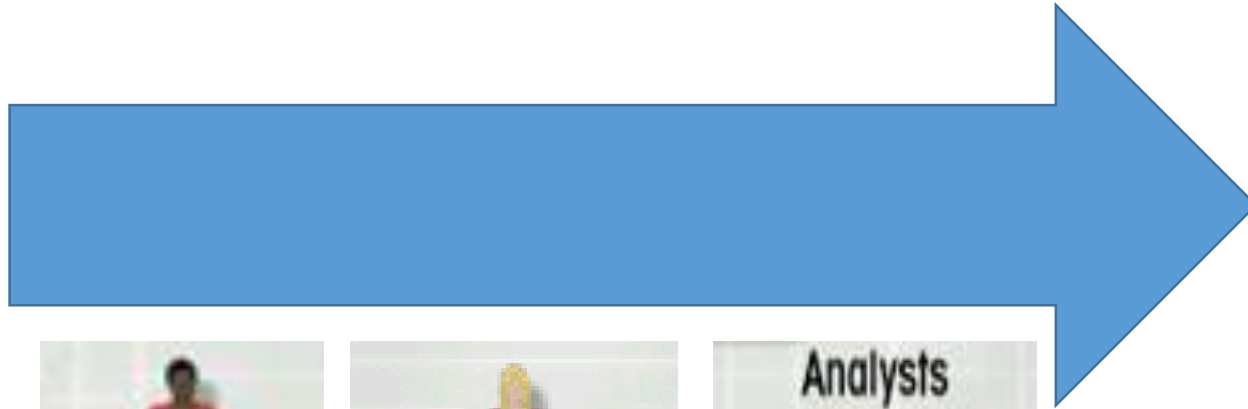
Aim of Data & Information



Defining Data, Information & Database

Who Builds Useful Information

Data



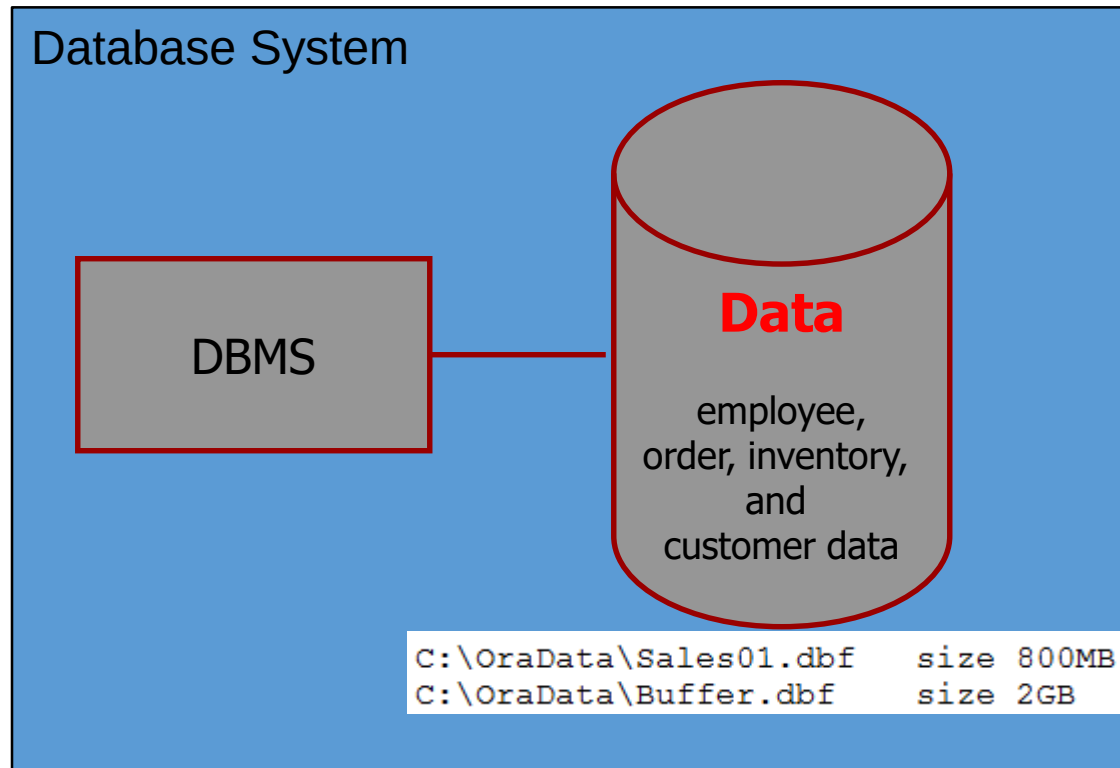
Information



Defining Data, Information & Database

A database is collection of stored operational data used by application systems of some particular enterprise (C.J. Date)

Database & DBMS Package



Chapter 2

DBMS, Role of DBA with Database

- Examples of DBMS
- Data Accessing using DBMS
- Forms, Reports & Queries
- DBMS Language
- DBMS & its Sub Schemas
- Role/ Jobs in DB Environment
- Components of DBMS Environment
- Interaction of DBMS with DB Files



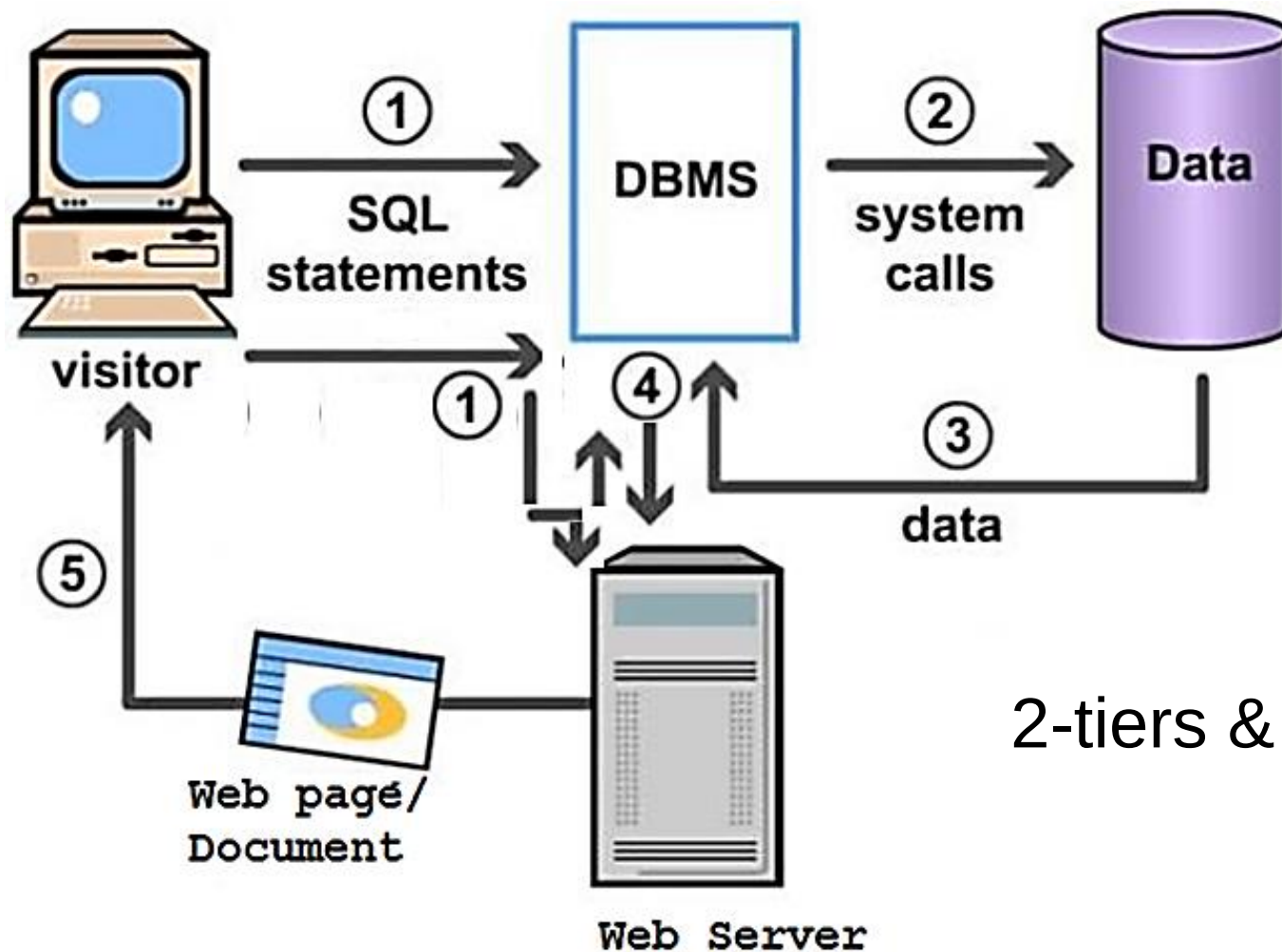
Introduction to DBMS & its environment for DBA

Examples of DBMS



- Oracle
- IBM DB2
- Ingress
- Teradata
- MS SQL Server
- **MS Access**
- MySQL etc.

Introduction to DBMS & its environment for DBA



**Data Accessing
using DBMS**

2-tiers & 3-tiers Architectures

Introduction to DBMS & its environment for DBA

Forms, Reports & Queries

CLASS

ClassNumber: 40
Name: Acct101
Term: F04
Section: 1

STUDENT

StudentNumber	StudentName	EmailAddress
100	Cooke	Cooke@OurU.edu
400	Greene	Greene@OurU.edu

Record: 3 of 3

Record: 4 of 5

PART

Number	PartName	SalesPrice	ROQ	QOH	CompanyName	City	Country
1000	Cedar Shakes	\$22.00	100	200			
					Bristol Systems	Machester	England
					ERS Systems	Vancouver	Canada
					Forrest Supplies	Denver	US
2000	Garage Heater	\$1,750.00	3	4			
					Bristol Systems	Machester	England
					ERS Systems	Vancouver	Canada
					Kyoto Importers	Kyoto	Japan
					Forrest Supplies	Denver	US
3000	Utility Cabinet	\$55.00	7	3			
					Ajax Manufacturing	Sydney	Australia

```
SQL> SELECT empno, ename, sal  
2 FROM EMP;
```

EMPNO	ENAME	SAL
7369	SMITH	800
7499	ALLEN	1600

Introduction to DBMS & its environment for DBA

DBMS Language

A Software Package that enables users to define, create, maintain, and control access to the database.

- Data Definition Language (DDL)

- Data Manipulation Language (DML)

- Control access (DCL):

 - Security, integrity, concurrent access, recovery, support for data communication, etc.

- Utility services (DCL)

 - File import/export, monitoring facilities, code generator, report writer, etc.

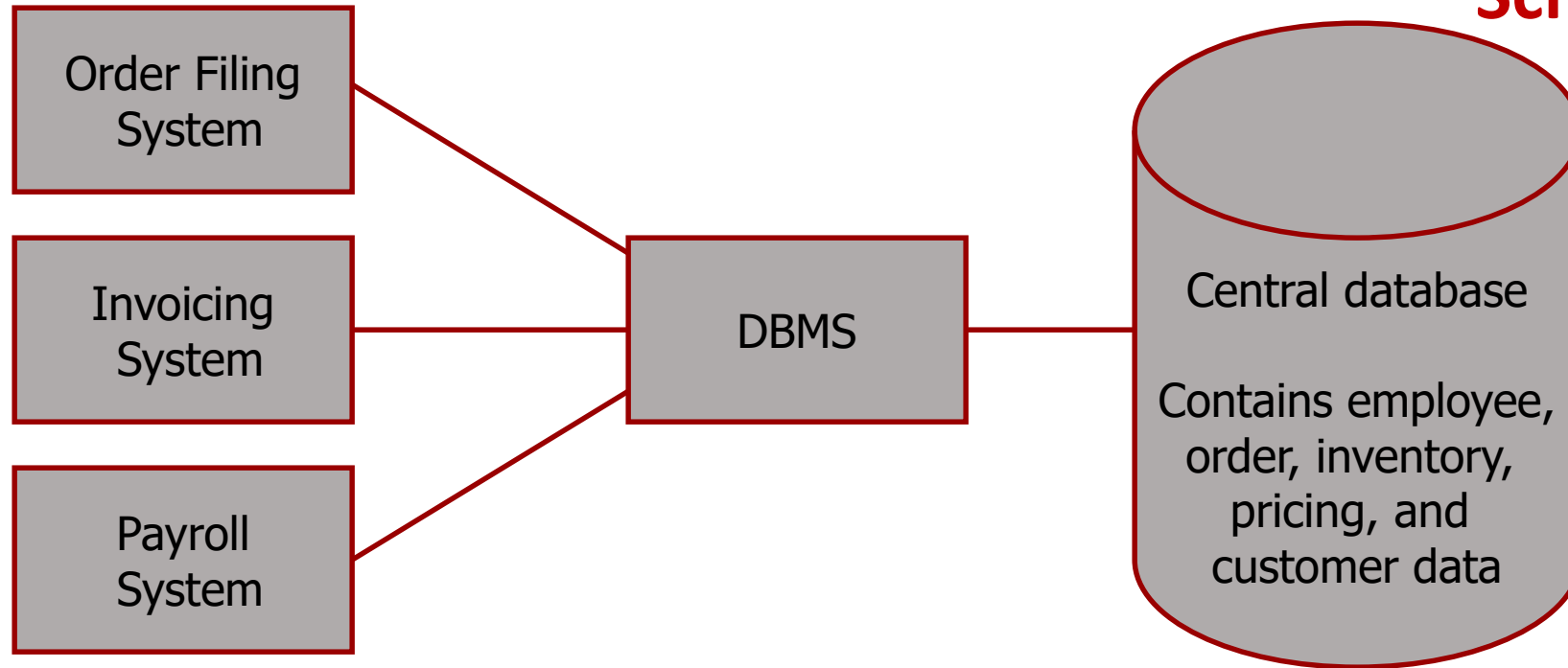
Support Ad Hoc queries

Database Languages are DML, DDL, DCL in SQL.

Introduction to DBMS & its environment for DBA

A software system that is used to create, maintain, and provide controlled access to user databases

DBMS and its Sub Schemas

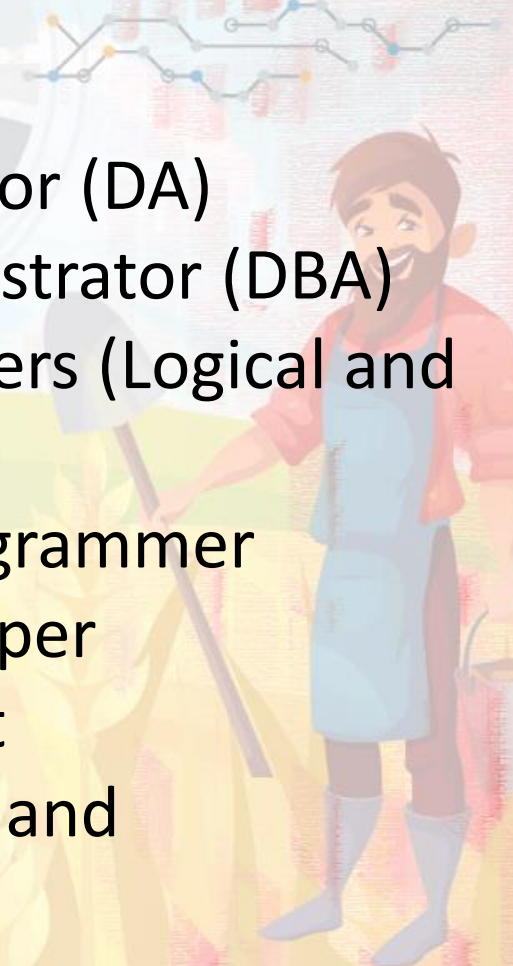


DBMS manages data resources like an operating system manages hardware resources

Introduction to DBMS & its environment for DBA

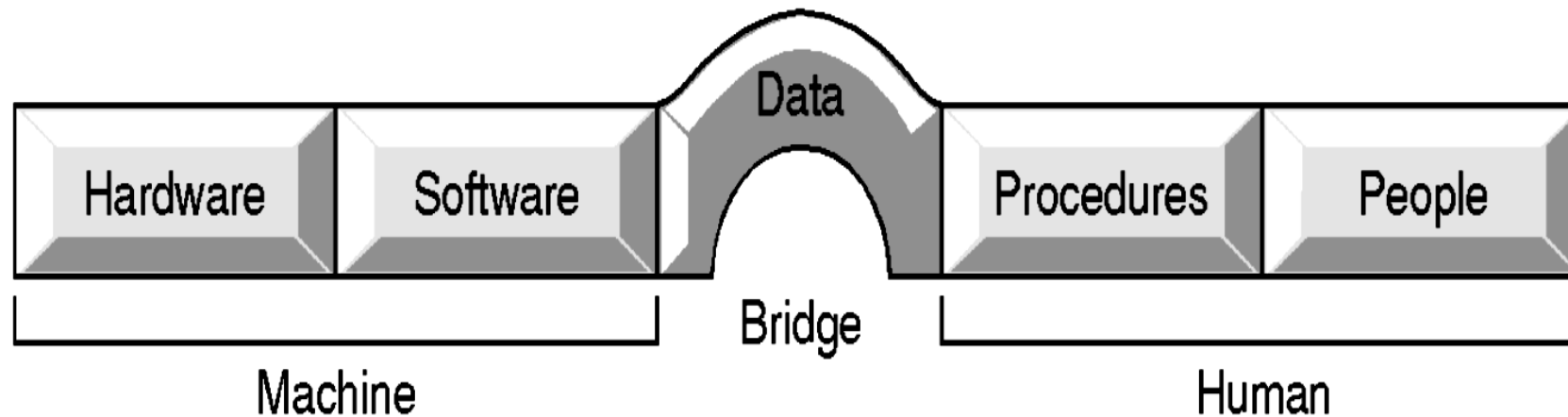
Roles/ Jobs in the Database Environment

- Data Administrator (DA)
- Database Administrator (DBA)
- Database Designers (Logical and Physical)
- Applications Programmer
- Database Developer
- Database Analyst
- End users (Naive and sophisticated)



Introduction to DBMS & its environment for DBA

Components of DBMS Environment



Introduction to DBMS & its environment for DBA

Interaction of DBMS with DB Files

The screenshot illustrates the interaction of a Database Management System (DBMS) with its underlying database files. It consists of three main components:

1. Oracle SQL*Plus Window: This window shows the SQL*Plus interface. The title bar indicates the address is `C:\oracle9i\ora92\oradata\ejazz`. The main text area displays the SQL*Plus version (9.2.0.1.0) and the connected database (Oracle9i Enterprise Edition Release 9.2.0.1.0). The user has executed the query `select * from dept;`, which returns the following results:

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

2. File Explorer Window: This window shows the file system structure of the database. The address bar indicates the path `C:\oracle9i\ora92\oradata\ejazz`. The file list includes various database files, such as `AE_DATA_01.DBF`, `AE_INDEX_01.DBF`, `CONTROL01.CTL`, `CONTROL02.CTL`, `CONTROL03.CTL`, `CONTROL04.CTL`, `CWMLITE01.DBF`, `DCLDEV01.DBF`, `DCLINDEX01.DBF`, `DRSYS01.DBF`, `EXAMPLE01.DBF`, `INDEX01.DBF`, `MCARE_INDEX01.DBF`, `MEDCARE01.DBF`, `MEDCARE02.DBF`, `MEDTEST01.DBF`, `ODM01.DBF`, `REDO01.LOG`, `REDO02.LOG`, `REDO03.LOG`, `SYSTEM01.DBF`, `TEMP01.DBF`, `TOOLS01.DBF`, and `UNDOTBS01.DBF`. A circle highlights the `CONTROL04.CTL` file.

3. Oracle Enterprise Manager Console, Standalone: This window provides a graphical view of the database. The left pane shows the tree structure of the database, including `Databases`, `Instance`, `Schema`, `Security`, `Storage`, `Controlfile`, `Datfiles`, `Rollback Segments`, `Redo Log Groups`, `Archive Logs`, and `Distributed`. The right pane displays a table of database files with columns for `Name`, `Type`, `Extent Ma...`, `Size (M)`, `Used (M)`, and `Used %`. The table lists various database files, including `AE_DATA_01`, `AE_INDEX_01`, `CWMLITE`, `DCL_DEV`, `DCL_INDEX`, `DRSYS`, `EXAMPLE`, `INDEX`, `MCARE_INDEX`, `MEDCARE`, `MEDTEST`, `ODM`, `SYSTEM`, `TOOLS`, `UNDOTBS1`, and `USERS`.

Chapter 3

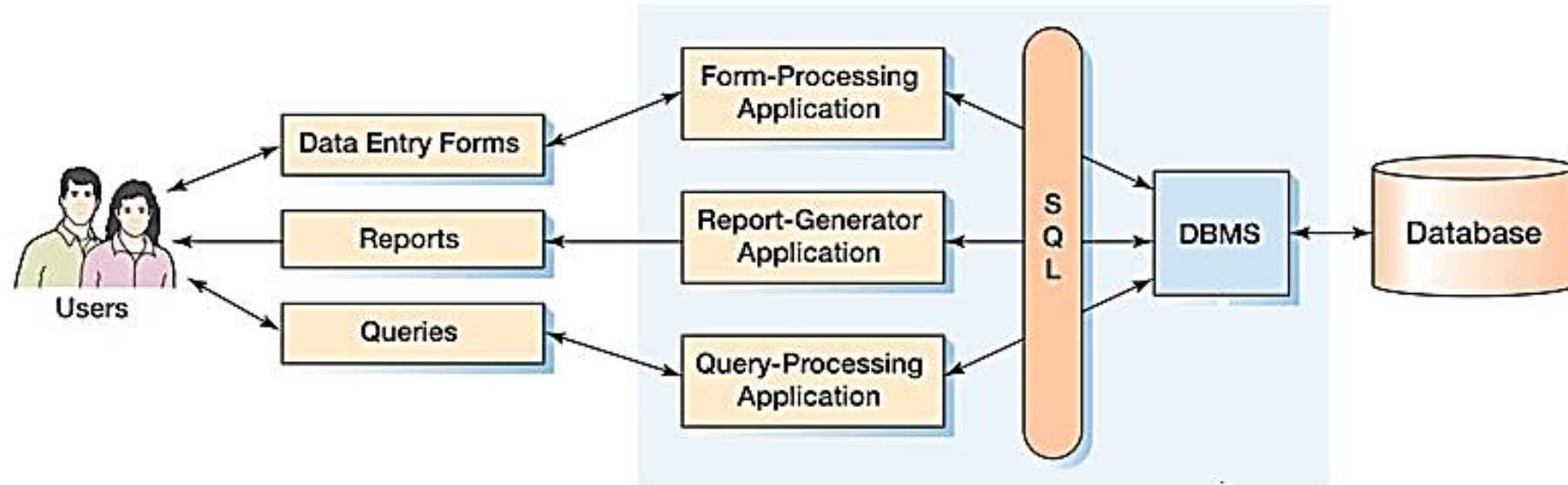
DBMS Components & Architecture

- Interaction of users with DBMS
- ANSI Spark three level architecture
- Multi user DBMS Architecture
- Teleprocessing
- DBMS Client-Server Architecture (2-tiers)
- DBMS Client-Server Architecture (3-tiers)
- Alternate Client-Server Architectures
- Paradigm of Database System



Introduction to DBMS Components & Architecture

Interaction of users with DBMS

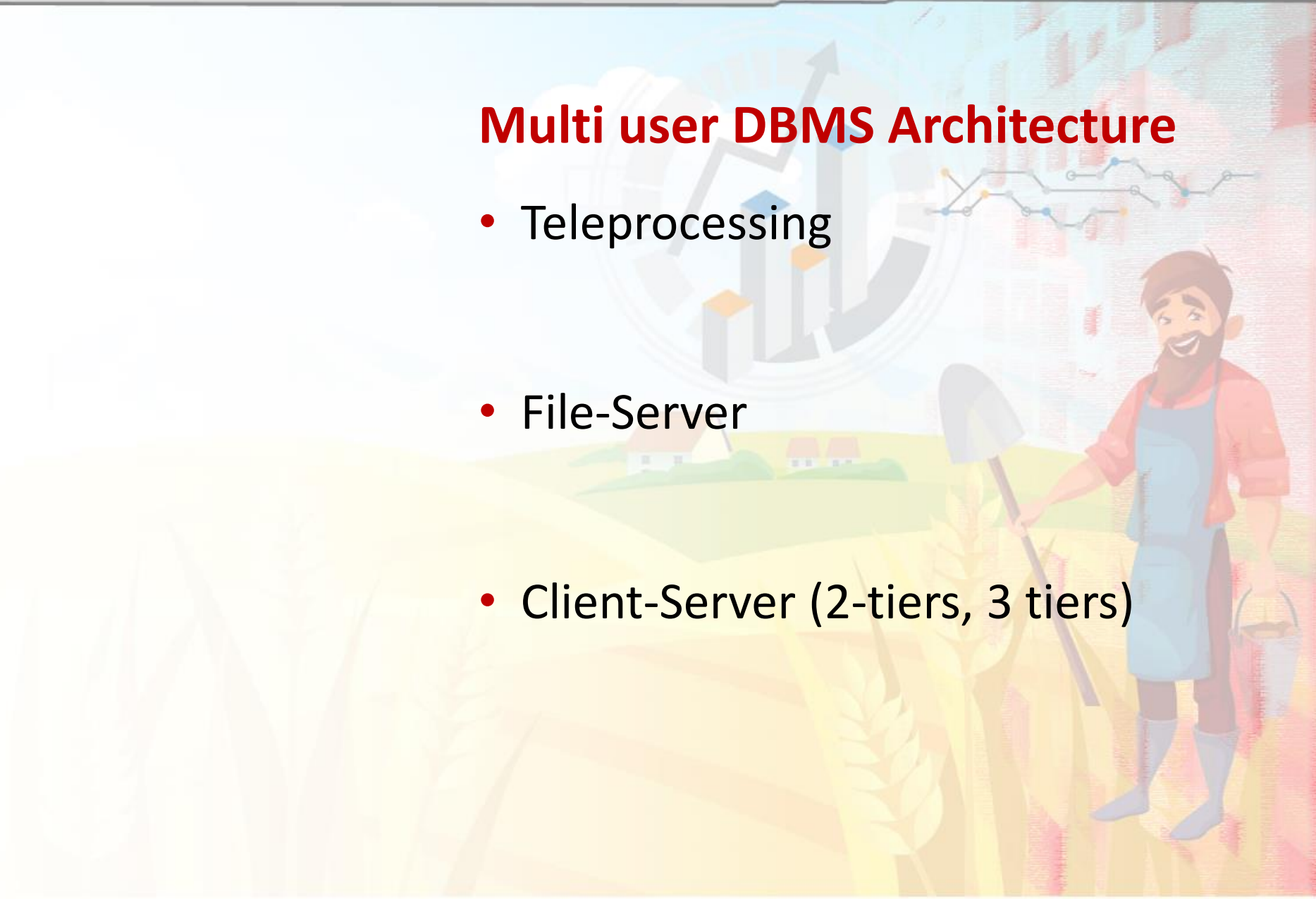


Form Processing & Report Processing Applications are built by using VB, DOT Net or PHP programming. Reports can be developed by using Crystal Reports tool. Query Processing can be managed by using vendors SQL tool or 3rd party tools such as TOAD, SQL Developer etc.

Introduction to DBMS Components & Architecture

Multi user DBMS Architecture

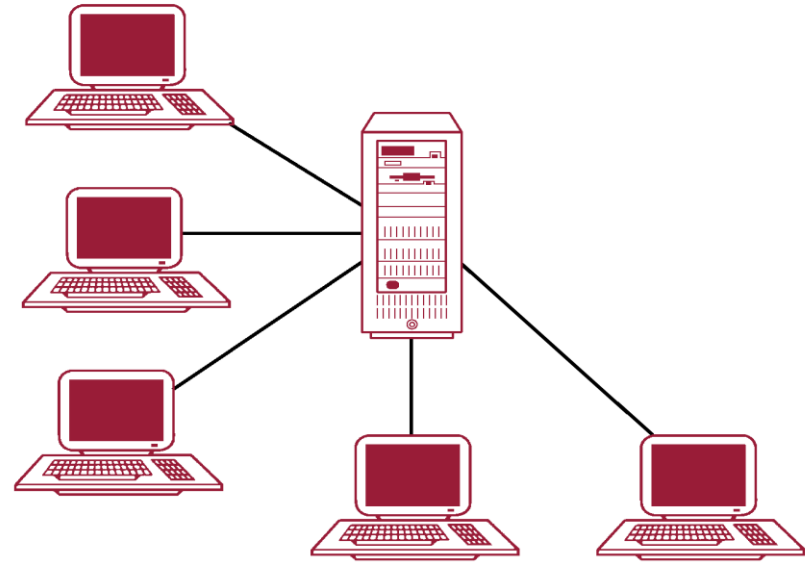
- Teleprocessing
- File-Server
- Client-Server (2-tiers, 3 tiers)



Introduction to DBMS Components & Architecture

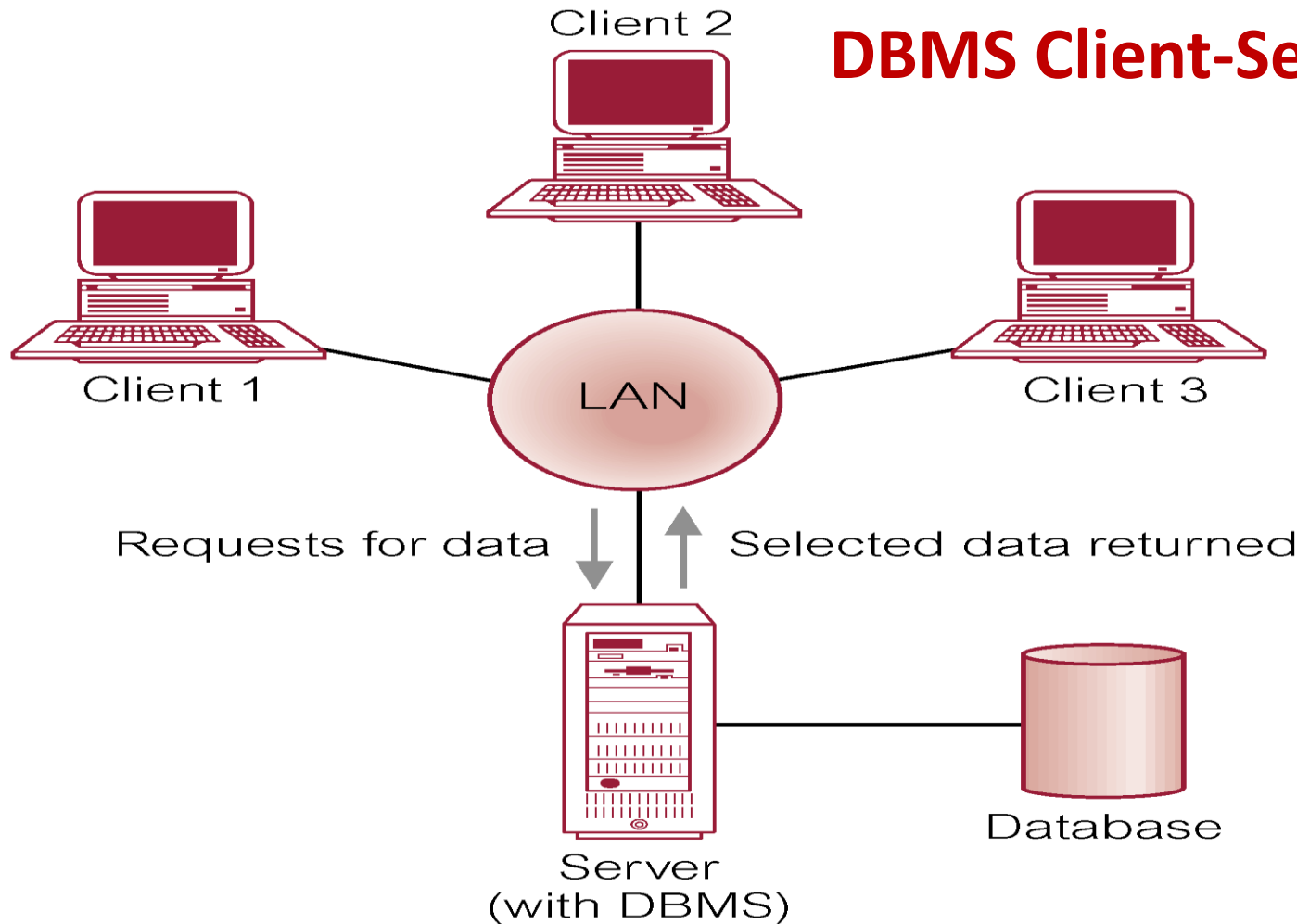
Teleprocessing

- Traditional architecture
- Single mainframe with a number of terminals attached
- Trend is now towards downsizing



Introduction to DBMS Components & Architecture

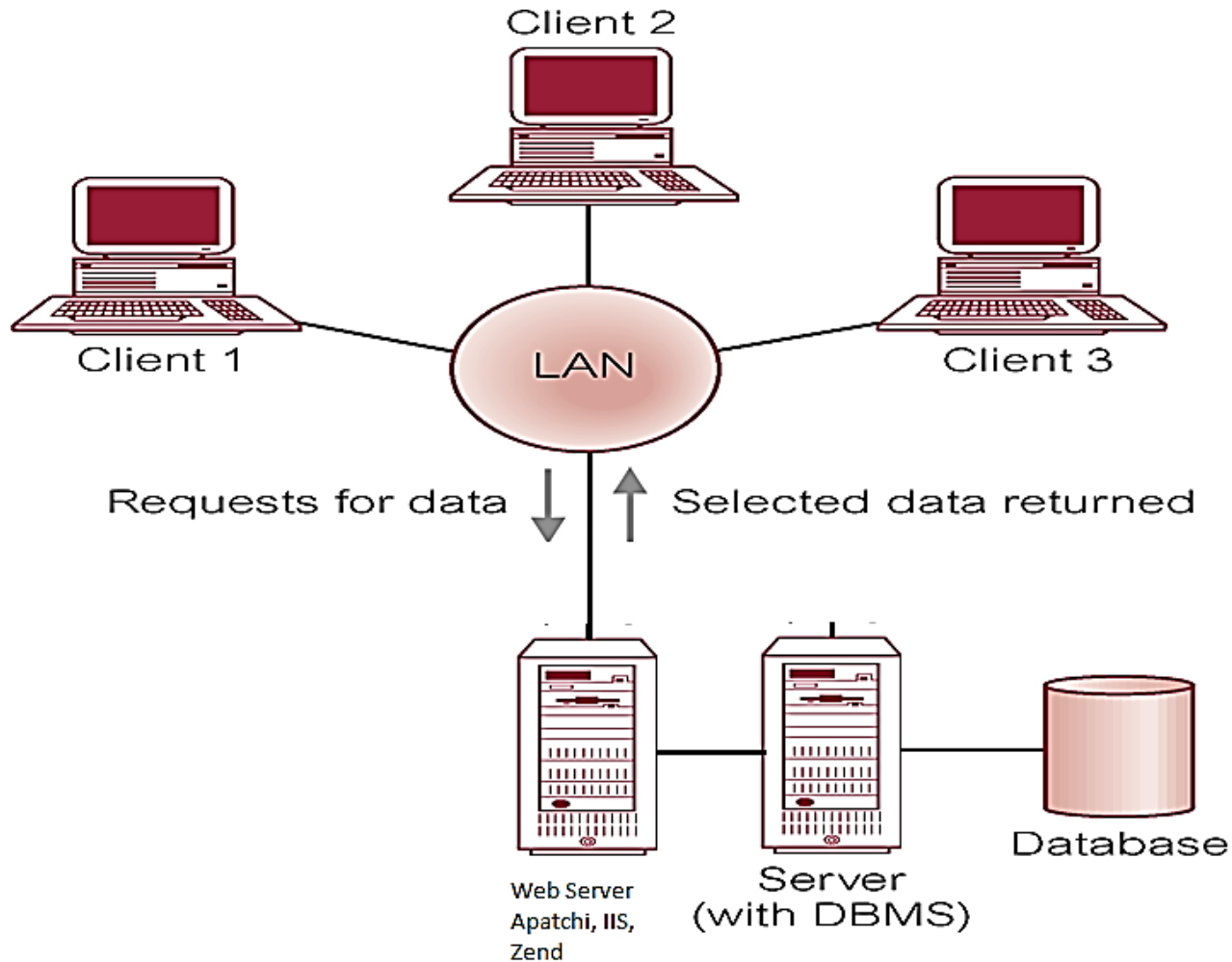
DBMS Client-Server Architecture (2-tiers)



DB Connections can be managed by
ODBC, JDBC

Introduction to DBMS Components & Architecture

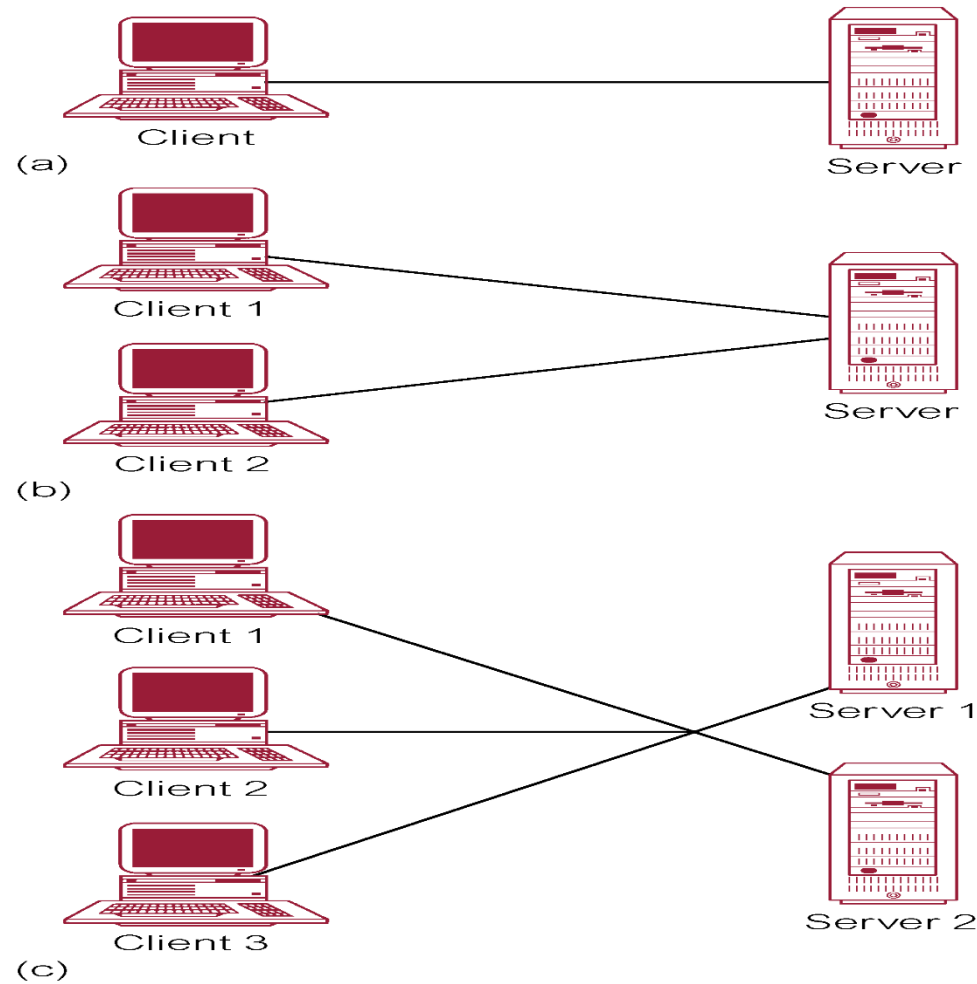
DBMS Client-Server Architecture (3-tiers)



DB Connections can be managed by using MS ODBC, JDBC

Introduction to DBMS Components & Architecture

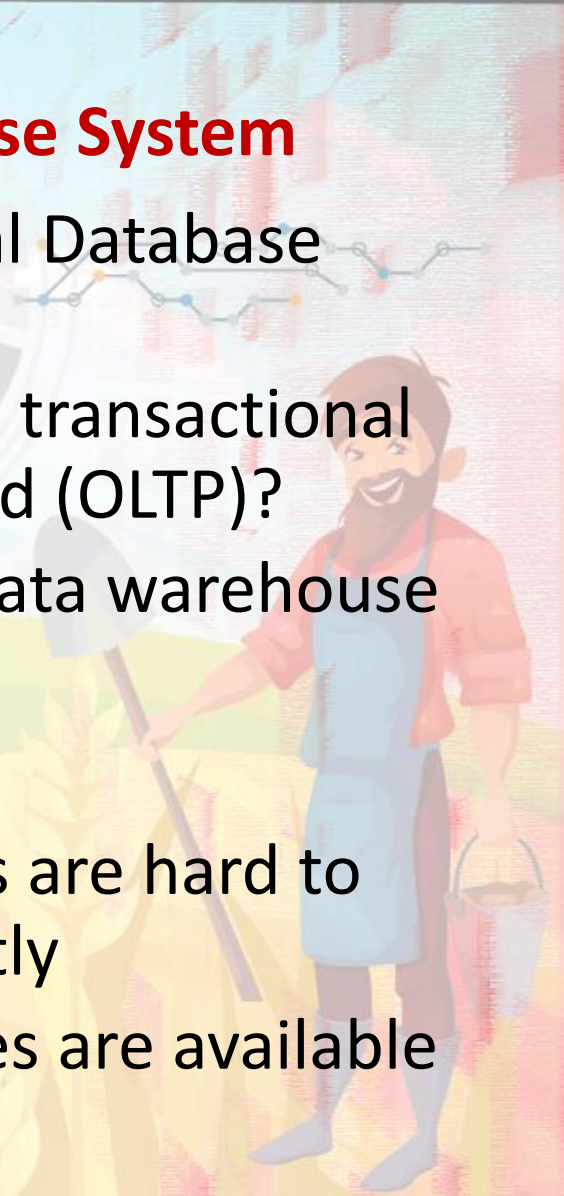
Alternate Client-Server Architectures



Introduction to DBMS Components & Architecture

Paradigm of Database System

- Logical/ Conceptual Database Design
 - How should the transactional data be modeled (OLTP)?
 - Designing the data warehouse schema (OLAP)
- Query Processing
 - Analysis queries are hard to answer efficiently
 - What techniques are available to the DBMS?



Introduction to DBMS Components & Architecture

Paradigm of Database System

- Physical Database Design (DBA)
 - How should the data be organized on disk?
 - What data structures should be used?
- Data Mining
 - What use is all this data?
 - Which questions should we ask our data warehouse OLAP?
- Data Science/Big Data

END

Chapter 4

DBMS Three Levels Architecture

- DBMS Three Levels Architecture
- ANSI-SPARC Three-Level Architecture
- Data Independence
- Degrees of Data Abstraction
- Schema & Instance
- Schema Objects
- Database State



Introduction to DBMS Three Levels Architecture

DBMS Three Levels Architecture

- **External Level**

User's view of the database

Describes that part of database that is relevant to a particular user

- **Conceptual Level**

Community view of the database

Describes what data is stored in database and relationships among the data

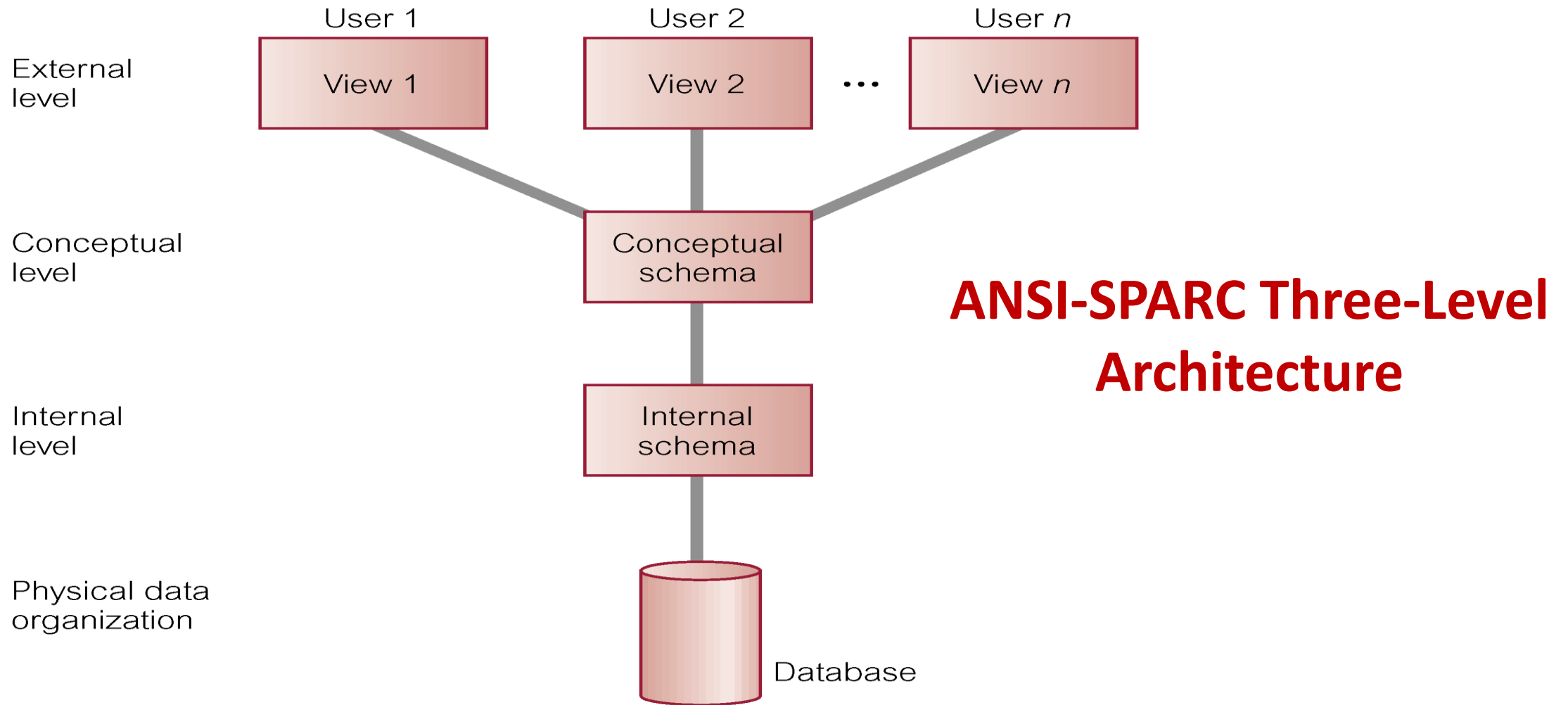
- **Internal Level**

Physical representation of the database on the computer

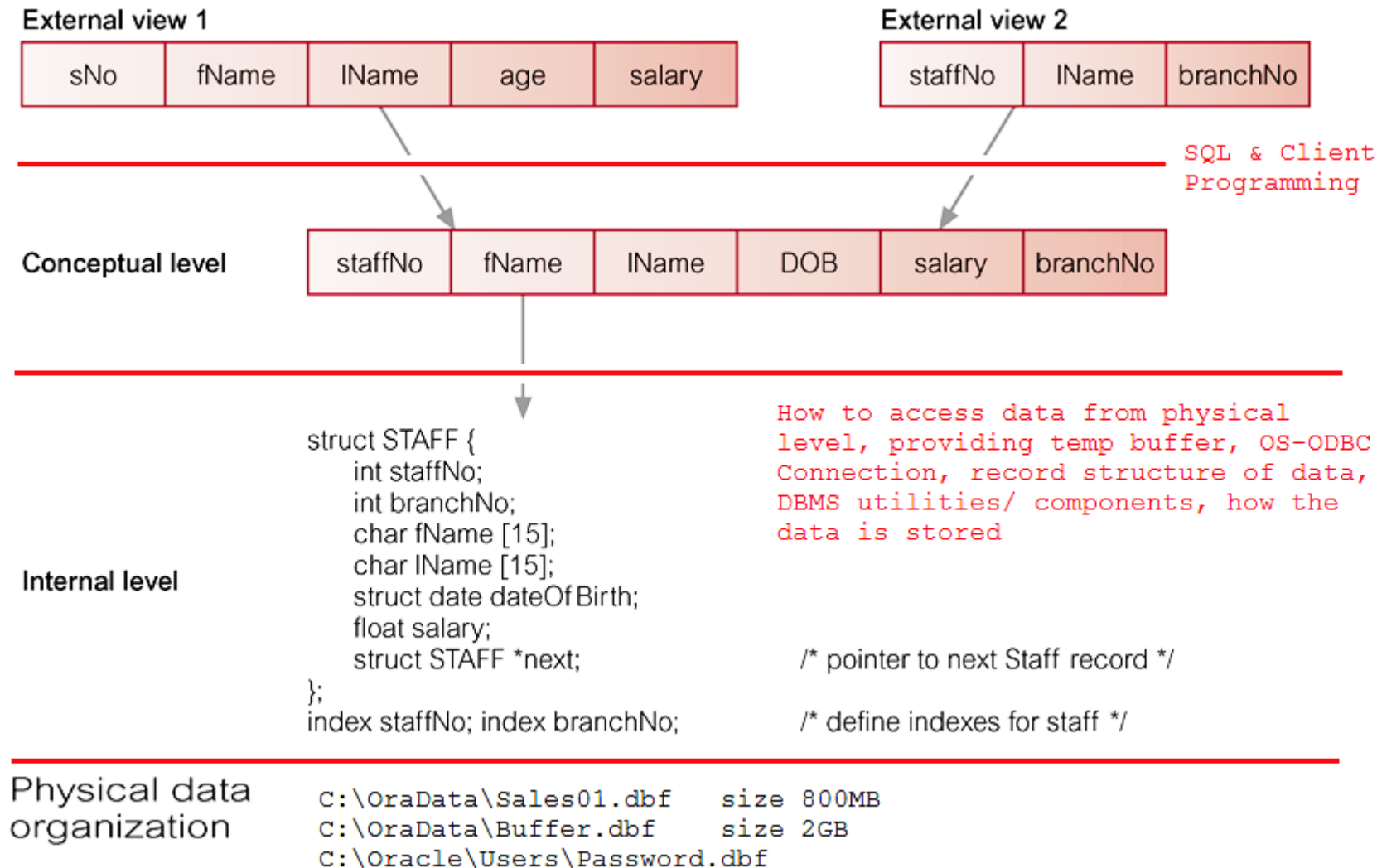
Describes how the data is stored in the database



Introduction to DBMS Three Levels Architecture



Introduction to DBMS Three Levels Architecture

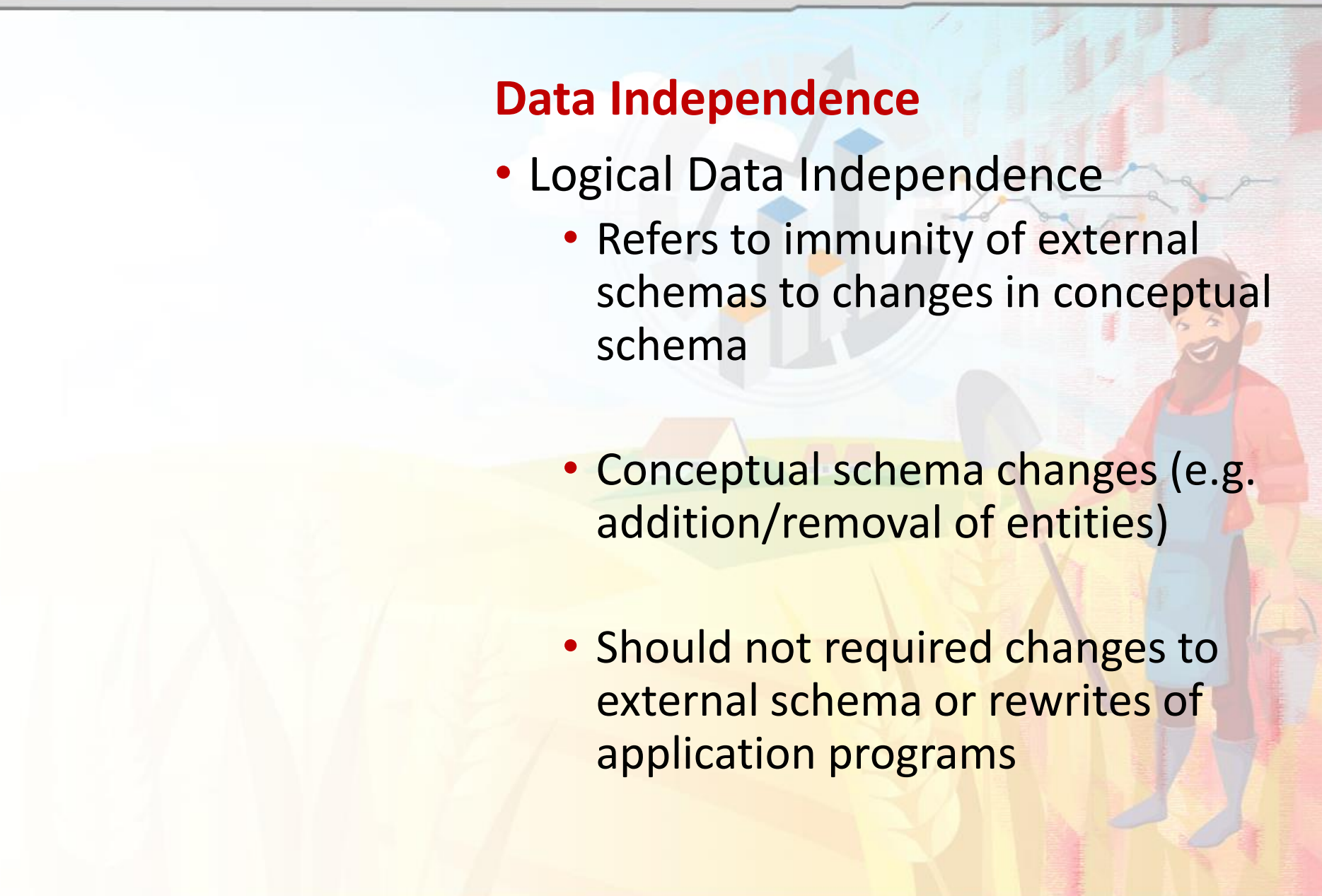


ANSI-SPARC Three-Levels Architecture

Introduction to DBMS Three Levels Architecture

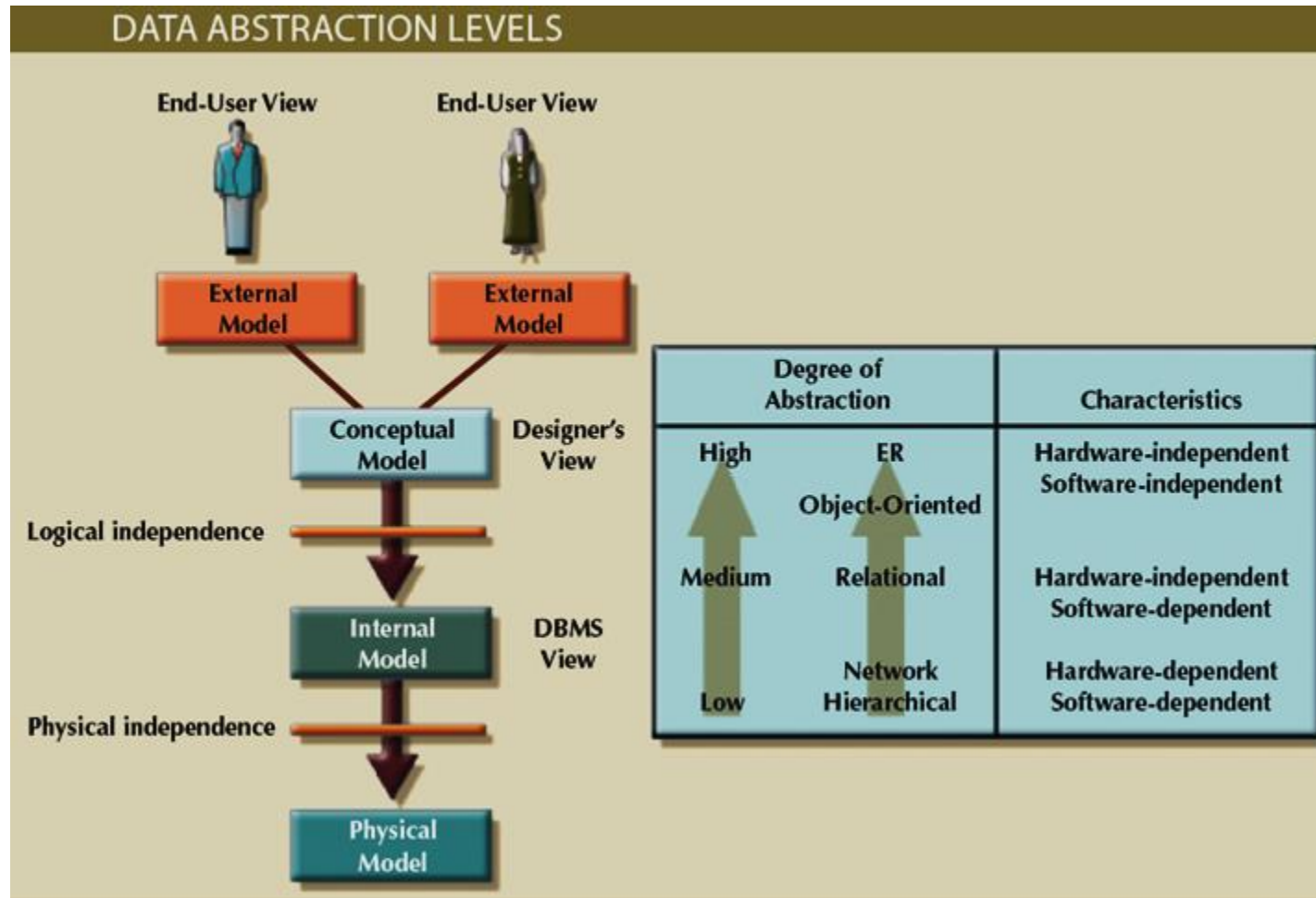
Data Independence

- Logical Data Independence
 - Refers to immunity of external schemas to changes in conceptual schema
- Conceptual schema changes (e.g. addition/removal of entities)
- Should not required changes to external schema or rewrites of application programs



Introduction to DBMS Three Levels Architecture

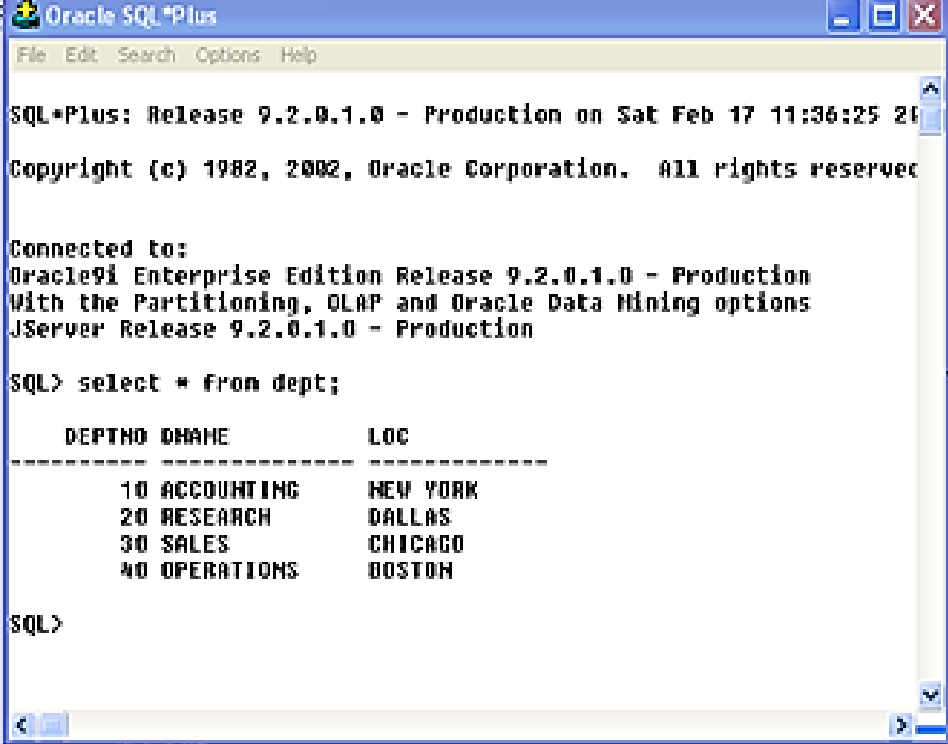
Degrees of Data Abstraction



Introduction to DBMS Three Levels Architecture

- Similar to types and variables in programming languages
- Schema – the logical structure of the database
 - E.g., the database consists of information about a set of customers and accounts and the relationship between them
 - Analogues to type information of a variable in a program
 - **Physical schema**: database design at the physical level
 - **Logical schema**: database design at the logical level
 - **Sub-Schema**: there may be several of these at the view level, each a different view of database
- **Instance** – the actual content of the database at a particular point in time
 - Analogous to the value of the variable

Schema & Instance



The screenshot shows the Oracle SQL*Plus command-line interface. It displays the version (9.2.0.1.0), release date (Sat Feb 17 11:36:25 2000), and copyright information. It also shows the connection details for Oracle9i Enterprise Edition. The user has executed the query 'select * from dept;', and the results are displayed in a table format with columns DEPTNO, DNAME, and LOC.

```
Oracle SQL*Plus
File Edit Search Options Help

SQL*Plus: Release 9.2.0.1.0 - Production on Sat Feb 17 11:36:25 2000
Copyright (c) 1982, 2002, Oracle Corporation. All rights reserved.

Connected to:
Oracle9i Enterprise Edition Release 9.2.0.1.0 - Production
With the Partitioning, OLAP and Oracle Data Mining options
JSERVER Release 9.2.0.1.0 - Production

SQL> select * from dept;

  DEPTNO DNAME          LOC
-----
10 ACCOUNTING        NEW YORK
20 RESEARCH          DALLAS
30 SALES              CHICAGO
40 OPERATIONS         BOSTON

SQL>
```

Introduction to DBMS Three Levels Architecture

Schema Objects

1. Tables
2. Indexes
3. Constraints
4. Sequences
5. Views
6. Clusters
7. Triggers
8. Links



Introduction to DBMS Three Levels Architecture

Schema Objects

9. User Defined

10. Types

11. Synonym

12. Dimensions

13. Procedural Program codes in
PL/SQL, Java, C etc.



Introduction to DBMS Three Levels Architecture

Database State

STUDENT	Name	StudentNumber	Class	Major
	Smith	17	1	CS
	Brown	8	2	CS

COURSE	CourseName	CourseNumber	CreditHours	Department
	Intro to Computer Science	CS1310	4	CS
	Data Structures	CS3320	4	CS
	Discrete Mathematics	MATH2410	3	MATH
	Database	CS3380	3	CS

SECTION	SectionIdentifier	CourseNumber	Semester	Year	Instructor
	85	MATH2410	Fall	98	King
	92	CS1310	Fall	98	Anderson
	102	CS3320	Spring	99	Knuth
	112	MATH2410	Fall	99	Chang
	119	CS1310	Fall	99	Anderson
	135	CS3380	Fall	99	Stone

GRADE_REPORT	StudentNumber	SectionIdentifier	Grade
	17	112	B
	17	119	C
	8	85	A
	8	92	A
	8	102	B
	8	135	A

PREREQUISITE	CourseNumber	PrerequisiteNumber
	CS3380	CS3320
	CS3380	MATH2410
	CS3320	CS1310

Chapter 5

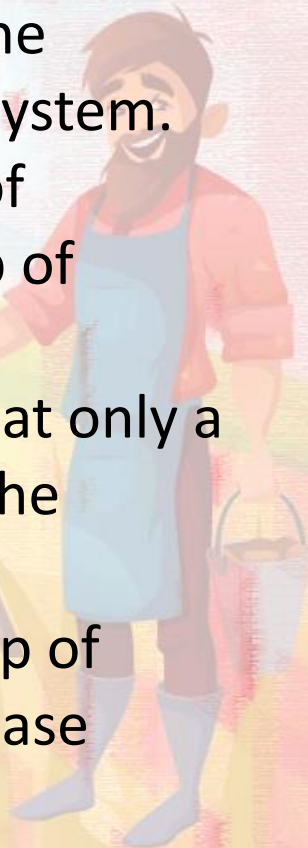
- Who is DBA?
- DBA & Databases
- Network Administrator
- Application Developers
- DBA's Tasks
- DBA's Responsibilities



Introduction to DBA Responsibilities & Tasks

Who is DBA?

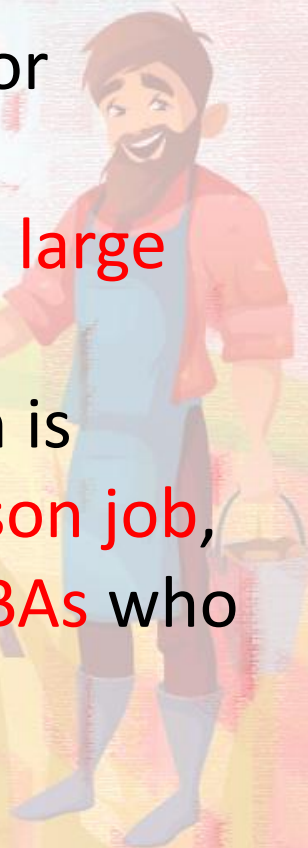
- A Database Administrator (DBA) is a person or a group of person who are responsible for managing all the activities related to database system.
- DBA job requires a high level of expertise by a person or group of persons.
- There are very rare chances that only a single person can manage all the database system activities so companies always have a group of people who take care of database system.



Introduction to DBA Responsibilities & Tasks

DBA & Databases

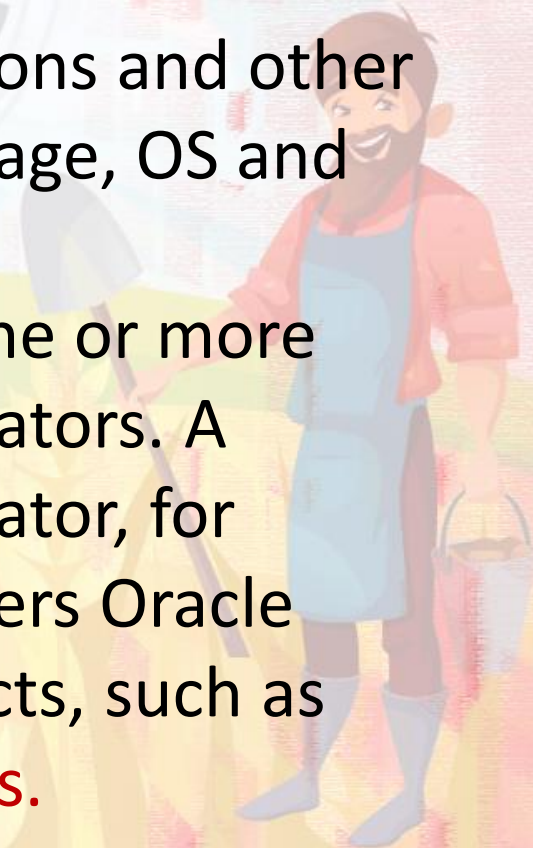
- Each database requires at **least one** database administrator (DBA).
- A Database system can be **large** and can have **many users**.
- A database administration is sometimes not a **one-person job**, but a job for a **group of DBAs** who share responsibility.



Introduction to DBA Responsibilities & Tasks

Network Administrator

- Network administrator coordinates with the DBA for database connections and other issues such as storage, OS and hardware.
- Some sites have one or more network administrators. A network administrator, for example, administers Oracle networking products, such as **Oracle Net Services**.



Introduction to DBA Responsibilities & Tasks

Application Developers

- Designing and developing the database application
- Designing the database structure for an application
- Estimating storage requirements for an application
- Specifying modifications of the database structure for an application



Introduction to DBA Responsibilities & Tasks

Application Developers

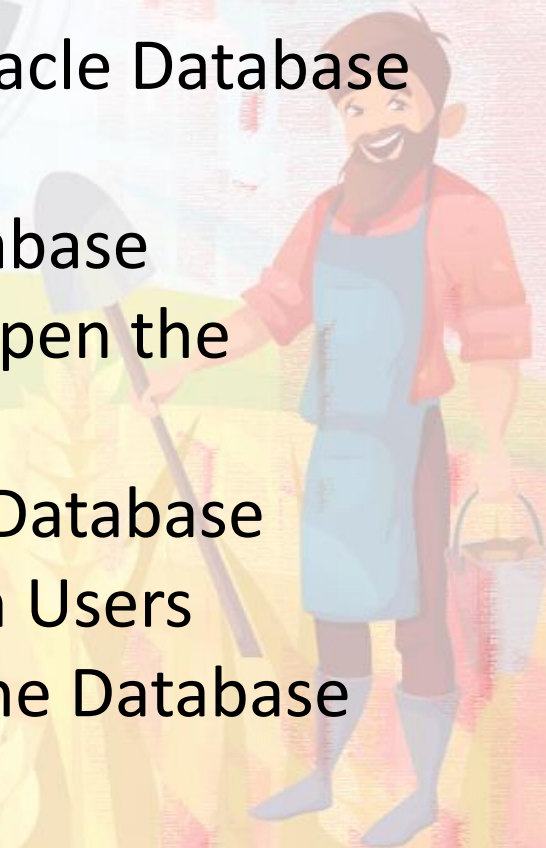
- Relaying this information to a database administrator
- Tuning the application during development
- Establishing security measures for an application during development
- Database Server Programming using Oracle PL/SQL



Introduction to DBA Responsibilities & Tasks

DBA's Tasks

- Task 1:** Evaluate the Database Server Hardware
- Task 2:** Install the Oracle Database Software
- Task 3:** Plan the Database
- Task 4:** Create and Open the Database
- Task 5:** Back Up the Database
- Task 6:** Enroll System Users
- Task 7:** Implement the Database Design



Introduction to DBA Responsibilities & Tasks

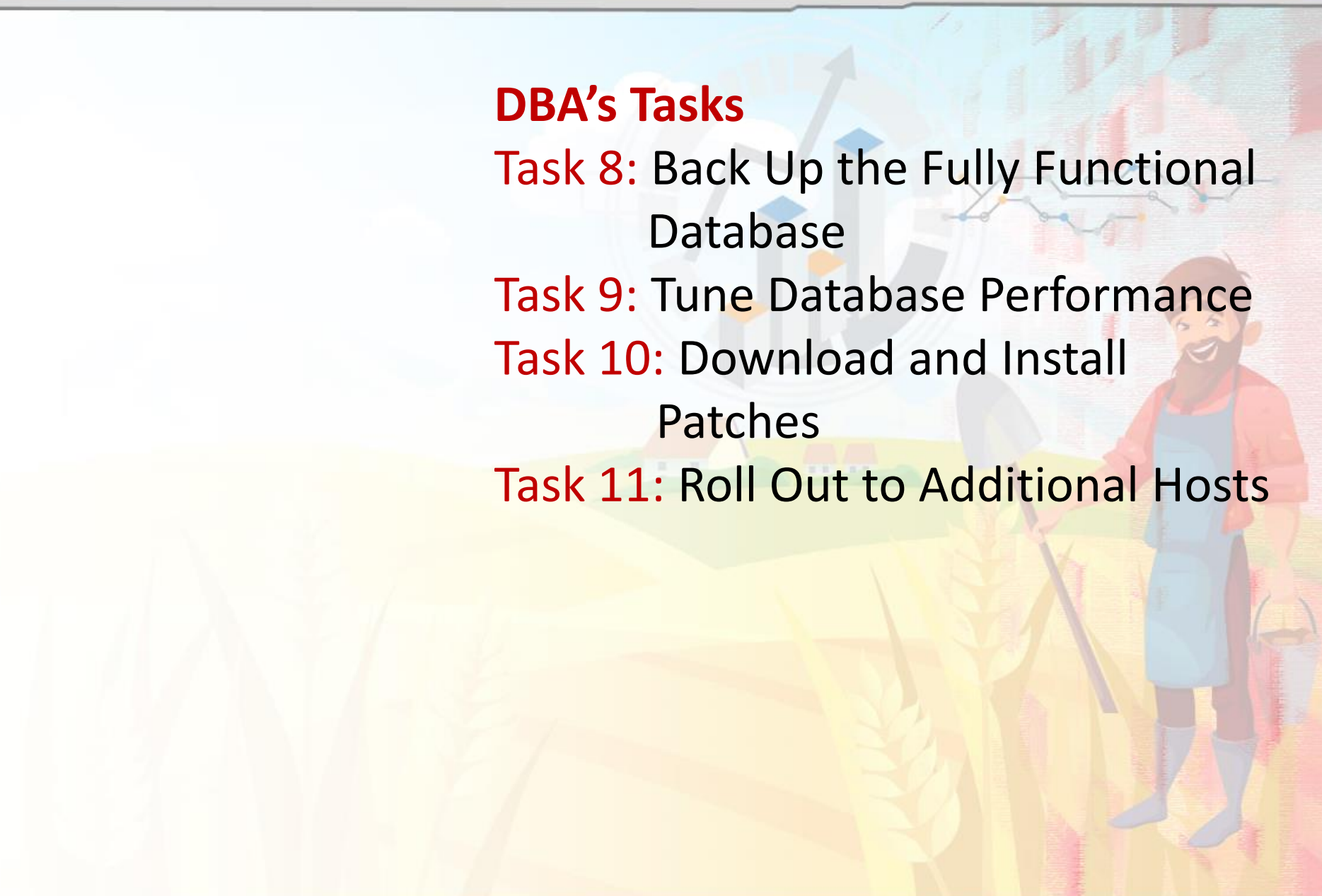
DBA's Tasks

Task 8: Back Up the Fully Functional Database

Task 9: Tune Database Performance

Task 10: Download and Install Patches

Task 11: Roll Out to Additional Hosts



Introduction to DBA Responsibilities & Tasks

DBA's Responsibilities

- Installing and upgrading the Oracle Database server and application tools
- Allocating system storage and planning future storage requirements for the database system
- Creating primary database storage structures (tablespaces) after application developers have designed an application



Introduction to DBA Responsibilities & Tasks

DBA's Responsibilities

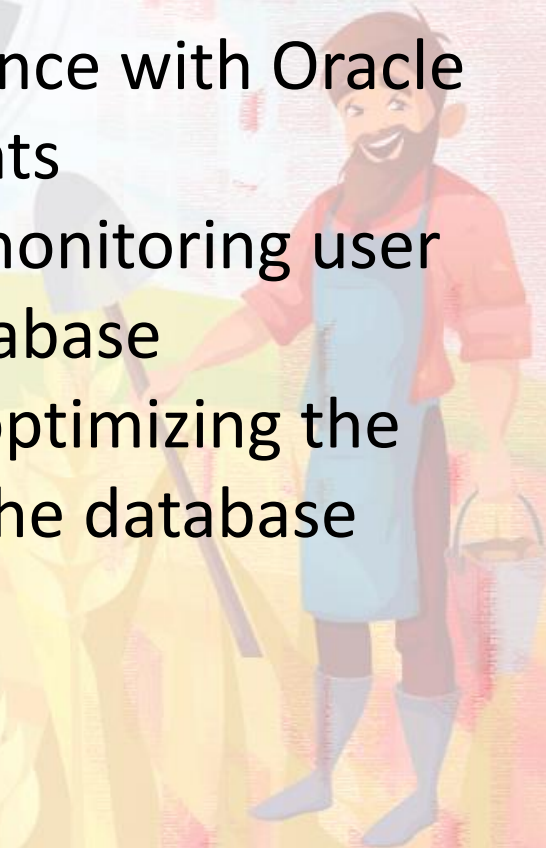
- Creating primary objects (tables, views, indexes) once application developers have designed an application
- Modifying the database structure, as necessary, from information given by application developers



Introduction to DBA Responsibilities & Tasks

DBA's Responsibilities

- Enrolling users and maintaining system security
- Ensuring compliance with Oracle license agreements
- Controlling and monitoring user access to the database
- Monitoring and optimizing the performance of the database



Introduction to DBA Responsibilities & Tasks

DBA's Responsibilities

- Planning for backup and recovery of database information
- Maintaining archived data on tape
- Backing up and restoring the database
- Contacting Oracle for technical support

END

Chapter 6

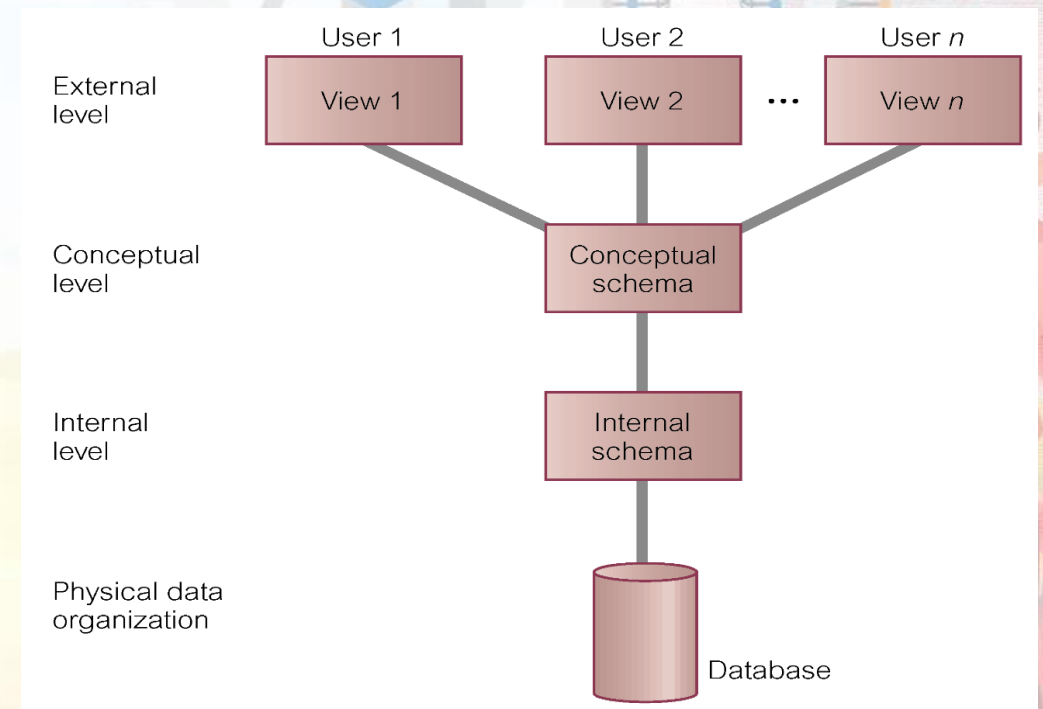
Physical Database Design

- Physical level in ANSI SPARC Arch.
- Purposes of Physical Database Design
- Factors that influence Physical Database Design
- Design decisions about Indexing
- Performance & Tuning
- Tuning Indexes



Overview of Physical Database Design

Physical Level in ANSI SPARC Architecture



Ref. DBA-4 (4)

Overview of Physical Database Design

Purposes

Meeting the expectations of Database Designer for the database, following are two main purposes of Physical Database Design for a DBA.

- Managing Storage Structure for database or DBMS
- Performance & Tuning



Overview of Physical Database Design

Factor (A): Analyzing the database queries and transactions

For each query, the following information is needed.

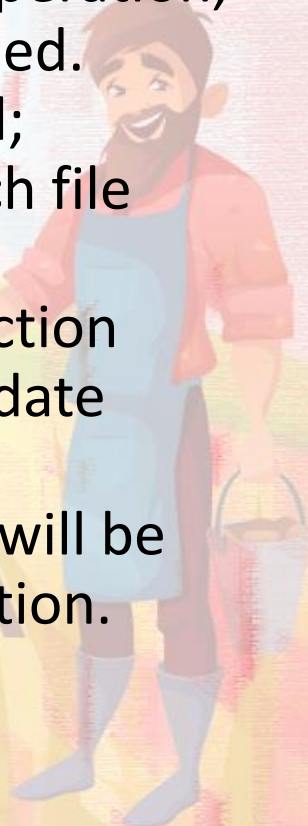
- The files that will be accessed by the query
- The attributes on which any selection conditions for the query are specified;
- The attributes on which any join conditions or conditions to link multiple tables or objects for the query are specified
- The attributes whose values will be retrieved by the query.

Overview of Physical Database Design

Factor (A): Analyzing the database queries and transactions

For each **update** transaction or operation, the following information is needed.

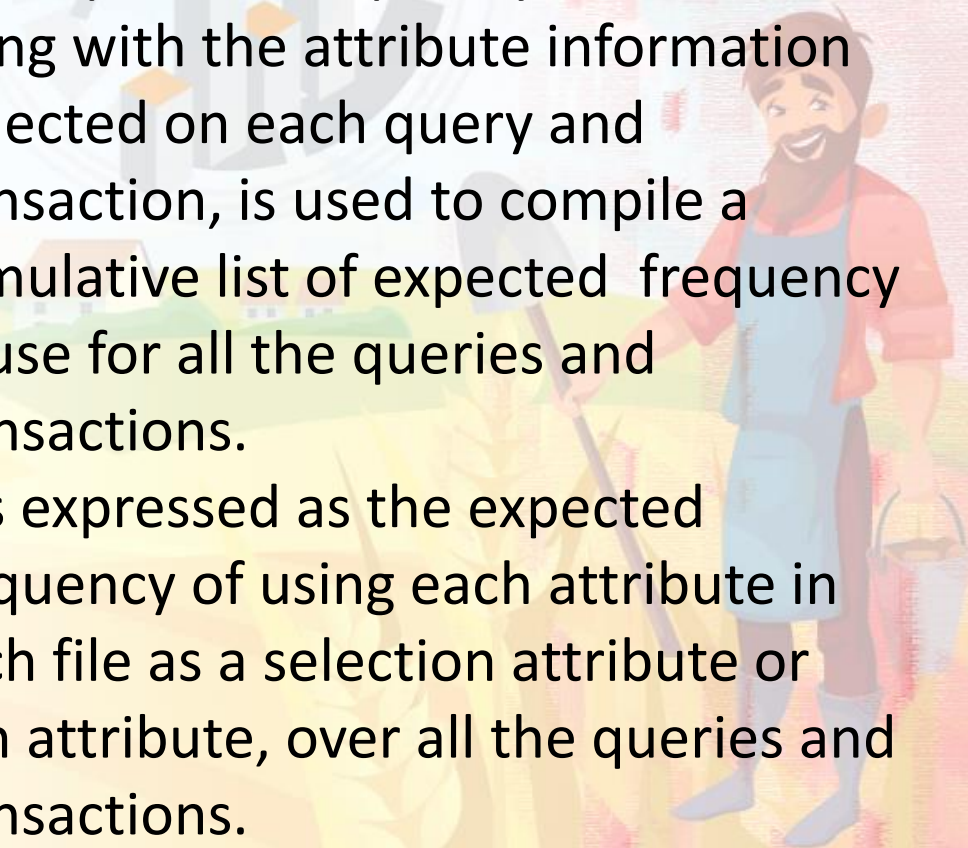
- The files that will be updated;
- The type of operation on each file (insert, update or delete);
- The attributes on which selection conditions for a delete or update operation are specified;
- The attributes whose values will be changed by an update operation.



Overview of Physical Database Design

Factor (B): Frequency with Queries and Transactions

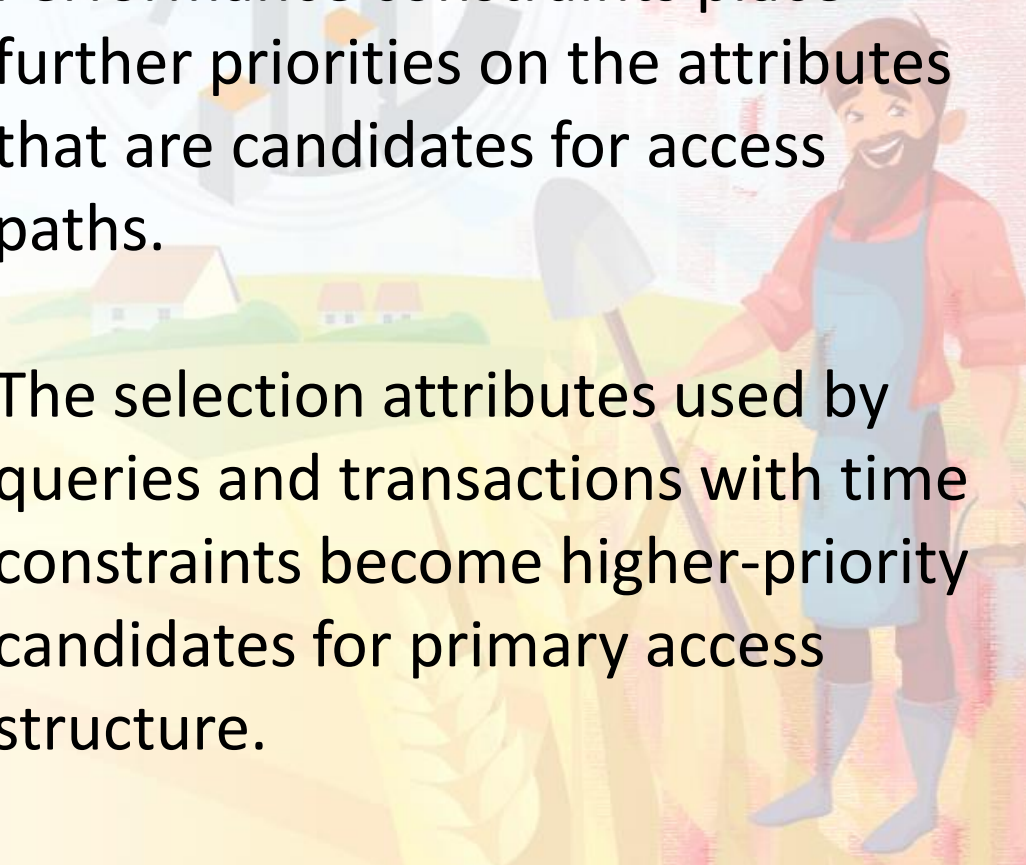
- The expected frequency information, along with the attribute information collected on each query and transaction, is used to compile a cumulative list of expected frequency of use for all the queries and transactions.
- It is expressed as the expected frequency of using each attribute in each file as a selection attribute or join attribute, over all the queries and transactions.



Overview of Physical Database Design

Factor (C): Time constraints of queries & transactions

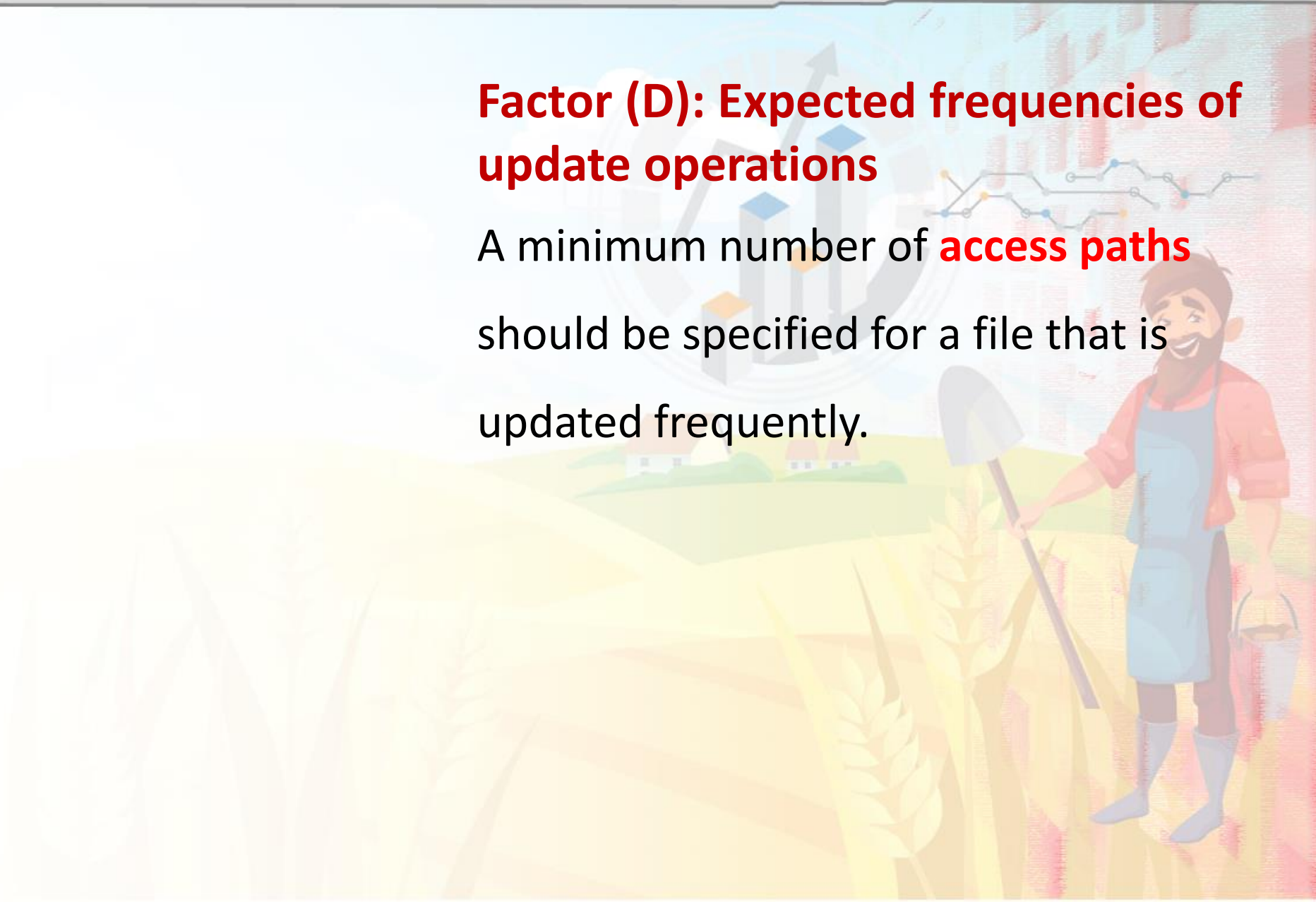
- Performance constraints place further priorities on the attributes that are candidates for access paths.
- The selection attributes used by queries and transactions with time constraints become higher-priority candidates for primary access structure.



Overview of Physical Database Design

Factor (D): Expected frequencies of update operations

A minimum number of **access paths** should be specified for a file that is updated frequently.



Overview of Physical Database Design

Factor (E): Uniqueness constraints on attributes

Access paths should be specified on all candidate key attributes

or set of attributes

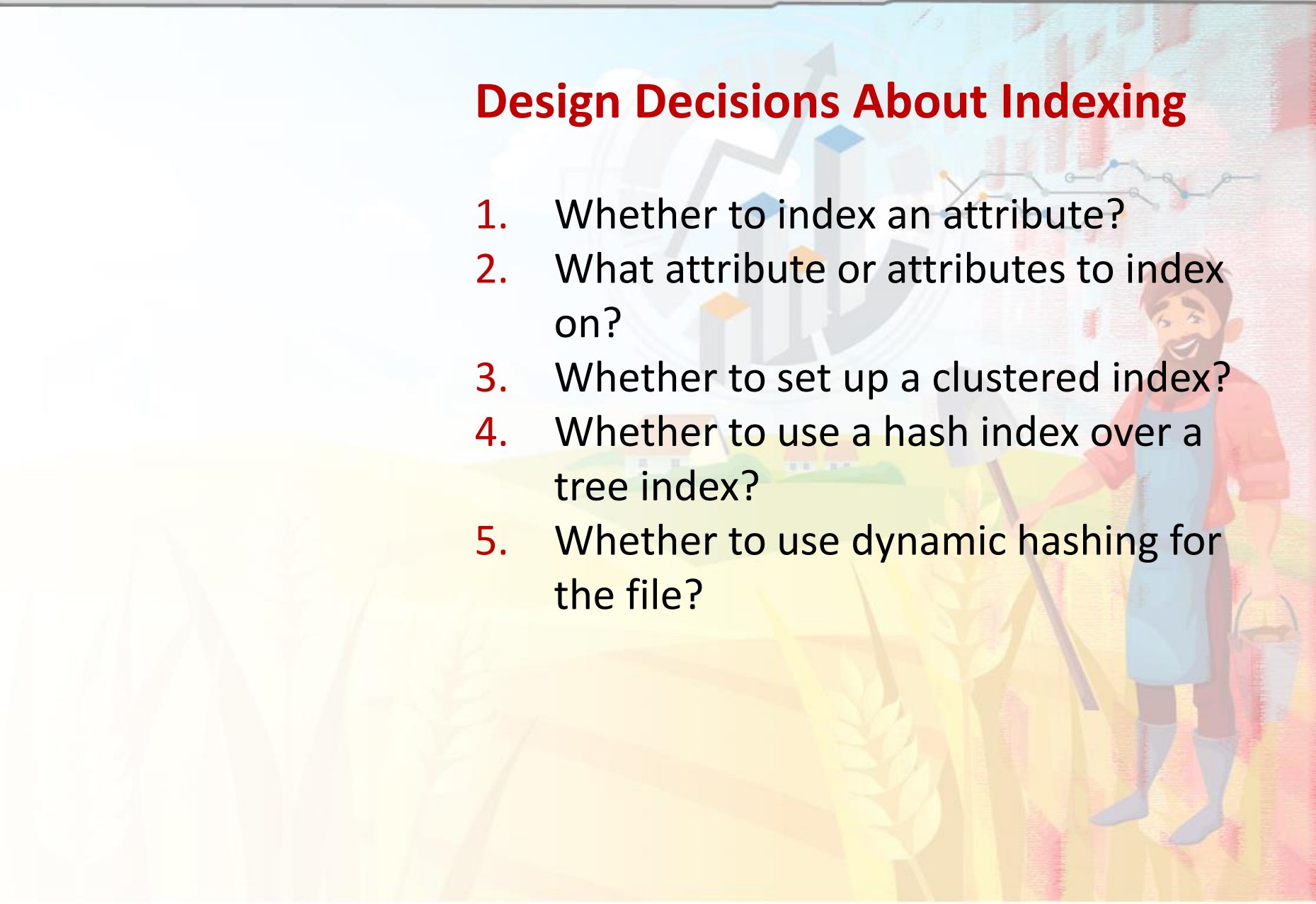
that are either the primary key or constrained to be unique.



Overview of Physical Database Design

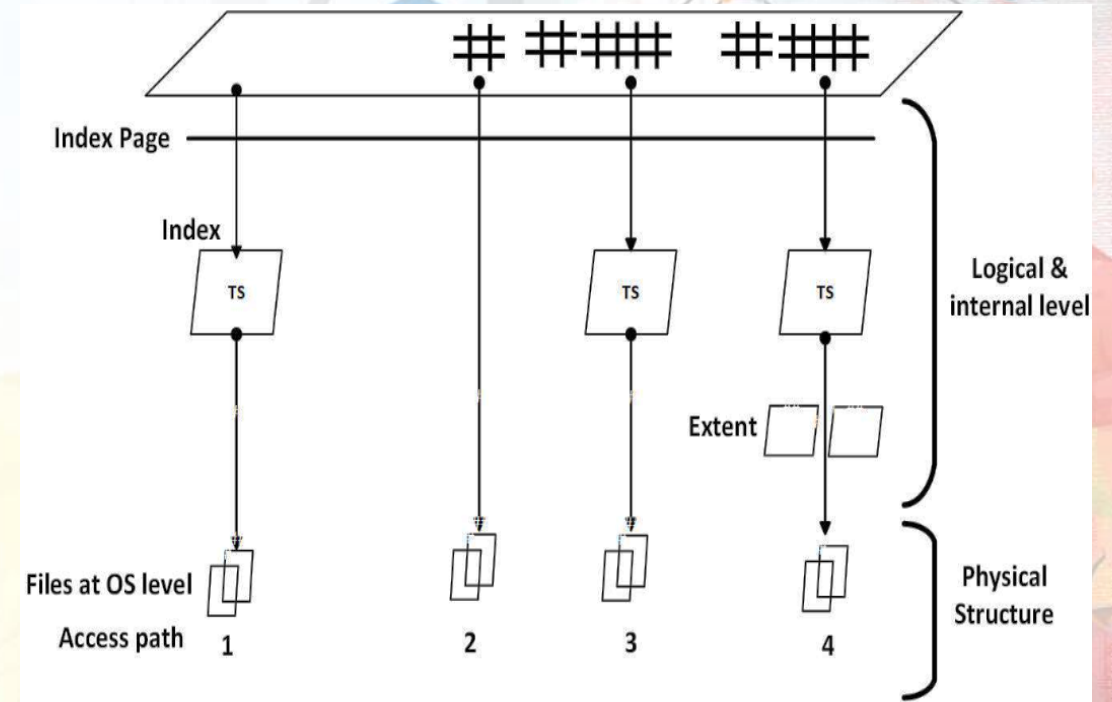
Design Decisions About Indexing

1. Whether to index an attribute?
2. What attribute or attributes to index on?
3. Whether to set up a clustered index?
4. Whether to use a hash index over a tree index?
5. Whether to use dynamic hashing for the file?



Overview of Physical Database Design

Access Path & Storage

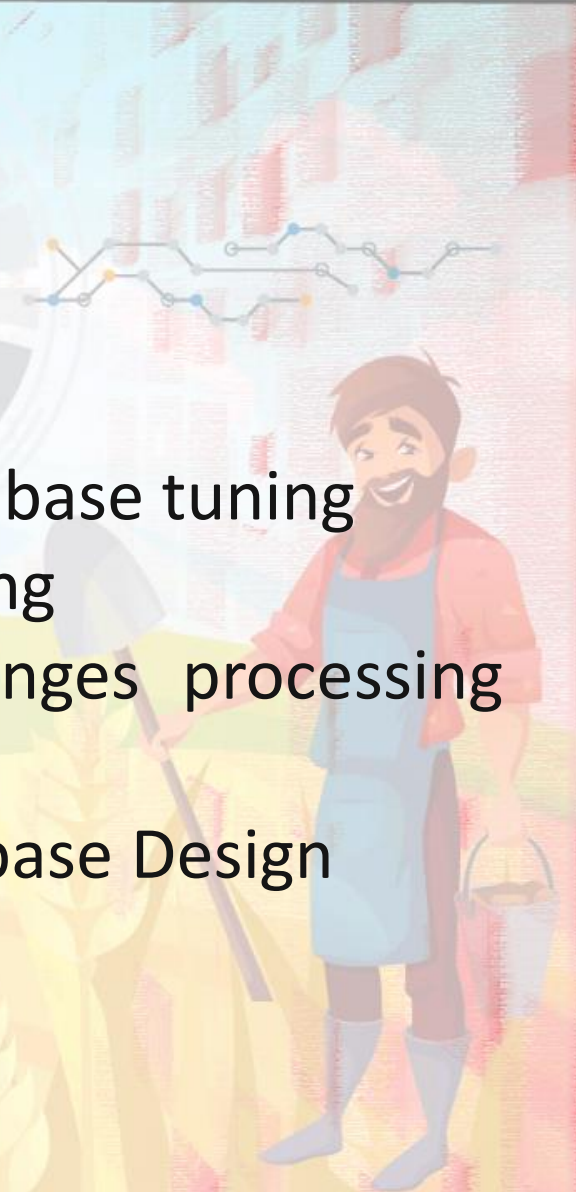


END

Chapter 7

Database Tuning

- Tuning & goals
- Tuning indexes
- Overview of database tuning
- Problems in tuning
- Dynamically changes processing requirements
- Changes in Database Design
- Tuning queries

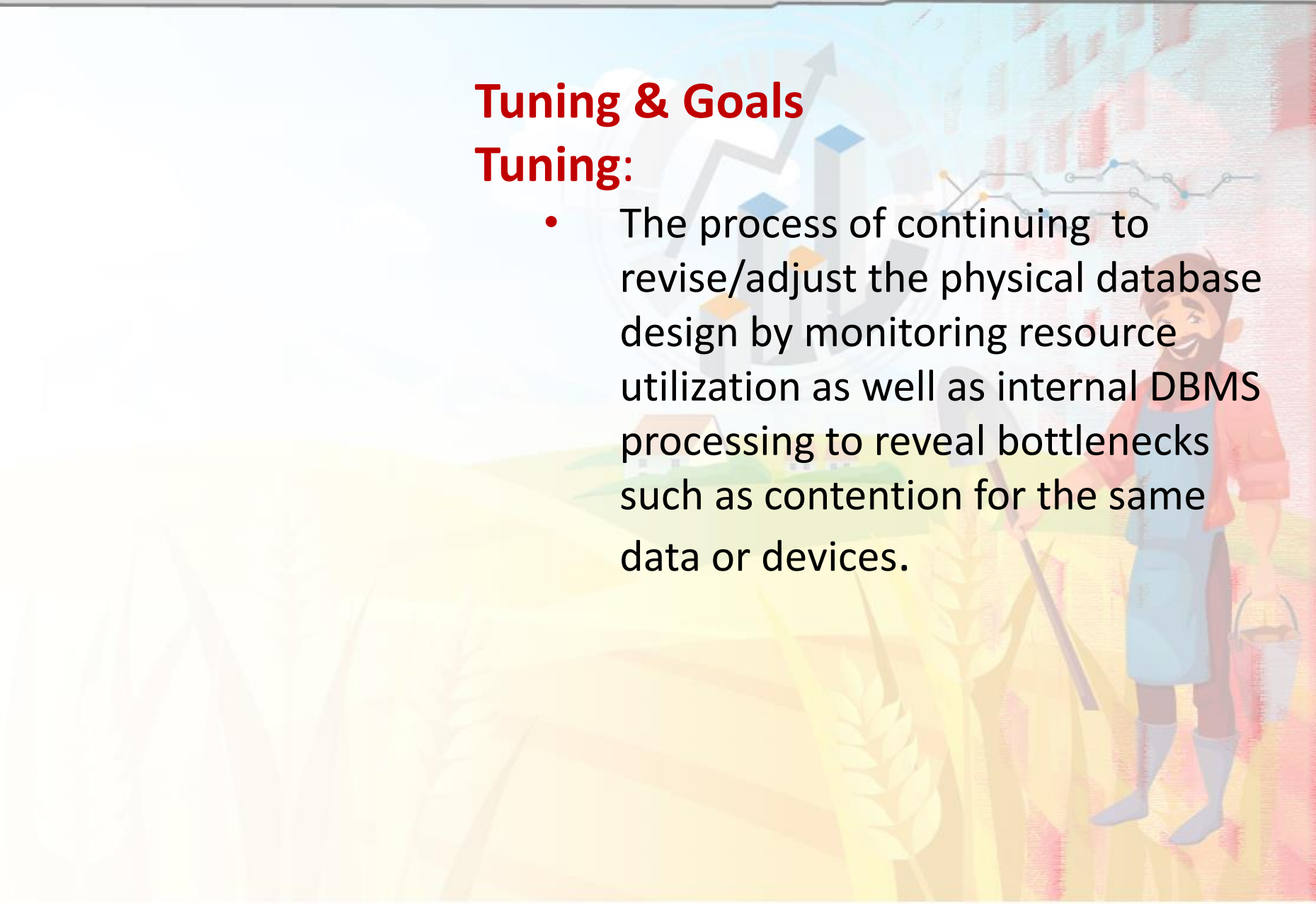


Introduction to Database Tuning

Tuning & Goals

Tuning:

- The process of continuing to revise/adjust the physical database design by monitoring resource utilization as well as internal DBMS processing to reveal bottlenecks such as contention for the same data or devices.

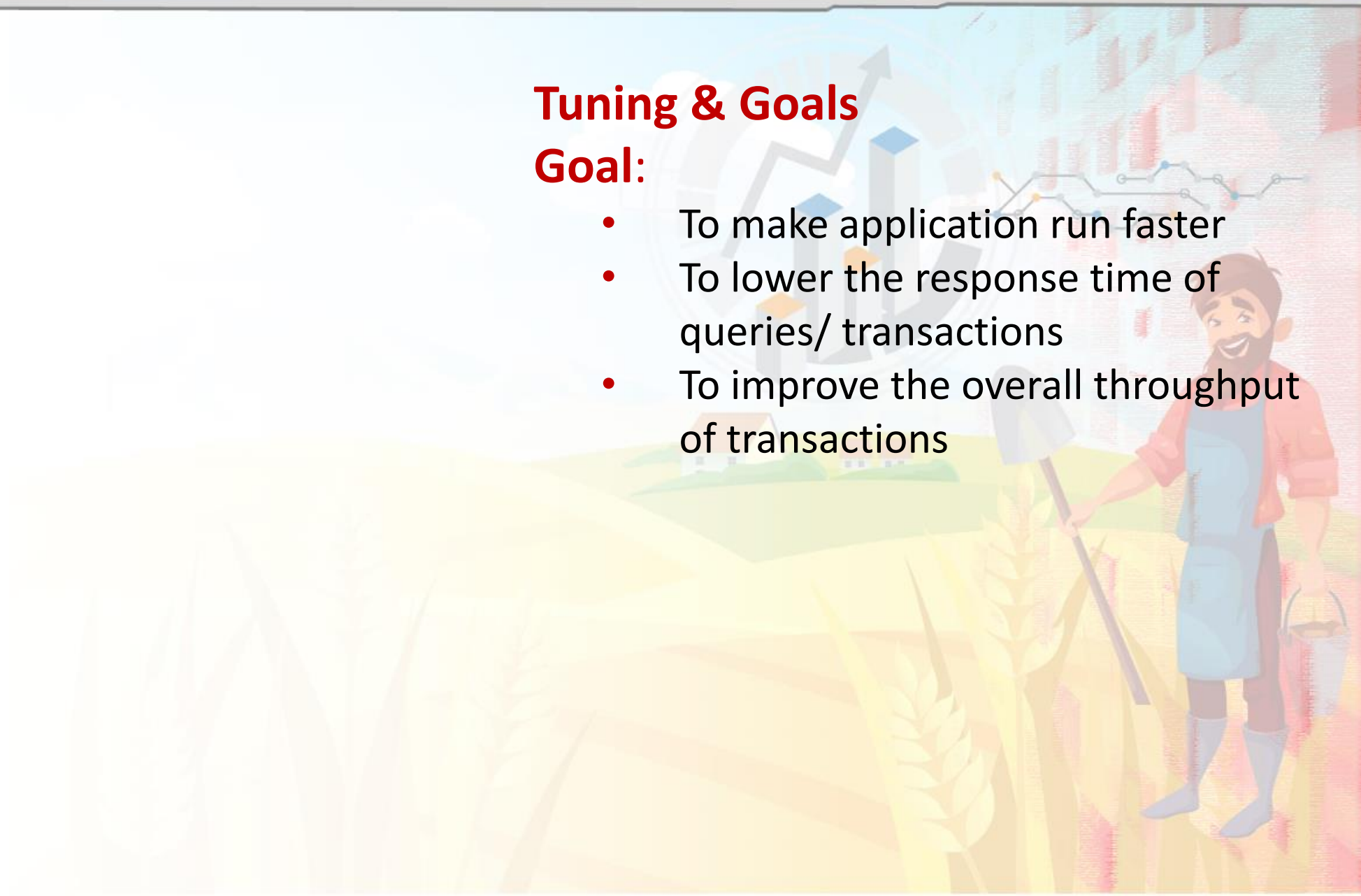


Introduction to Database Tuning

Tuning & Goals

Goal:

- To make application run faster
- To lower the response time of queries/ transactions
- To improve the overall throughput of transactions

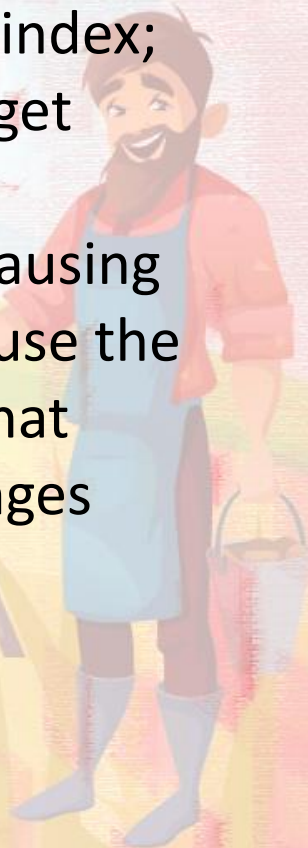


Introduction to Database Tuning

Tuning Indexes

Reasons to tuning indexes

- Certain queries may take too long to run for lack of an index;
- Certain indexes may not get utilized at all;
- Certain indexes may be causing excessive overhead because the index is on an attribute that undergoes frequent changes

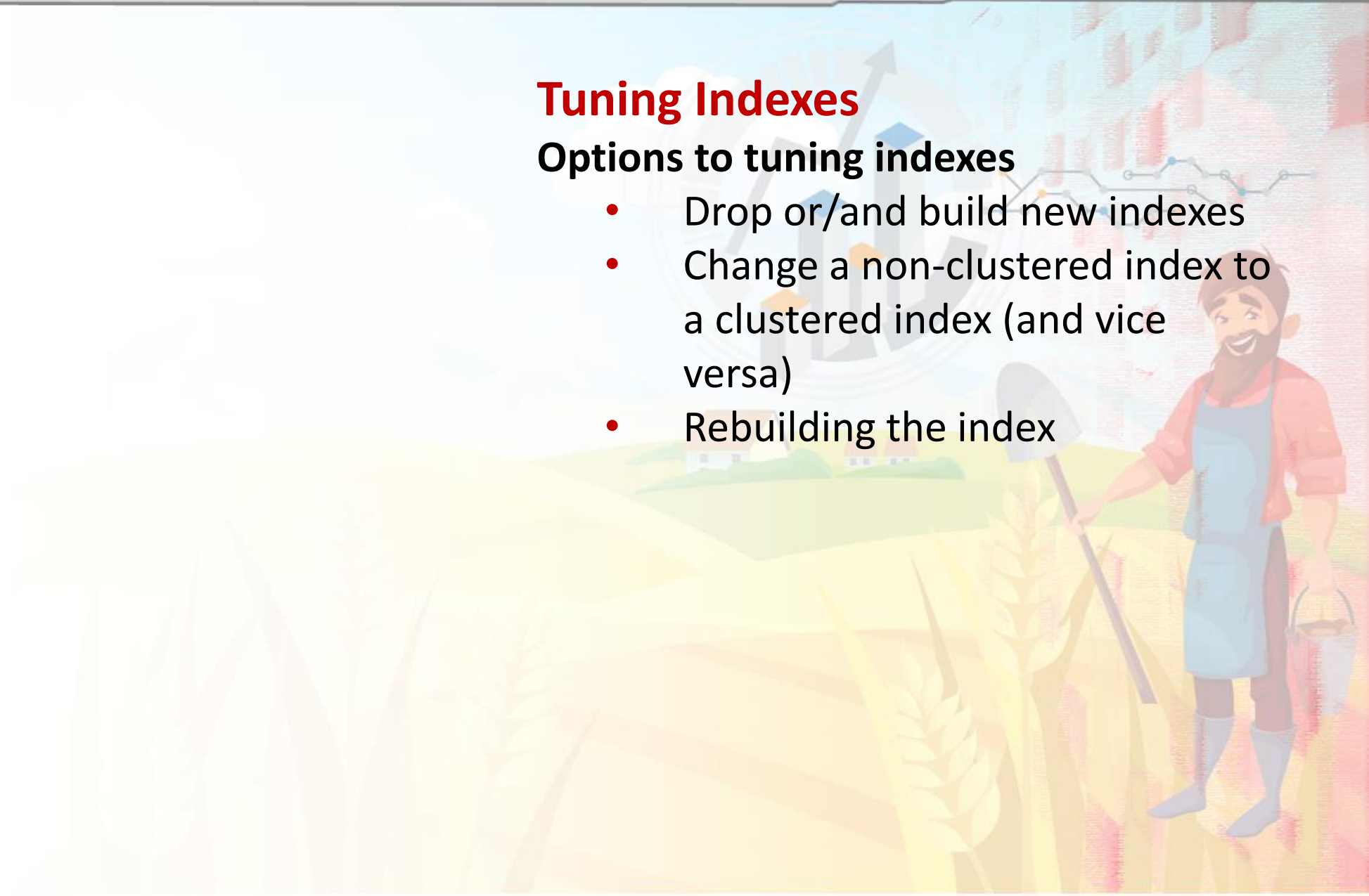


Introduction to Database Tuning

Tuning Indexes

Options to tuning indexes

- Drop or/and build new indexes
- Change a non-clustered index to a clustered index (and vice versa)
- Rebuilding the index

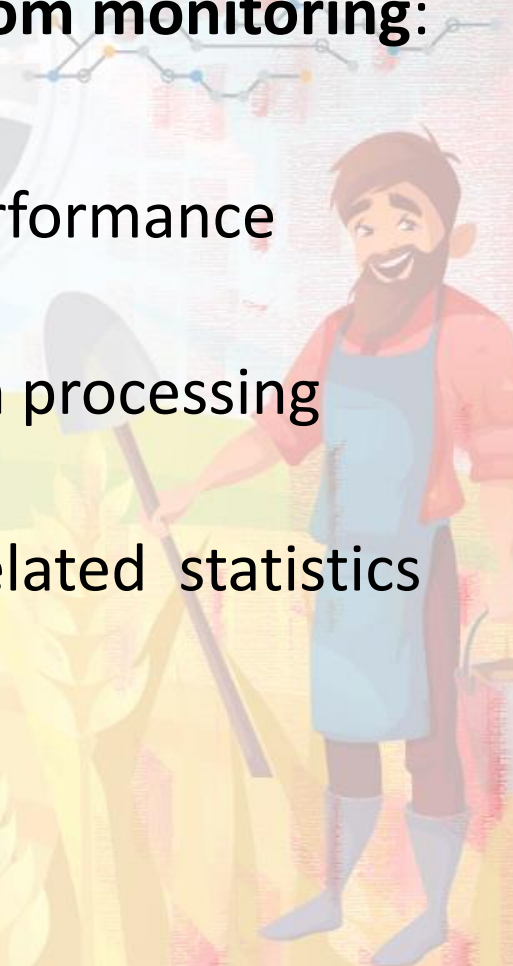


Introduction to Database Tuning

Overview of Database Tuning

Statistics obtained from monitoring:

- Storage statistics
- I/O and device performance statistics
- Query/transaction processing statistics
- Locking/logging related statistics
- Index statistics



Introduction to Database Tuning

Overview of Database Tuning

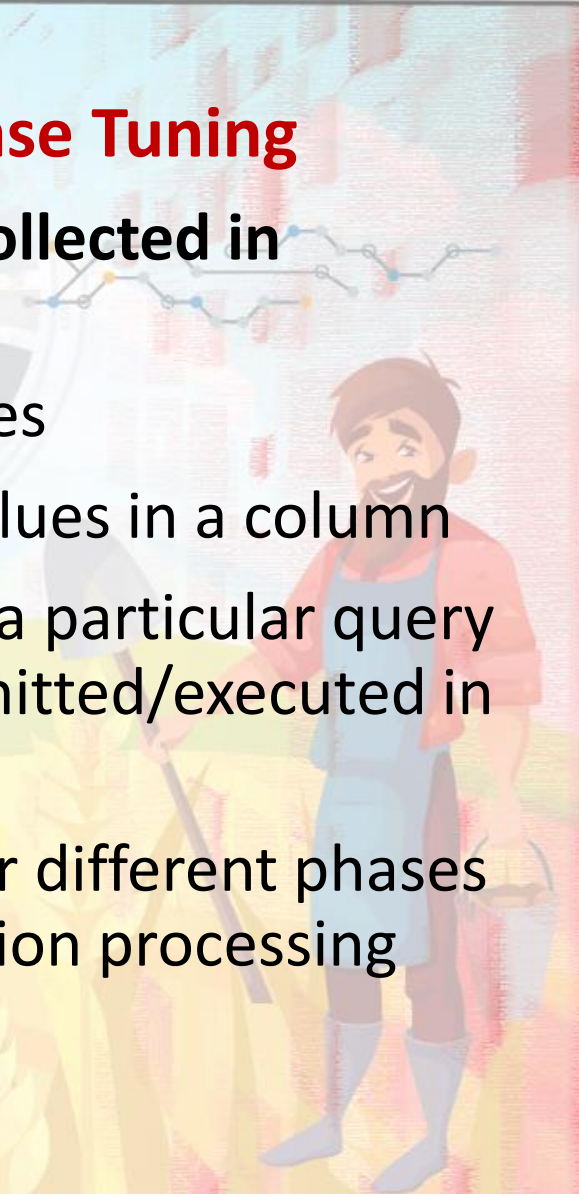
Statistics internally collected in DBMSs:

Size of individual tables

Number of distinct values in a column

The number of times a particular query or transaction is submitted/executed in an interval of time

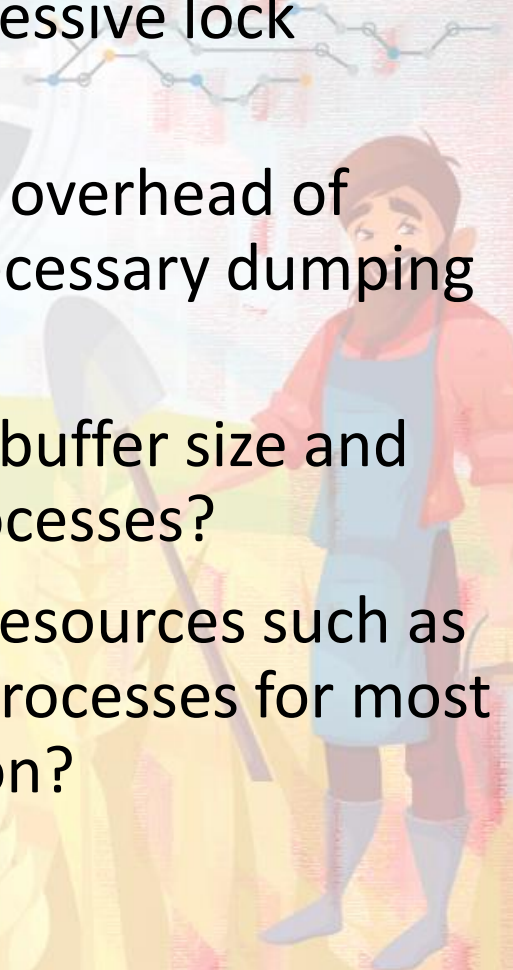
The times required for different phases of query and transaction processing



Introduction to Database Tuning

Problems in Tuning

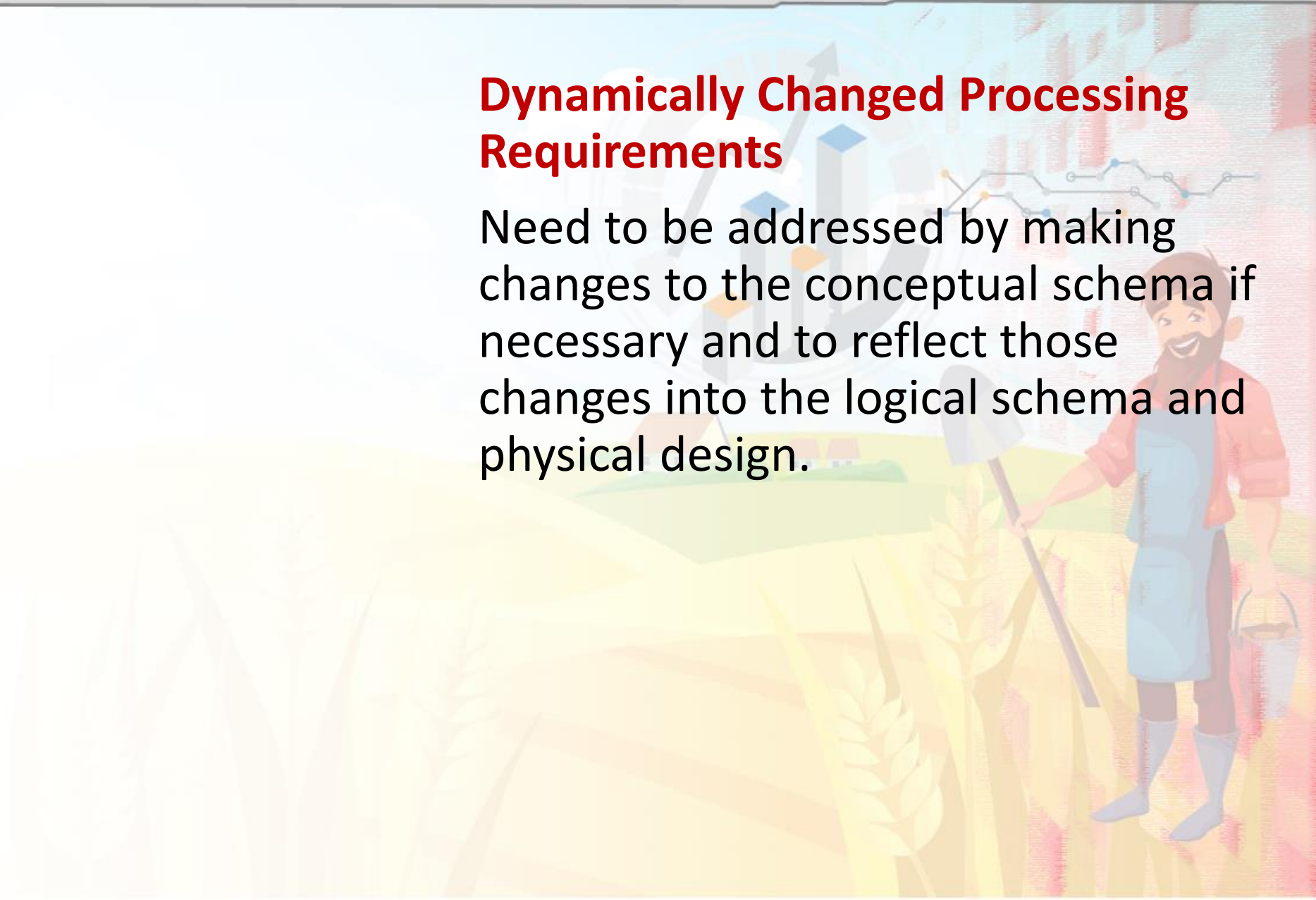
- How to avoid excessive lock contention?
- How to minimize overhead of logging and unnecessary dumping of data?
- How to optimize buffer size and scheduling of processes?
- How to allocate resources such as disks, RAM and processes for most efficient utilization?



Introduction to Database Tuning

Dynamically Changed Processing Requirements

Need to be addressed by making changes to the conceptual schema if necessary and to reflect those changes into the logical schema and physical design.



Introduction to Database Tuning

Changes in Database Design

- Existing tables may be joined (de-normalized) because certain attributes from two or more tables are frequently needed together.
- For the given set of tables, there may be alternative design choices, all of which achieve 3NF or BCNF. One may be replaced by the other.

Introduction to Database Tuning

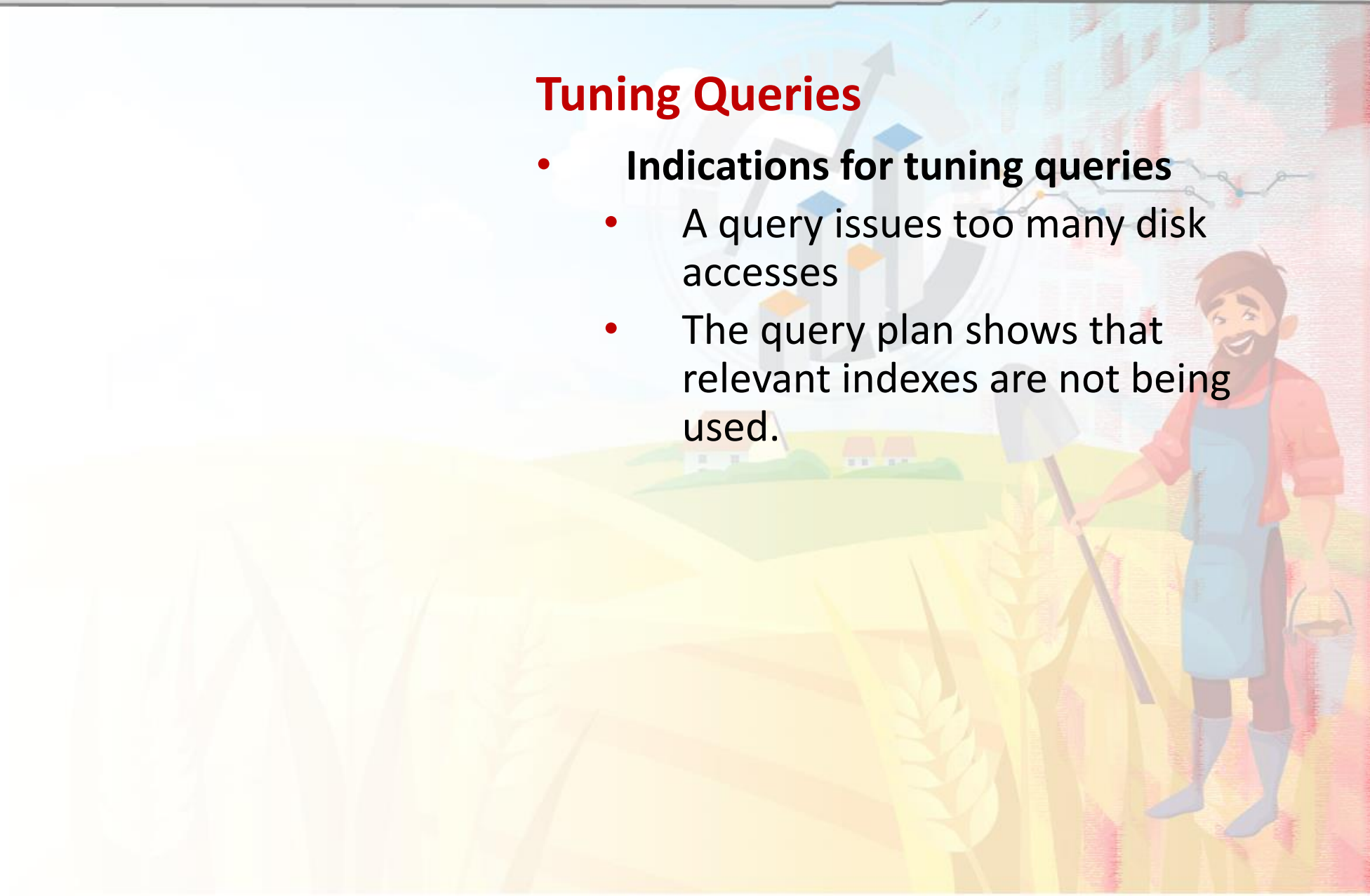
Changes in Database Design

- A relation of the form $R(\underline{K}, A, B, C, D, \dots)$ that is in BCNF can be stored into multiple tables that are also in BCNF by replicating the key K in each table.
- Attribute(s) from one table may be repeated in another even though this creates redundancy and potential anomalies.
- Apply **horizontal partitioning** as well as **vertical partitioning** if necessary.

Introduction to Database Tuning

Tuning Queries

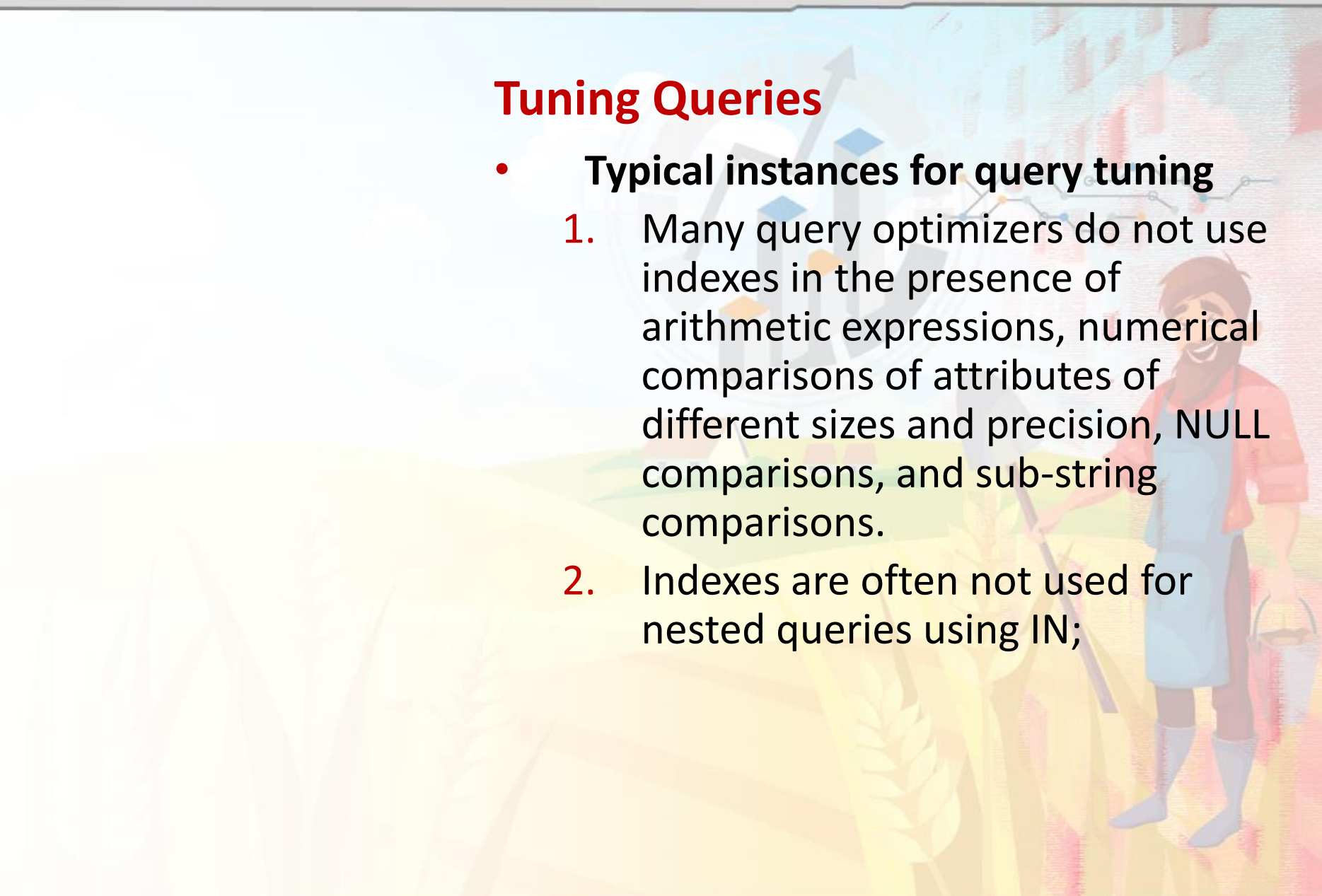
- **Indications for tuning queries**
 - A query issues too many disk accesses
 - The query plan shows that relevant indexes are not being used.



Introduction to Database Tuning

Tuning Queries

- **Typical instances for query tuning**
 1. Many query optimizers do not use indexes in the presence of arithmetic expressions, numerical comparisons of attributes of different sizes and precision, NULL comparisons, and sub-string comparisons.
 2. Indexes are often not used for nested queries using IN;



Introduction to Database Tuning

Tuning Queries

3. Some *DISTINCT*s may be redundant and can be avoided without changing the result.
4. Unnecessary use of temporary result tables can be avoided by collapsing multiple queries into a single query unless the temporary relation is needed for some intermediate processing.
5. If multiple options for join condition are possible, choose one that uses a clustering index and avoid those that contain string comparisons.

Introduction to Database Tuning

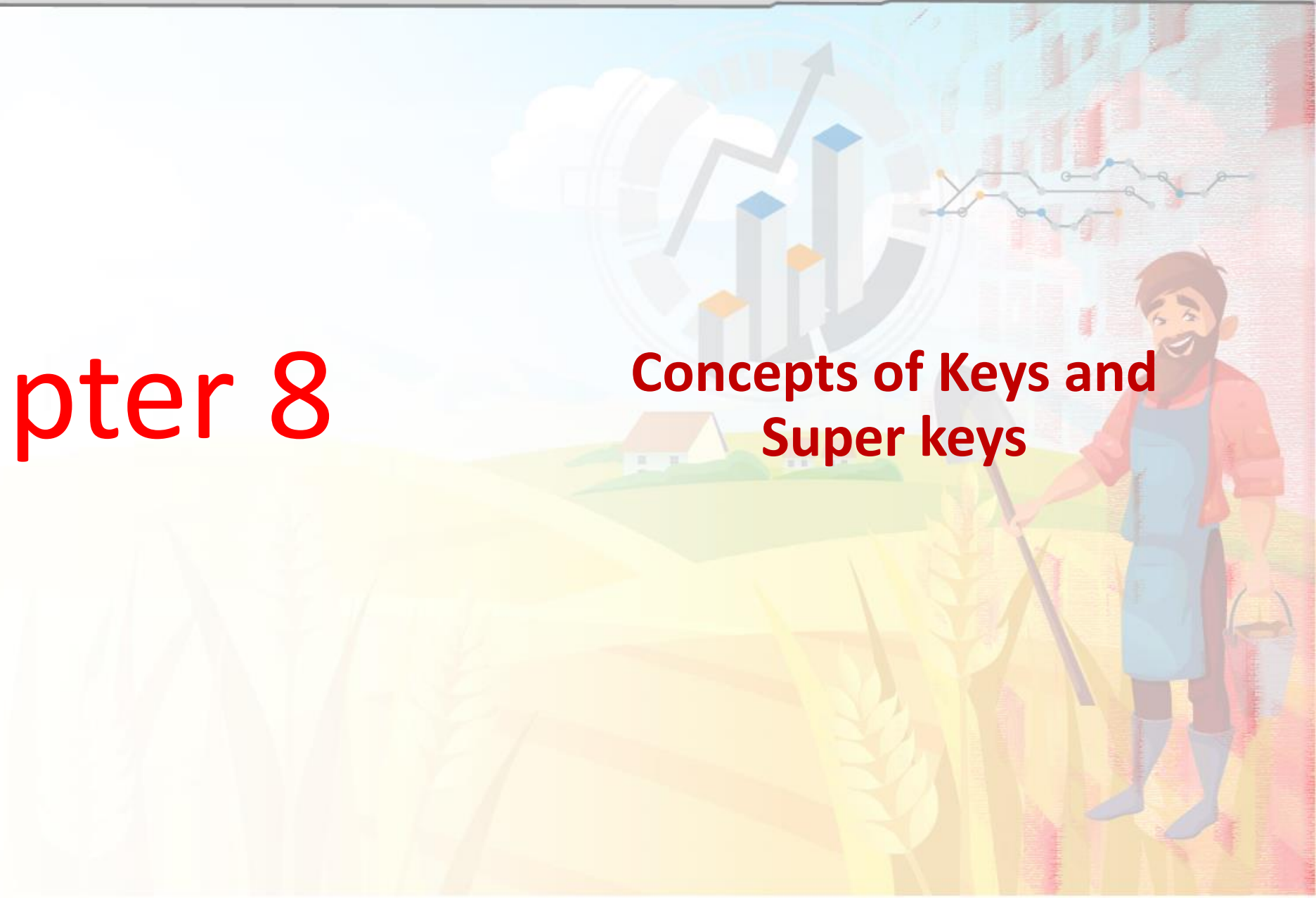
Tuning Queries

6. In some situations involving using of correlated queries, temporaries are useful.
7. The order of tables in the *FROM* clause may affect the join processing.
8. Some query optimizers perform worse on nested queries compared to their equivalent un-nested counterparts.
9. Many applications are based on views that define the data of interest to those applications. Sometimes these views become an overkill.

END

Chapter 8

**Concepts of Keys and
Super keys**



Concepts of Keys & Super Keys

Keys & Super Keys

- Concepts of Keys
- Super Keys
- Example – 1
- Example – 2



Concepts of Keys & Super Keys

Concepts of Keys

- A **key** is a combination of one or more columns that is used to identify rows in a relation
- A **composite key** is a key that consists of two or more columns



Concepts of Keys & Super Keys

Super Keys

- A super key is a combination of columns that uniquely identifies any row within a relational database management system (RDBMS) table.

Roll#	FirstName	LastName	Address	City	NIC#	Deptno
...						
12i-2565	...					CS
	Ali					CS
	Ali					EE
...						

{Roll#}, {Roll#, NIC#}, **{NIC#}**, {Roll#, FirstName, LastName}
{Roll#, FirstName, LastName, Address, City, NIC#, Deptno}

Concepts of Keys & Super Keys

Super Keys Example - 1

Roll Number	First Name	Last Name
CSU0001	Shabbir	Bhimani
CSU0002	Jain	Rao
CSU0003	Nitin	Uday

Now we have the following as super keys

Roll Number	First Name	
Roll Number	First Name	Last Name

Concepts of Keys & Super Keys

Super Keys Example - 2

Following table consists of four columns

EmployeeID, Name, Job, DeptID

- Examples of superkeys in this table would be {employeeID, Name}, {employeeID, Name, job}, and {employeeID, Name, job, departmentID}
- In a real database we don't need values for all of those columns to identify a row

Concepts of Keys & Super Keys

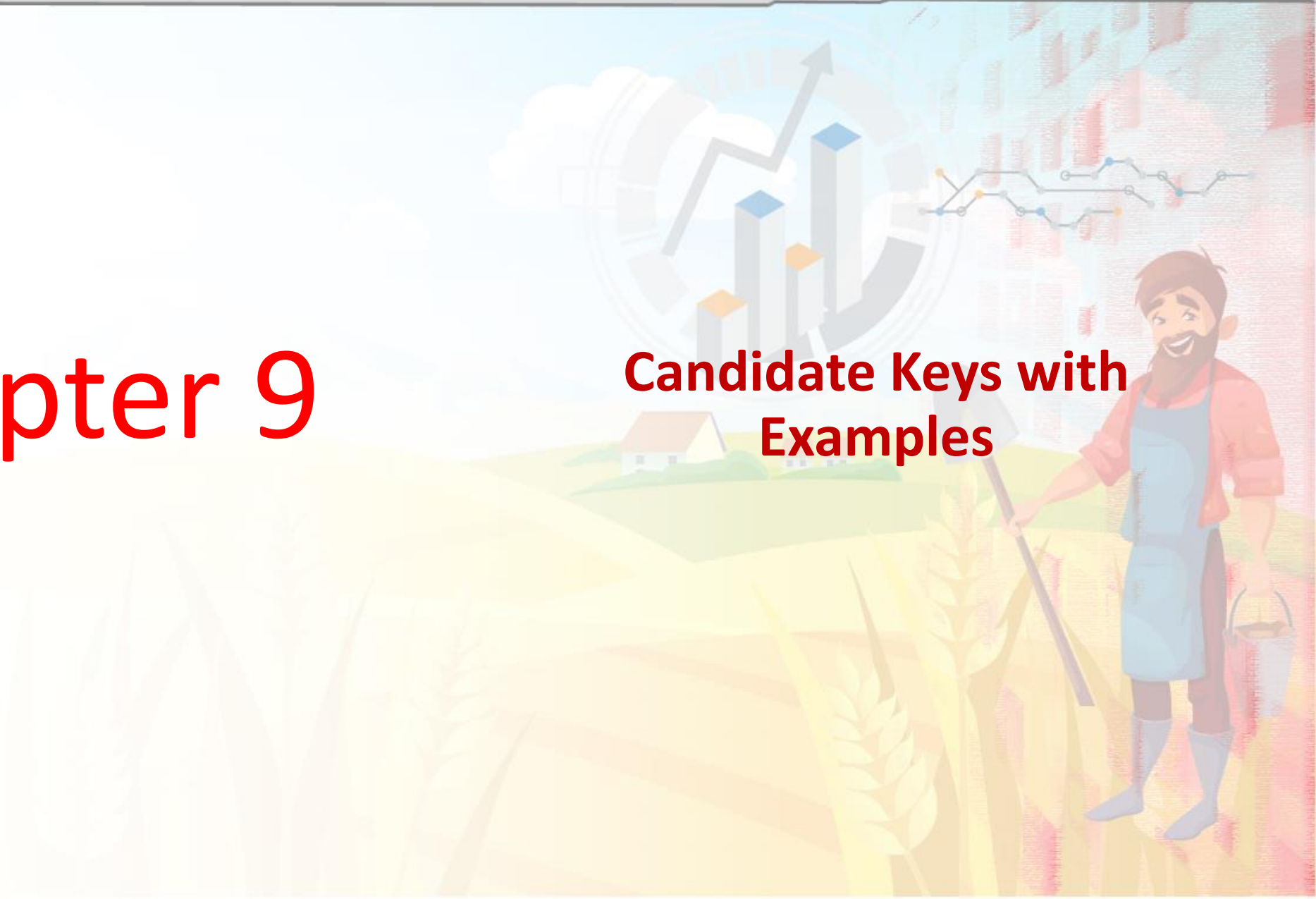
Super Keys Example - 2

- We only need, per our example, the set {EmployeeID}.
 - This is a minimal superkey
 - So, employeeID is a candidate key.
 - EmployeeID can also uniquely identify the tuples.

END

Chapter 9

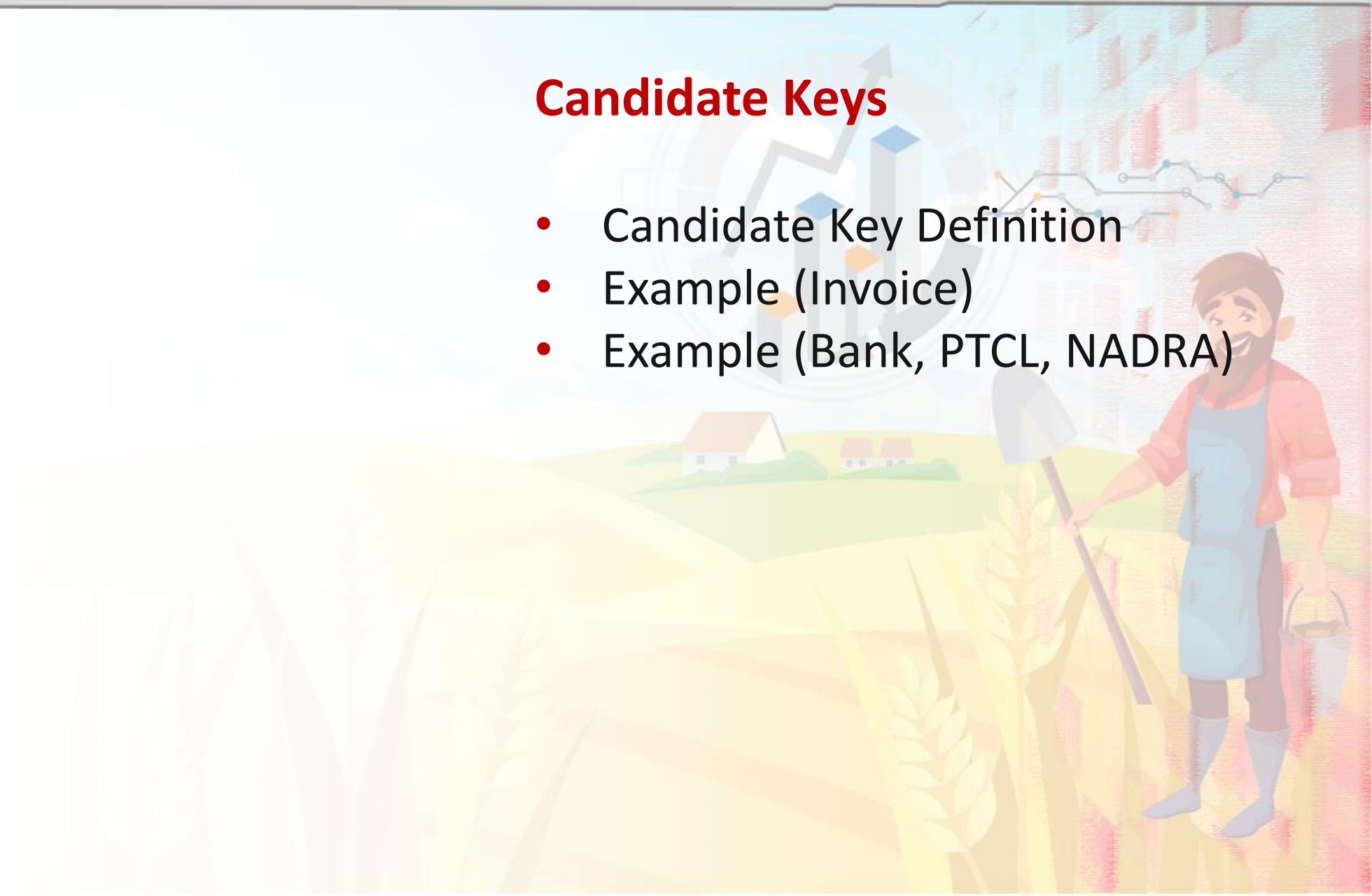
**Candidate Keys with
Examples**



Candidate Keys with Examples

Candidate Keys

- Candidate Key Definition
- Example (Invoice)
- Example (Bank, PTCL, NADRA)



Candidate Keys with Examples

Candidate Key Definition

- A **candidate key** is a key that determines all of the other columns in a relation
- Candidate key columns help in searching **fewer duplicated** or unique records.
- Examples

In PRODUCT relation

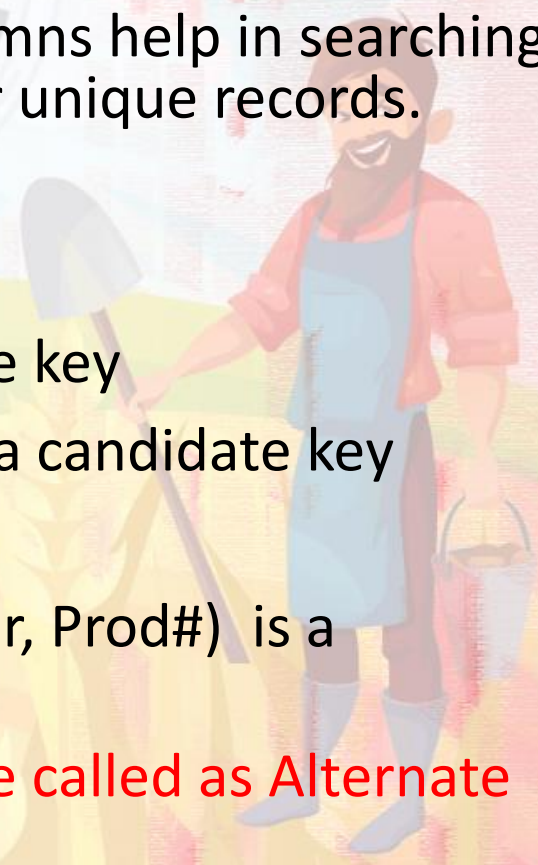
Prod# is a candidate key

Prod_Name is also a candidate key

In ORDER_PROD

(OrderNumber, Prod#) is a candidate key

(Candidate key can be called as Alternate Key)



Candidate Keys with Examples

Candidate Key Example Invoice (Buying Items)

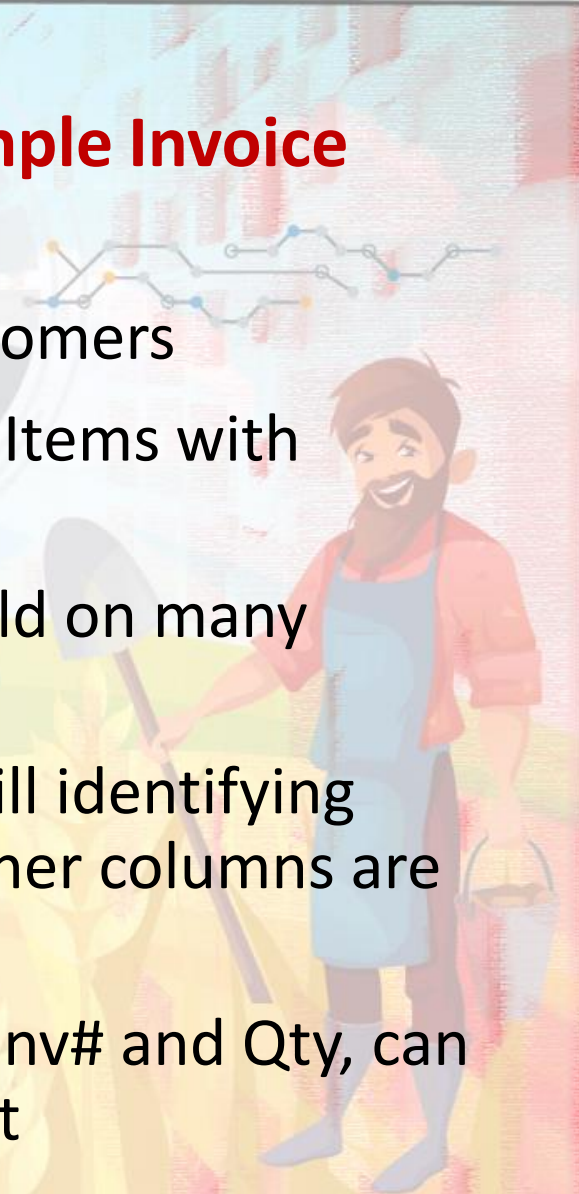
Items are sold to Customers

Customer should buy Items with Invoice

Same items can be sold on many invoices

Invoice# and Item# will identifying exact record, what other columns are required

If some one tells you inv# and Qty, can you find exact product



Candidate Keys with Examples

Candidate Key Examples

For a Bank as an Enterprise: **AccountHolder**
(or **Customer**)

ACC#, Fname, Lname, DOB, **CNIC#**, Addr,
City, **TelNo**, **Mobile#**, **DriveLic#**

For PTCL: **Customer** (single telno holders)

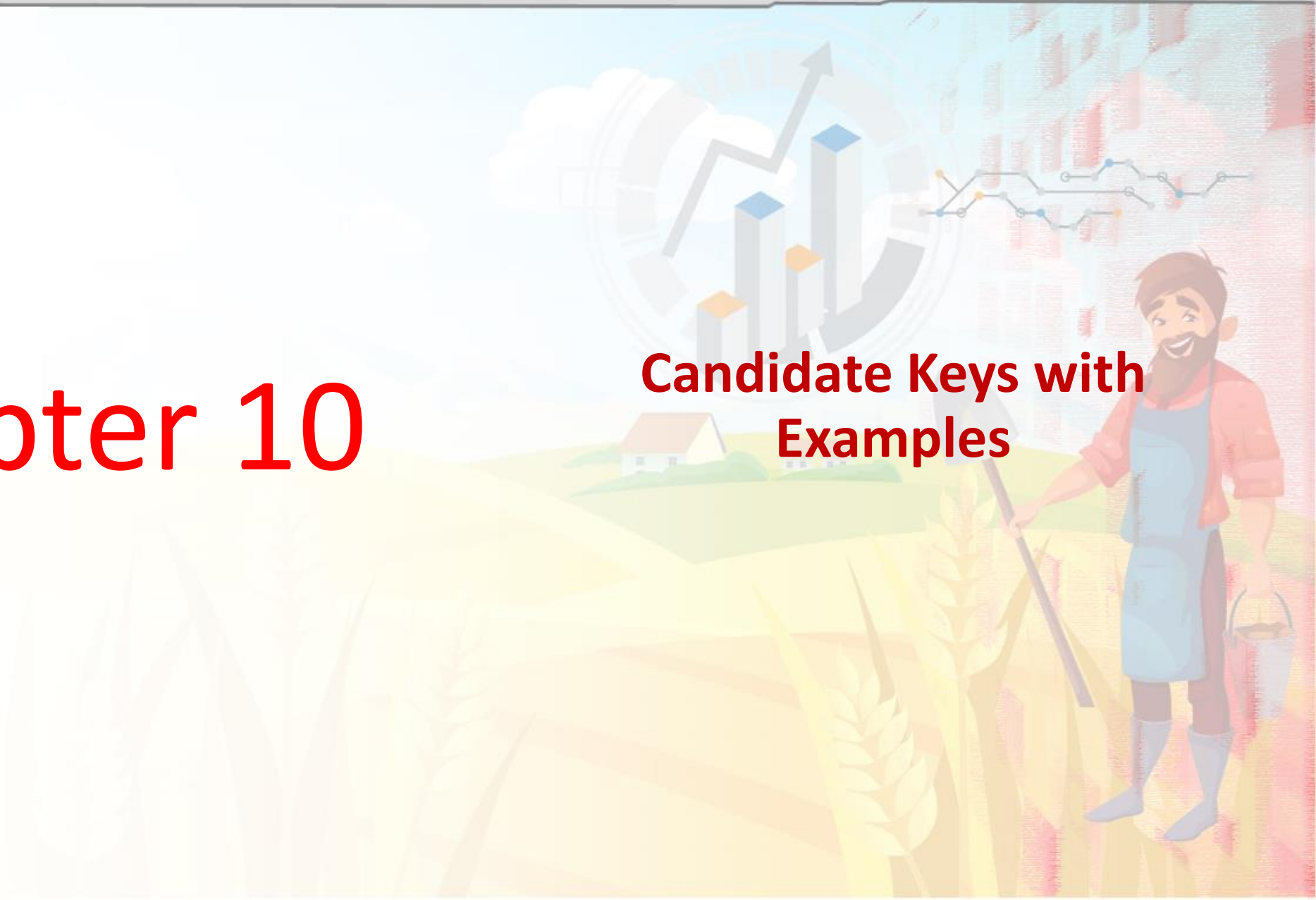
Consumer#, Fname, Lname, DOB, **CNIC#**,
Addr, City, **TelNo**, **Mobile#**

For NADRA: Citizen (**CNIC#**, Fname,
Lname, FatherName, DOB, **OldCNIC#**,
PAddr, PCity, TAddr, TCity, **TelNo**,
Mobile#)

END

Chapter 10

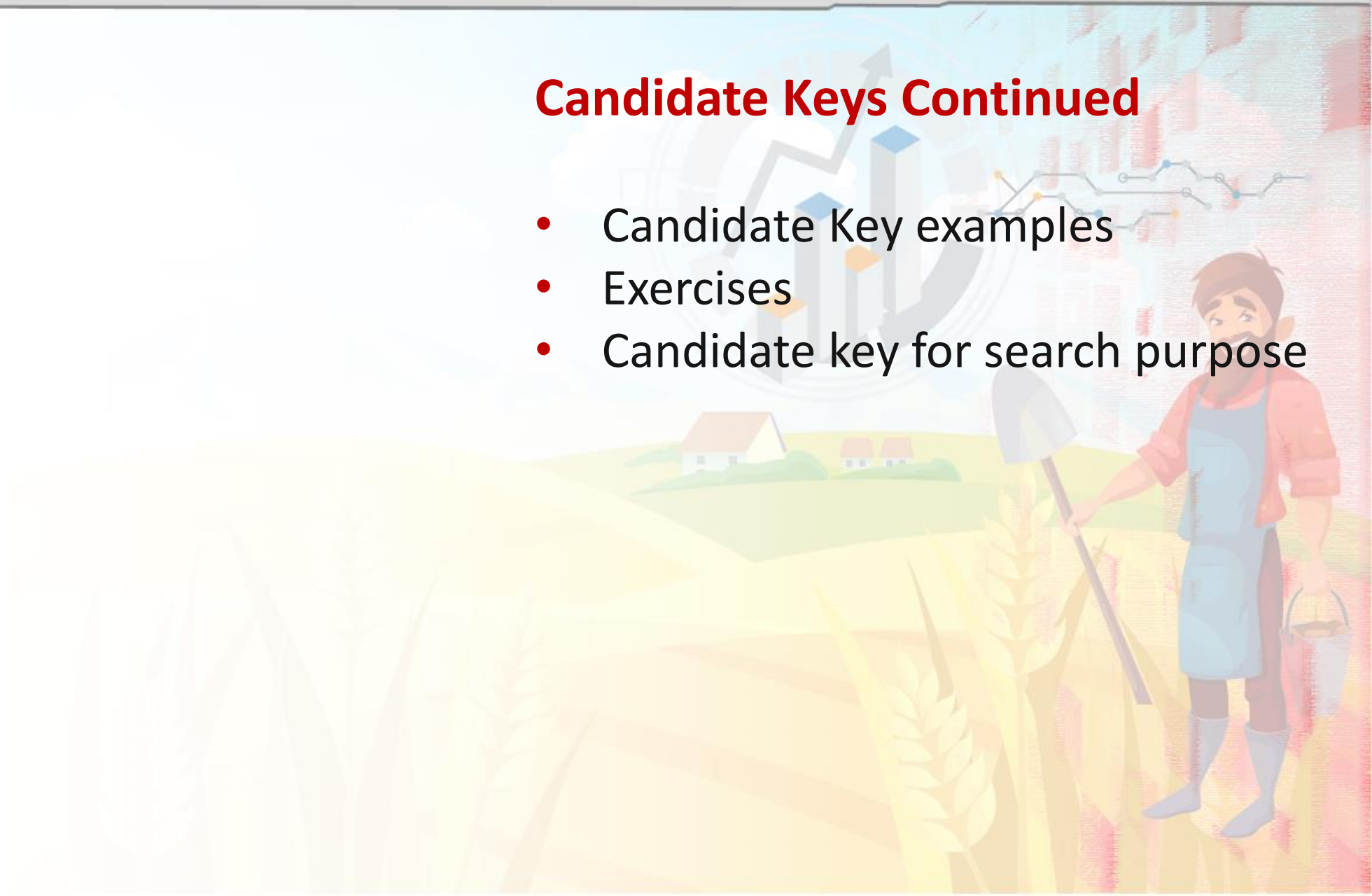
**Candidate Keys with
Examples**



Candidate Keys with Examples

Candidate Keys Continued

- Candidate Key examples
- Exercises
- Candidate key for search purpose



Candidate Keys with Examples

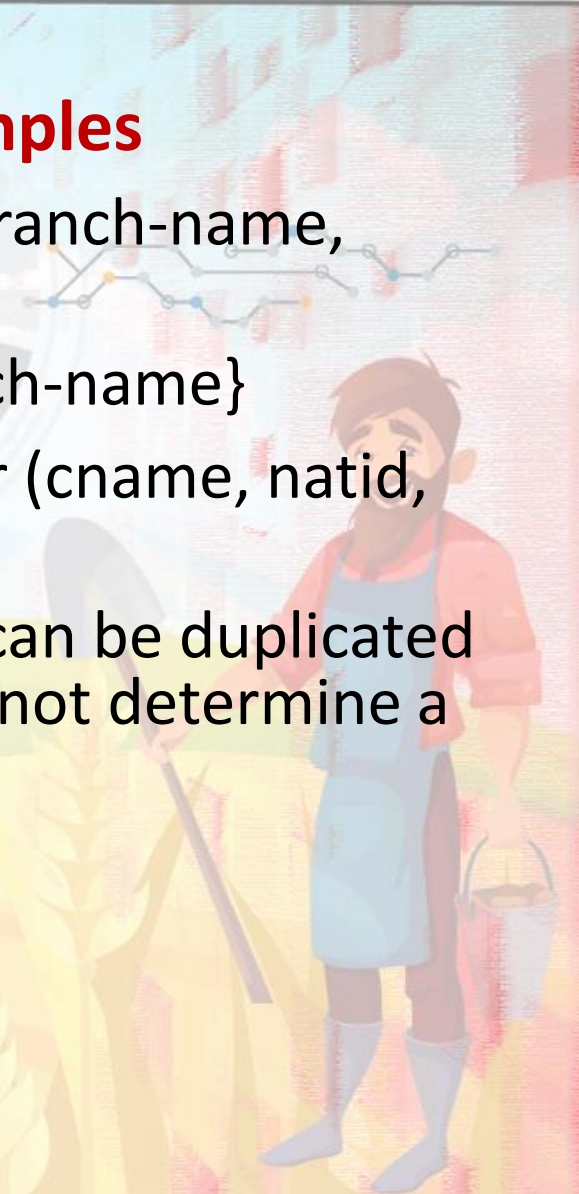
Candidate Key Examples

Example-1: Branch (branch-name, assets, branch-city)

Candidate Key: {branch-name}

Example-2: Customer (cname, naticid, address, city, telno)

cname, address, city can be duplicated individually and cannot determine a record.



Candidate Keys with Examples

Candidate Key Examples

The following combinations distinguish customer records or tuples.

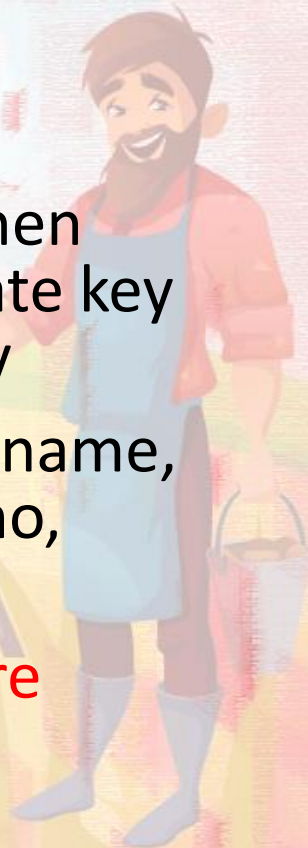
{cname, telno}, {natid}

{natid, cname}

As $\{\text{natid}\} \subseteq \{\text{natid}, \text{cname}\}$, then $\{\text{natid}, \text{cname}\}$ is not candidate key and $\{\text{natid}\}$ is a candidate key

Example-3: Employee(empno, name, birth_date, address, city, telno, citizenship_id)

empno, telno, citizenship_id are possible candidate keys



Candidate Keys with Examples

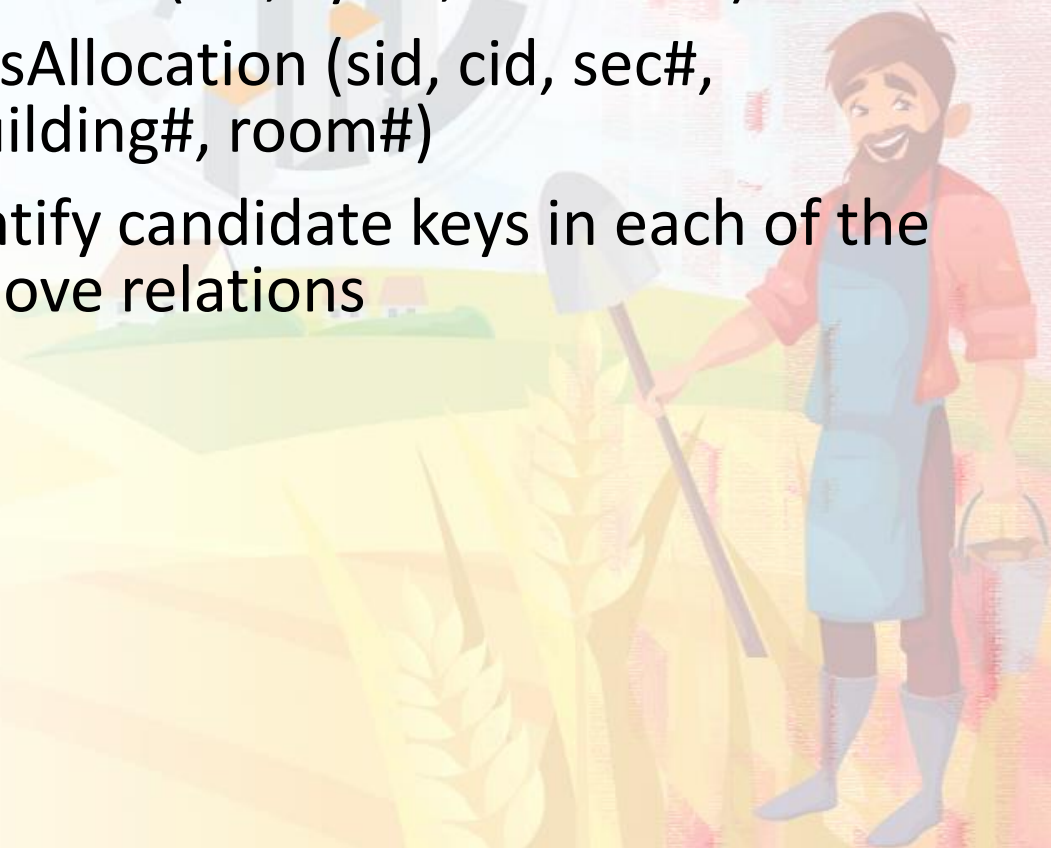
Exercises

Course (cid, cname, deptno)

Semester (sid, syear, startdate)

ClassAllocation (sid, cid, sec#,
building#, room#)

Identify candidate keys in each of the
above relations



Candidate Keys with Examples

Candidate Key for Search purpose

5 - Medicine

STOCK_NUMBER: X Find MANUFACTURER*: Enter Name for search Exit

BRAND_NAME*: X

GENERIC_NAME: X

MEDICINE_TYPE:

STOCK_TYPE: 01 01: Medicine, 02: Non medicine

MG_ML: MG UNITS_PERITEM*: 0

BARCODE:

DOSAGE_DESC:

AVAILABLE_STATUS: Y Y/N UNIT_TYPE: T: Tablet, P: Pack/ Strip

Brand Name X para X New Save Manufacture List 5676 Remove

Location: 21: Others

stock_nu	brand_name	generic_name	medicine_type	stock_ty	MG_ML	Barcode	Dosage_desc	available	m_name	manu
3902982	CALPOL 6 PLUS 90ML 1S	PARACETAMOL		01	MG			Y		1
3902985	DISPROL 90ML	PARACETAMOL		01	MG			Y		73
3901844	SAMEROL-N FORTE	PARACETAMOL + ORPHANADRIN	PK/10TAB		MG			Y		87
3901850	SAMEROL-N	PARACETAMOL + ORPHANADRIN	PK/72TAB		MG			Y		87
3903400	PANADOL TAB 500MG	PARACETAMOLE		01	MG			Y		1
3903401	PANADOL CF 500MG 100S	PARACETAMOLE		01	MG			Y		1
3903283	FEBROL DS 60ML	PARACETAMOL	BOT/120MG	01	MG					35
3901595	NUBEROL	PARACETAMOL BP & ORPHENAD	PK/100TAB		MG			Y		32
3901596	NUBEROL FORTE	PARACETAMOL BP & ORPHENAD	PK/15TAB		MG			Y		32
3900455	CALPOL SYP 100ML 1S	PARACETAMOL		01	ML			Y		1
3903285	TEMPRAMINE 60ML	PARACETAMOL		01						116
3902971	TEMPOL 60ML	PARACETAMOL	BOT/120MG		MG			Y		117
3902972	TEMPOL 6+ 60ML	PARACETAMOL	BOT/		MG			Y		117
3902973	FEBROL 60ML	PARACETAMOL	BOT/120MG	01	MG			Y		35
3902974	PANADOL SYRUP 100ML 1S	PARACETAMOL		01	MG			Y		1

DBA can create Index on searching attributes (candidate Keys)

Chapter 11

Primary Key



Primary Key

Primary Keys

- Primary key definition
- An alternate definition
- Primary key examples & issues



Primary Key

Primary Key Definition

A **primary key** is a candidate key selected as the primary means of identifying rows in a relation:

- There is one and only one primary key per relation
- The primary key is NOT NULL, UNIQUE
- The ideal primary key is short, numeric(alpha), indexed, fixed length and never changes
- Key that has its ownership of an Enterprise

The primary key may be a **composite key**

Primary Key

Primary key

An Alternate Definition

A primary key is a minimal identifier that is used to identify tuples uniquely.

This means that no subset of the primary key is sufficient to provide unique identification of tuples.

NULL values is not allowed in primary key attribute.



Primary Key

Primary key - Example & Issues

Example:

- STU_ID

(Practically PK must have UNIQUE, NOT NULL, INDEXED on an attribute)

STUDENT

STU_ID	NAME	CID_ID	DOB	DEPTCODE
--------	------	--------	-----	----------

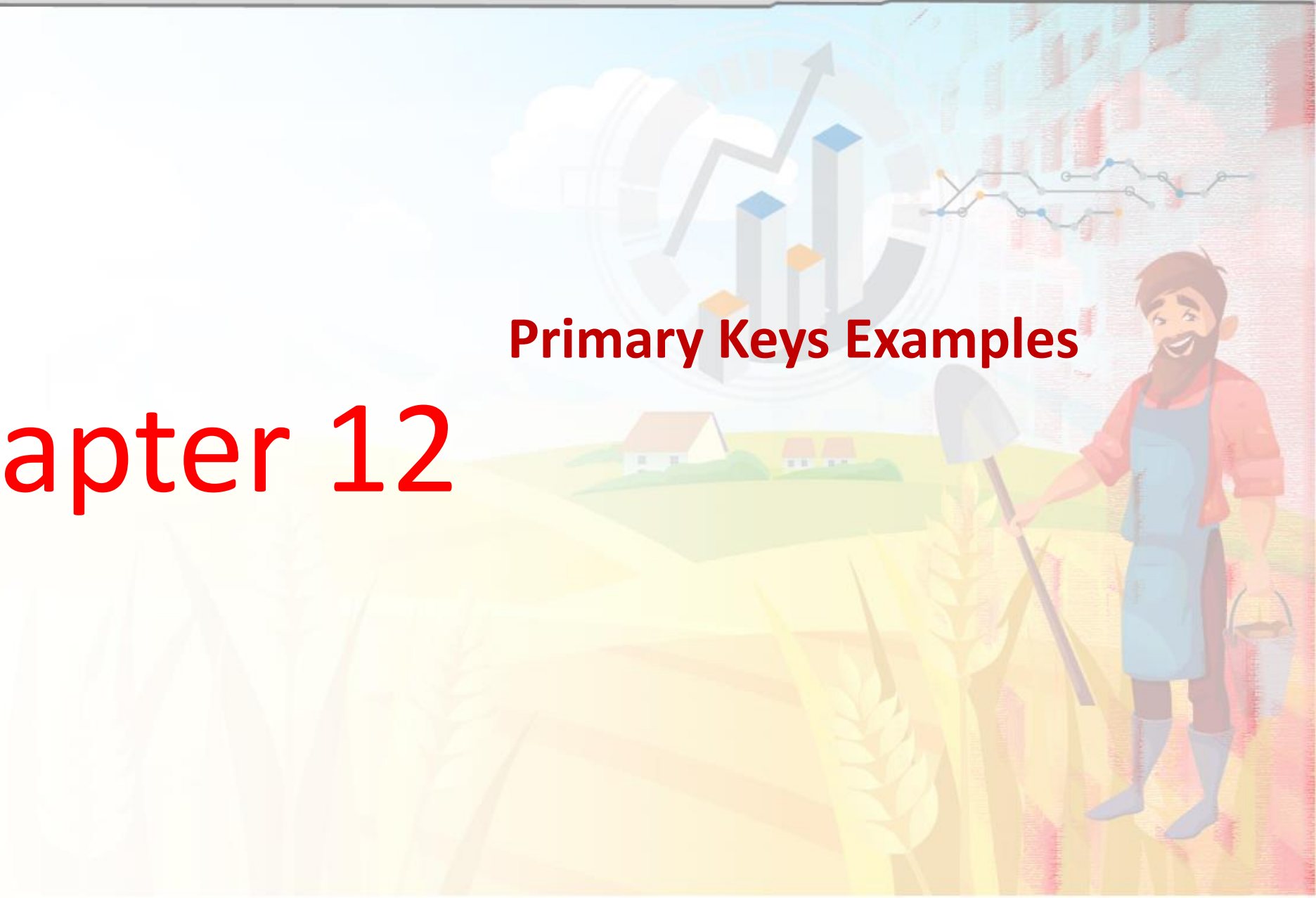
Although, candidate keys are STU_ID, CIT_ID. Organization ITSELF issues its own identifier PK. PK is a property of an organization. Driving license# is a property of Ministry of Motors/ transportation, it must not be used PK in university or other organizations. It is a candidate key for other organization for search purposes.

Note: It is not recommended to create PKs for columns like Quantity, Salary, Price and column that has datatype DATE



Primary Keys Examples

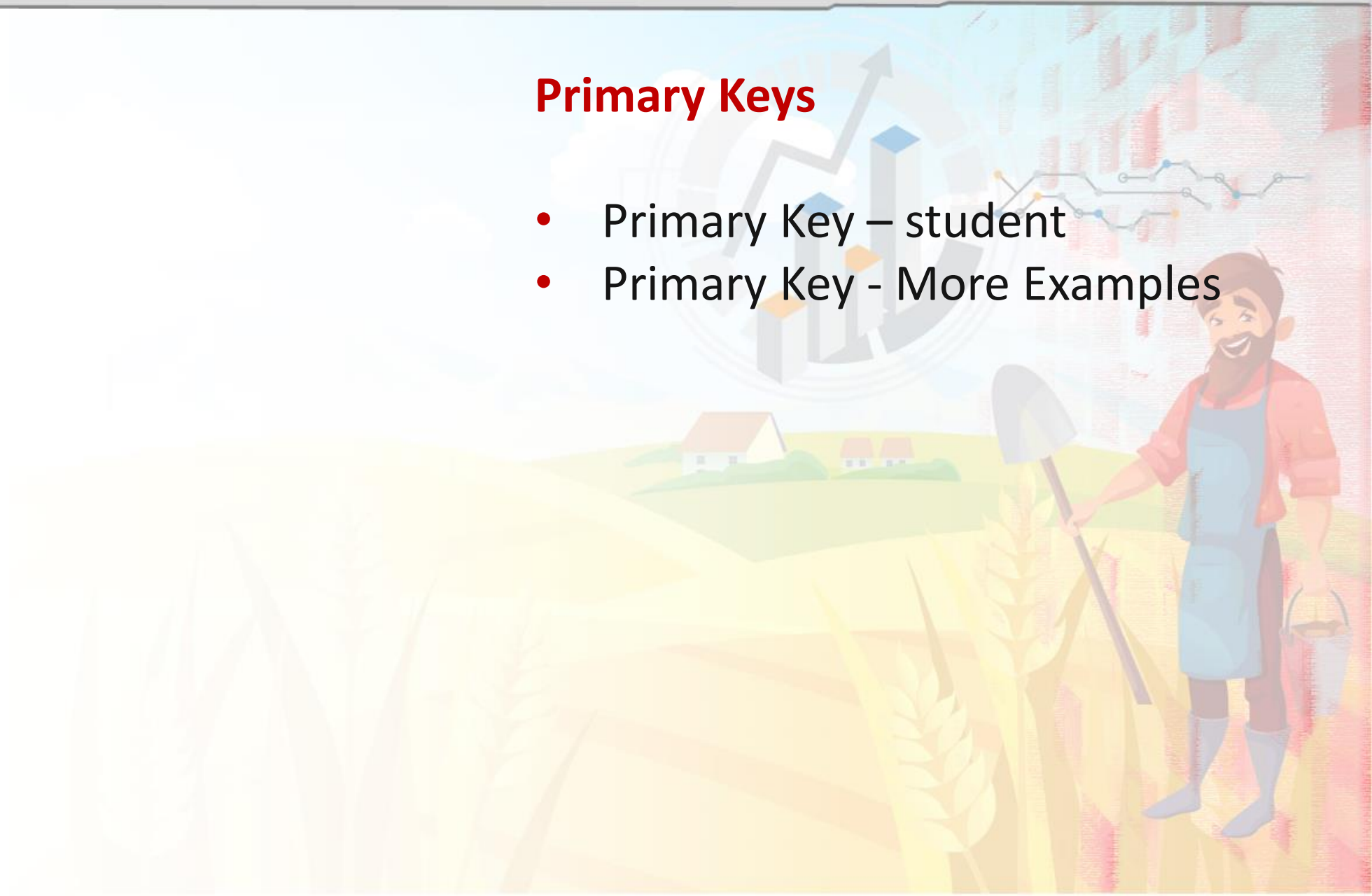
Chapter 12



Primary Keys Examples

Primary Keys

- Primary Key – student
- Primary Key - More Examples



Primary Keys Examples

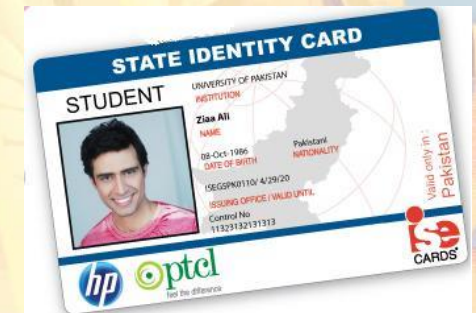
Primary key - Student

Example-1:

STUDENT(StuID, FirstName, FamilyName, DOB, ...)

Example-2: Building (B#, BName, Location, Region),

B# is a primary key. Although, BName is unique, not null but it is not short.



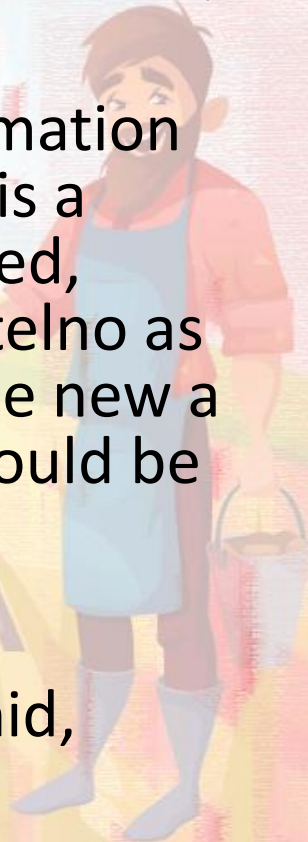
Primary Keys Examples

Primary key - More Examples

Example-3: Customer (cname, citizenid, address, city, telno)

This relation indicates the information about personal details. There is a chance that cname is duplicated, some may have citizenid and telno as null. This forces us to introduce new attribute such as cust# that would be a primary key.

Customer (cust#, cname, citizenid, address, city, telno)



Primary Keys Examples

Primary key - More Examples

Example-4:

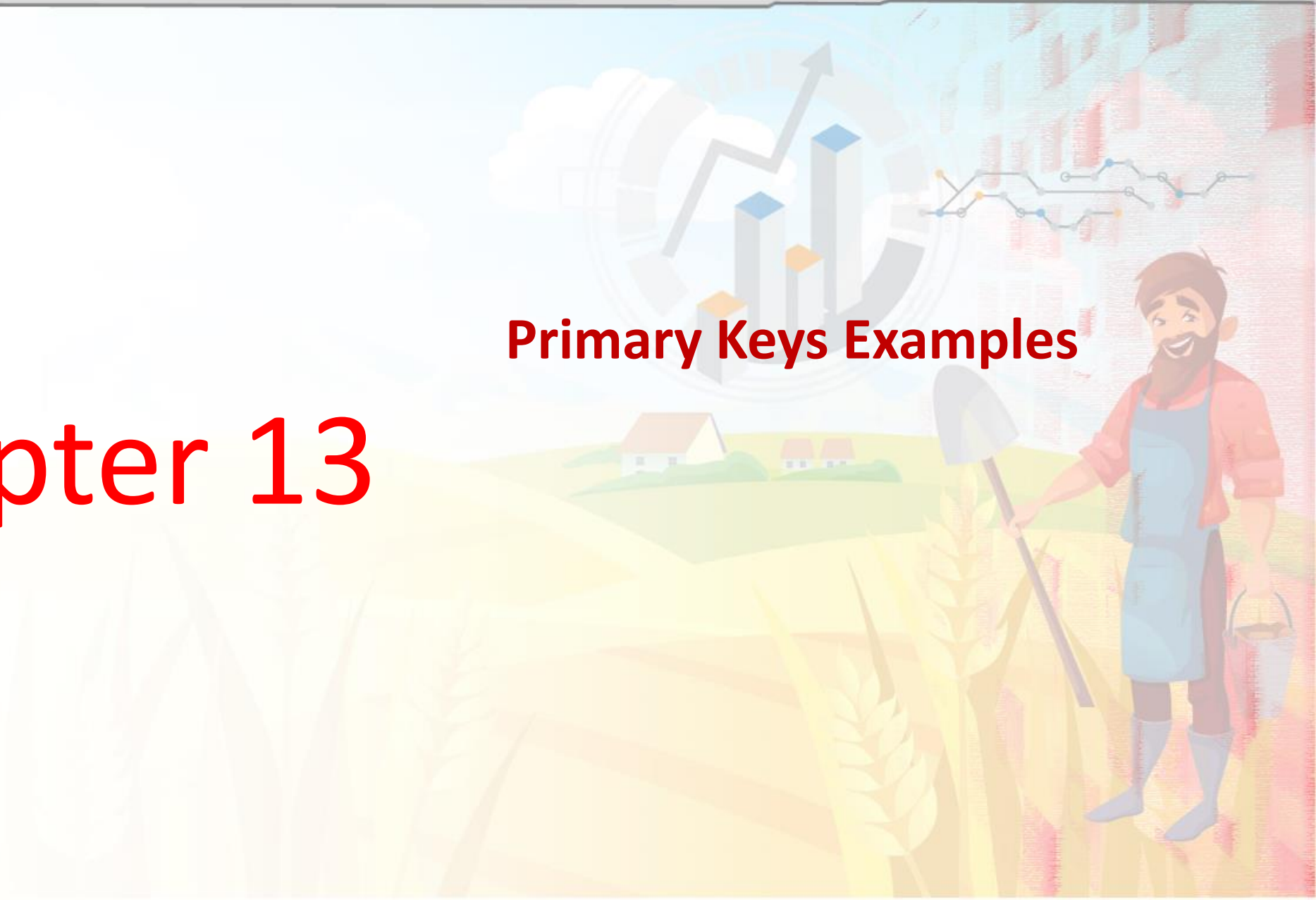
BankBranch(Branch-Name, City, TotalAsset)

- What will be a PK that you will suggest?
- What is a candidate key?
- Which attribute is unique?
- There may be many branches in one city then finalize this relation with possible constraints

End

Primary Keys Examples

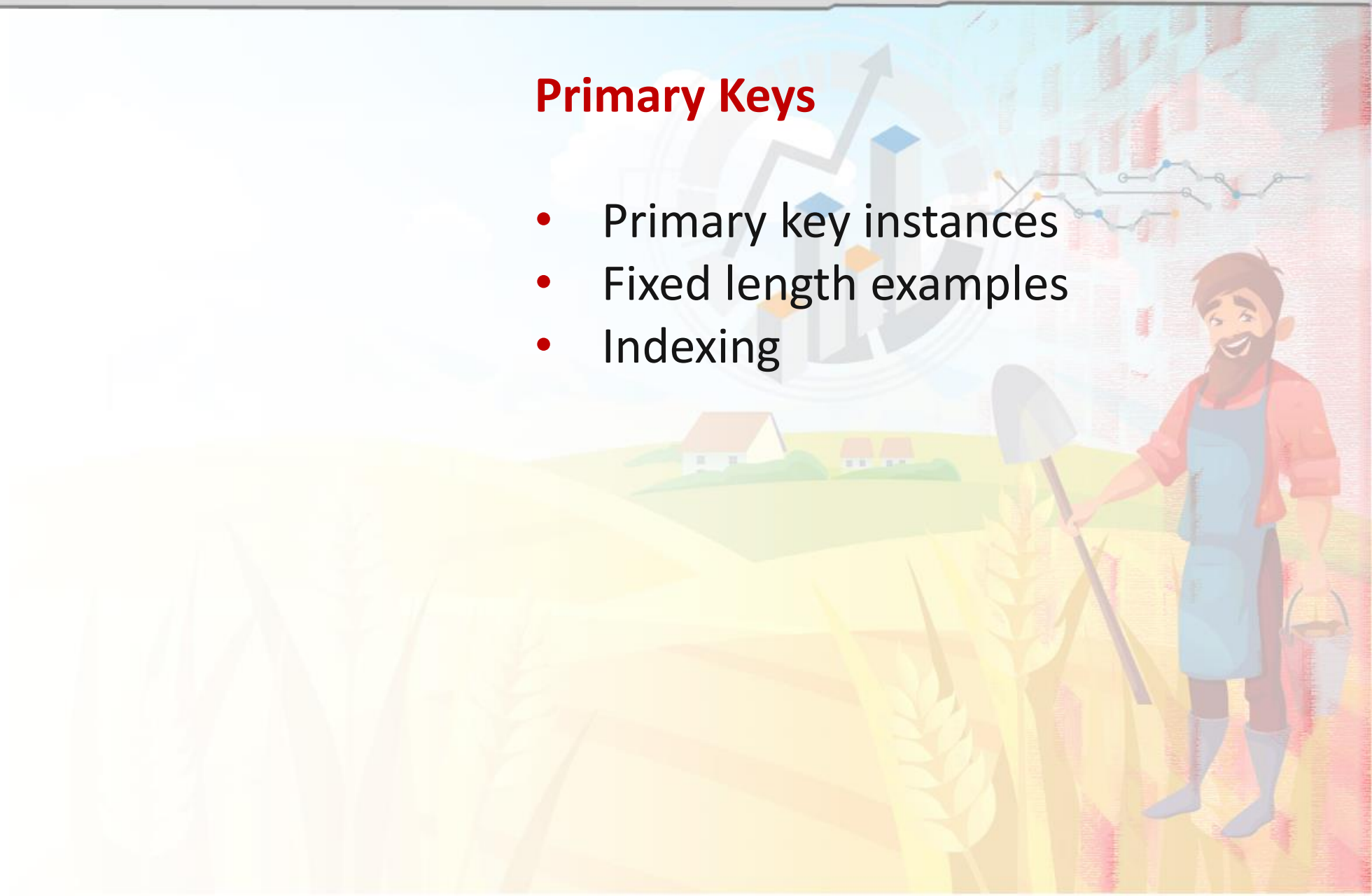
Chapter 13



Primary Keys Examples

Primary Keys

- Primary key instances
- Fixed length examples
- Indexing



Primary Keys Examples

Primary key Instances

512235665 Social Security Number

610112654 (By Birth Asian, European, Homeland)

200338900 Student ID (With Registration year)

LHR051 Product ID (with Manufacturing Loc)

31266885 Bank Acc# (followed by Branch Code)



Primary Keys Examples

Primary key Instances

Serial# starts with 1 and incremented by 1

005384 Company has many branch codes
005

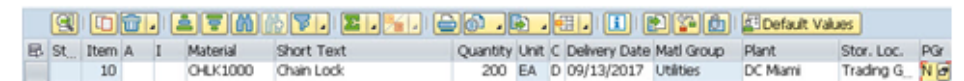
007004 invoices issued from branches



Primary Keys Examples

Primary key Fixed Length

After clicking on  compare your screen with the one below.



Ep	St	Item	A	I	Material	Short Text	Quantity	Unit	C	Delivery Date	Mat Group	Plant	Stor. Loc	PGr
		10			CHUK1000	Chain Lock	200	EA	D	09/13/2017	Utilities	DC Miami	Trading G...	N

Then, click on  to save your purchase requisition. The system will create a unique document number.  Purchase requisition number **0010000000** created

Click on the exit icon  to return to the SAP Easy Access screen

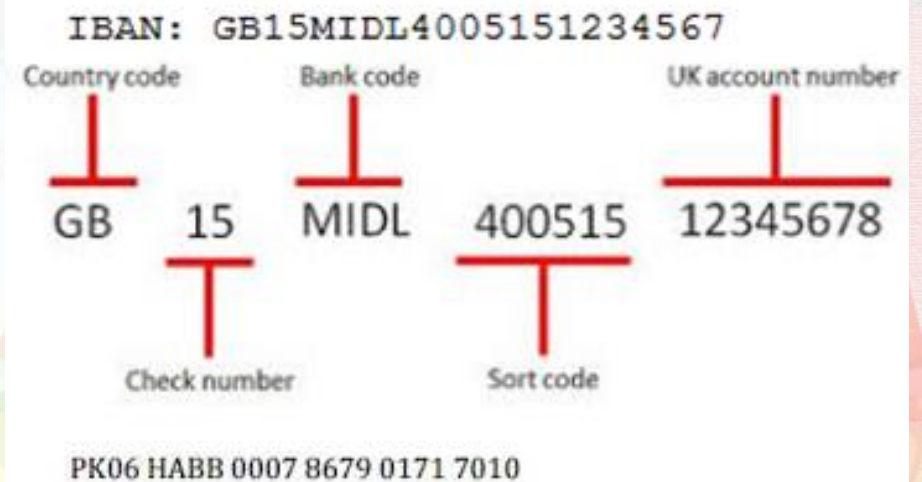


SearchTerm	Cty	PostalCode	City	Name 1	Vendor
055	US	68516	LINCOLN	MID-WEST SUPPLY	125005
055	US	68528	LINCOLN	MID-WEST SUPPLY	125094
056	US	68528	LINCOLN	MID-WEST SUPPLY	125096
056	US	68528	LINCOLN	MID-WEST SUPPLY	125100
056	US	68528	LINCOLN	MID-WEST SUPPLY	125112
057	US	68528	LINCOLN	MID-WEST SUPPLY	125115
059	US	68528	LINCOLN	MID-WEST SUPPLY	125128

Encircled Fixed Length Key Identifiers

Primary Keys Examples

Primary key - Fixed Length



Primary Keys Examples

Primary key - Indexing

Table: PATIENT_VISIT
Index: Yes, PK
Test Query execution time: less than 1 second

Table: PATIENT_VISIT1
Index: NO
Test Query execution time: 12 seconds

```
SQL> select count(*) Total_Records  
2 from PATIENT_VISIT;
```

Total_Records
625721

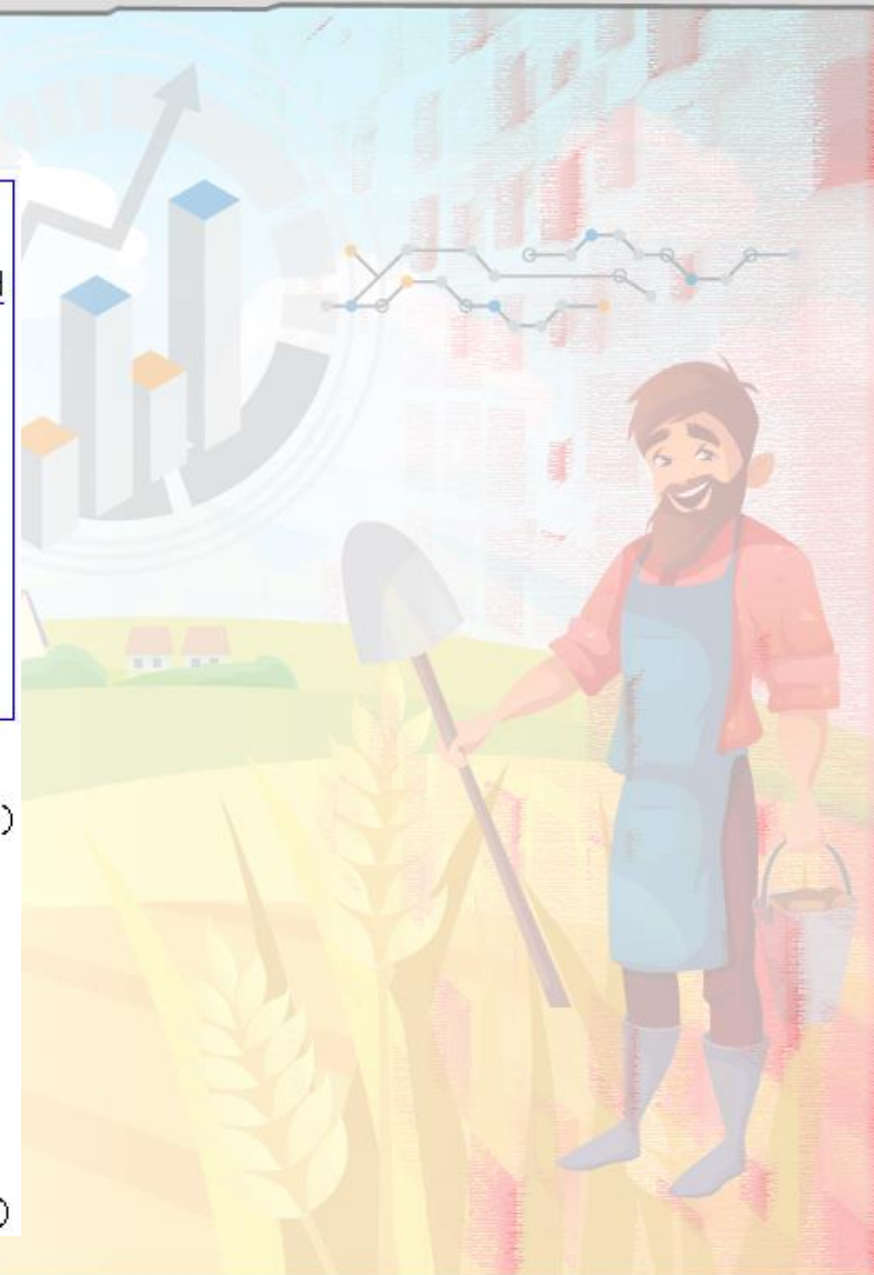
Elapsed: 00:00:00.04

```
SQL> select PATIENT_ID, max(PATIENT_VISITNO)  
2 from PATIENT_VISIT1  
3 where PATIENT_VISITNO = (select max(PATIENT_VISITNO)  
4 from PATIENT_VISIT1  
5 where DEPEND_SNO='0')  
6 group by PATIENT_ID  
7 having count(*)>=1;
```

PATIENT_ID	MAX(PATIENT_VISITNO)
9999993	1316

Elapsed: 00:00:11.54

PRIMARY KEY (PATIENT_ID, DEPEND_SNO, PATIENT_VISITNO)



Chapter 14

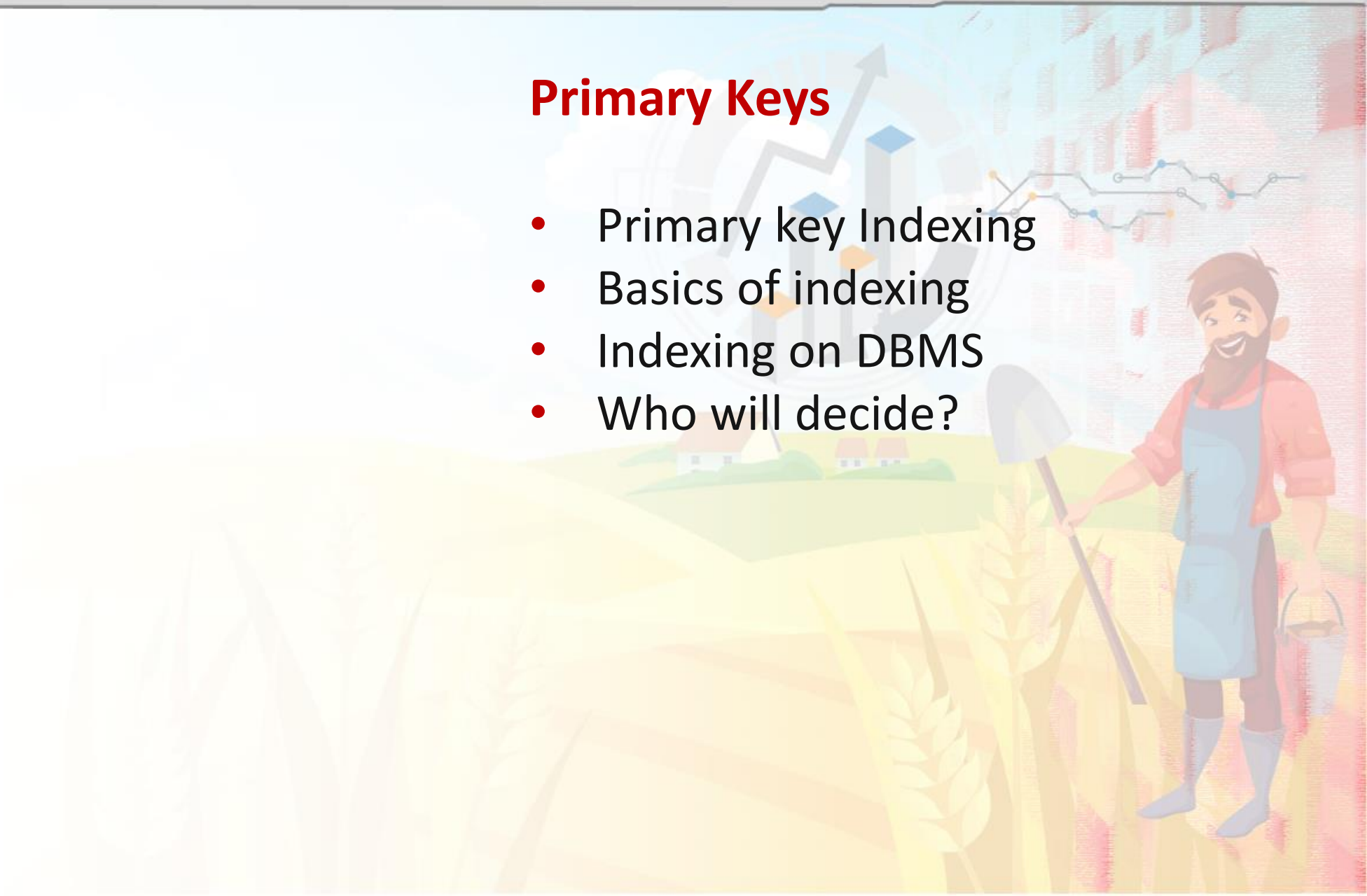
Primary Key Formats



Primary Keys Formats

Primary Keys

- Primary key Indexing
- Basics of indexing
- Indexing on DBMS
- Who will decide?

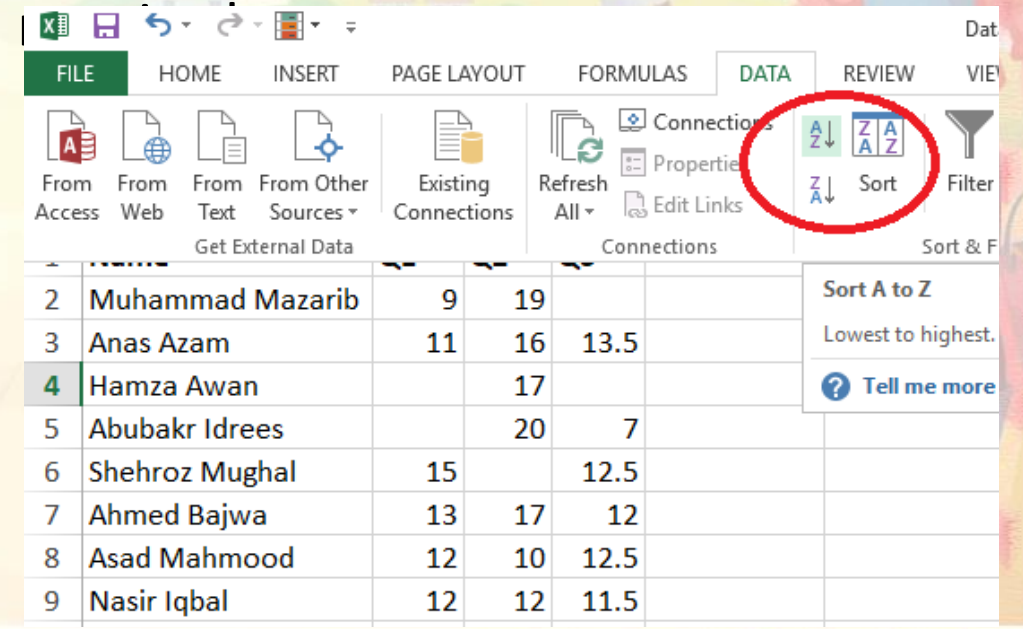


Primary Keys Formats

Basics of Indexing

Paging a key's data to access fast.
It may includes sorting from A to Z or Z to A.

Indexing is applied on non-key columns where mostly search is



The screenshot shows the Microsoft Excel interface with the 'DATA' tab selected on the ribbon. The 'Sort' button, represented by an icon with 'A-Z' and 'Z-A' arrows, is circled in red. Below the ribbon, a table of student data is visible. The table has columns for an index, a name, and three numerical values. The row for 'Hamza Awan' is highlighted in blue.

2	Muhammad Mazarib	9	19	
3	Anas Azam	11	16	13.5
4	Hamza Awan		17	
5	Abubakr Idrees		20	7
6	Shehroz Mughal	15		12.5
7	Ahmed Bajwa	13	17	12
8	Asad Mahmood	12	10	12.5
9	Nasir Iqbal	12	12	11.5

Primary Keys Formats

Primary key – Indexing on DBMS

```
Table: PATIENT_VISIT  
Index: Yes, PK  
Test Query execution time: less than 1 second
```

```
Table: PATIENT_VISIT1  
Index: NO  
Test Query execution time: 12 seconds
```

```
SQL> select count(*) Total_Records  
2 from PATIENT_VISIT;
```

```
Total_Records  
-----|  
625721
```

```
Elapsed: 00:00:00.04
```

```
SQL> select PATIENT_ID, max(PATIENT_VISITNO)  
2 from PATIENT_VISIT1  
3 where PATIENT_VISITNO = (select max(PATIENT_VISITNO)  
4 from PATIENT_VISIT1  
5 where DEPEND_SNO='0')  
6 group by PATIENT_ID  
7 having count(*)>=1;
```

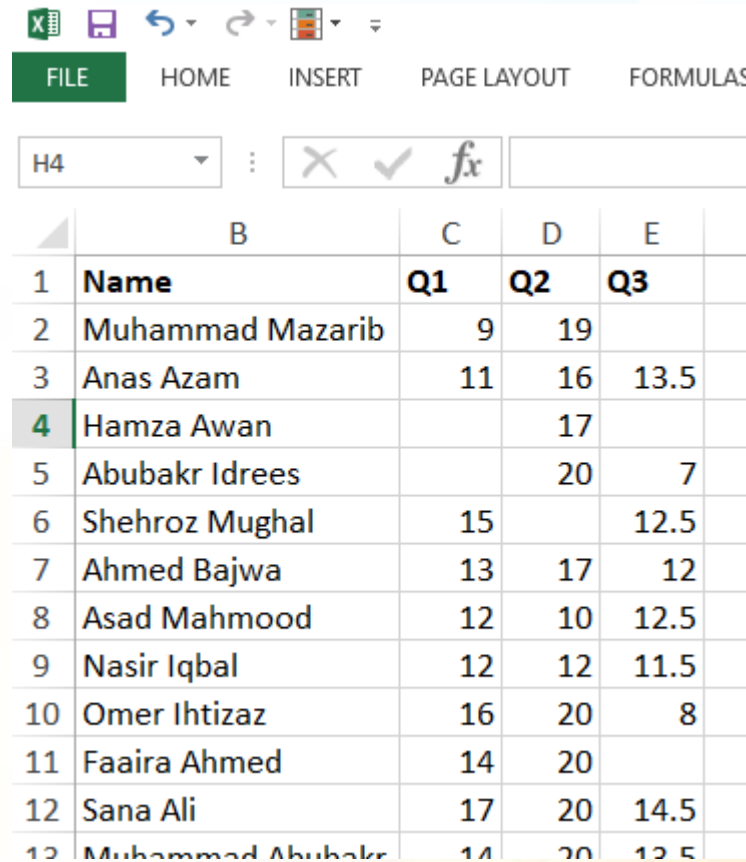
```
PATIENT_ID MAX(PATIENT_VISITNO)  
-----  
9999993 1316
```

```
Elapsed: 00:00:11.54
```

```
PRIMARY KEY ( PATIENT_ID, DEPEND_SNO, PATIENT_VISITNO )
```

Primary Keys Formats

Primary key – Who will decide?



The image shows a screenshot of an Excel spreadsheet. The ribbon at the top includes FILE, HOME, INSERT, PAGE LAYOUT, and FORMULAS. The active cell is H4. The spreadsheet contains a table with 13 rows and 5 columns. The columns are labeled B, C, D, and E, with the first column (A) containing row numbers 1 through 13. The data in the table is as follows:

	B	C	D	E
1	Name	Q1	Q2	Q3
2	Muhammad Mazarib	9	19	
3	Anas Azam	11	16	13.5
4	Hamza Awan		17	
5	Abubakr Idrees		20	7
6	Shehroz Mughal	15		12.5
7	Ahmed Bajwa	13	17	12
8	Asad Mahmood	12	10	12.5
9	Nasir Iqbal	12	12	11.5
10	Omer Ihtizaz	16	20	8
11	Faaira Ahmed	14	20	
12	Sana Ali	17	20	14.5
13	Muhammad Abubakr	14	20	12.5

Roll# is issued by some University or college Format or instance, meaningful column for PK

Chapter 15

Surrogated Keys



Surrogated Keys

Surrogated Keys

- Definition
- Examples
- Exercises

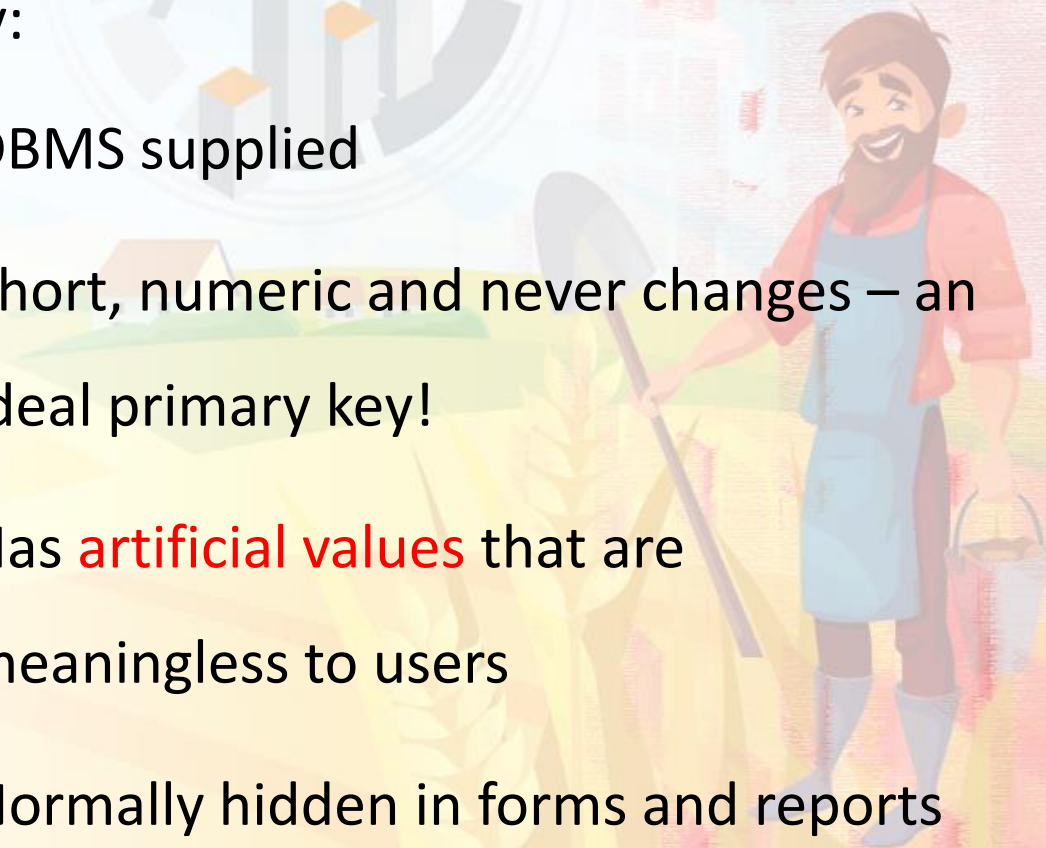


Surrogated Keys

Surrogated Key Definition

A **surrogate key** as an artificial column added to a relation to serve as a primary key:

- DBMS supplied
- Short, numeric and never changes – an ideal primary key!
- Has **artificial values** that are meaningless to users
- Normally hidden in forms and reports



Surrogated Keys

Example

- RENTAL_PROPERTY **without** surrogate key:

RENTAL_PROPERTY (Street, City, State/Province, Zip/PostalCode, Country, Rental_Rate)

- RENTAL_PROPERTY with surrogate key:
RENTAL_PROPERTY (PropertyID, Street, City, State/Province, Zip/PostalCode, Country, Rental_Rate)



Surrogated Keys

Exercises

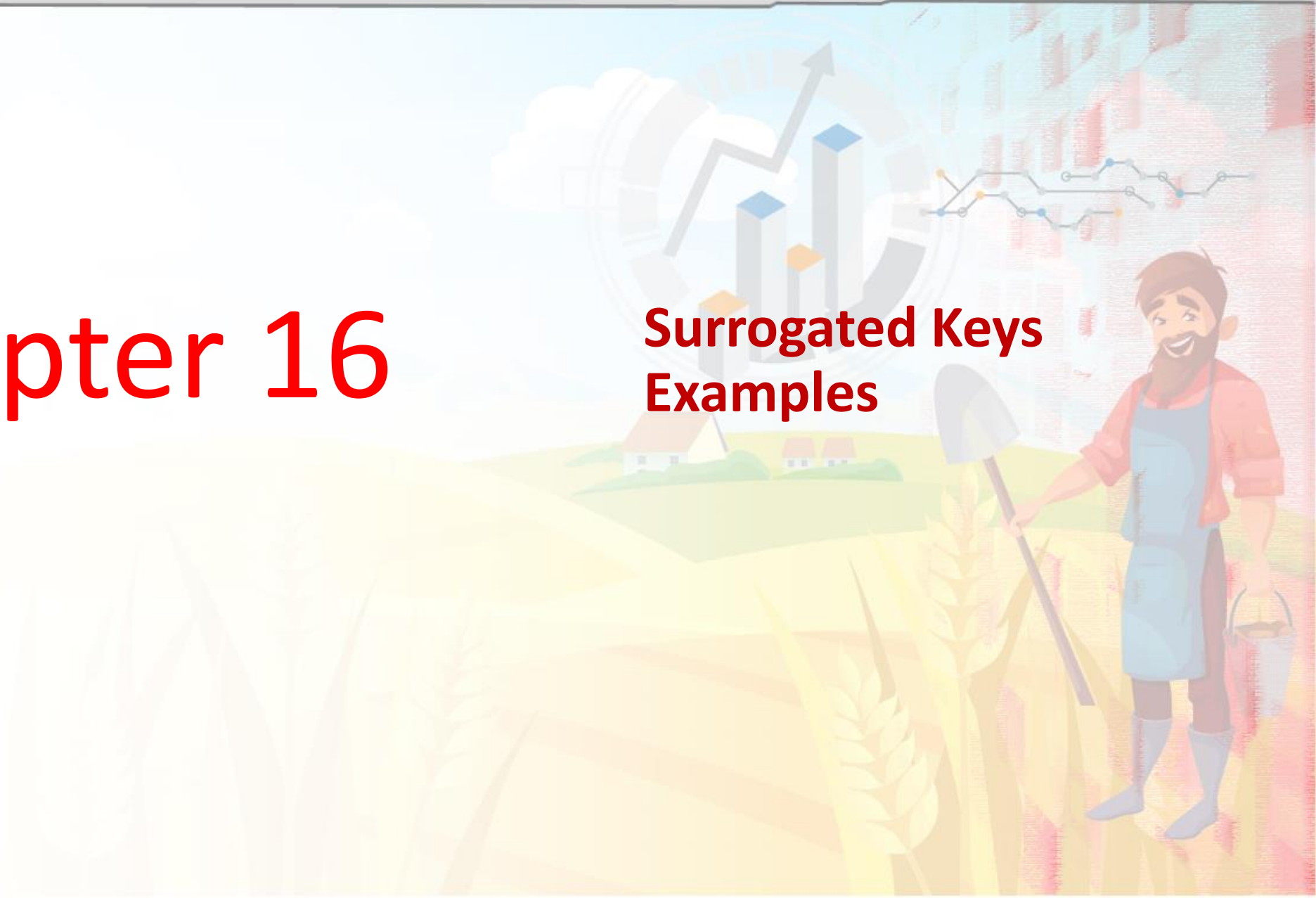
Lets solve following examples

- When to put up a complaint by using phone?
- ComplaintRef# is normally generated
- Needs attributes?

End

Chapter 16

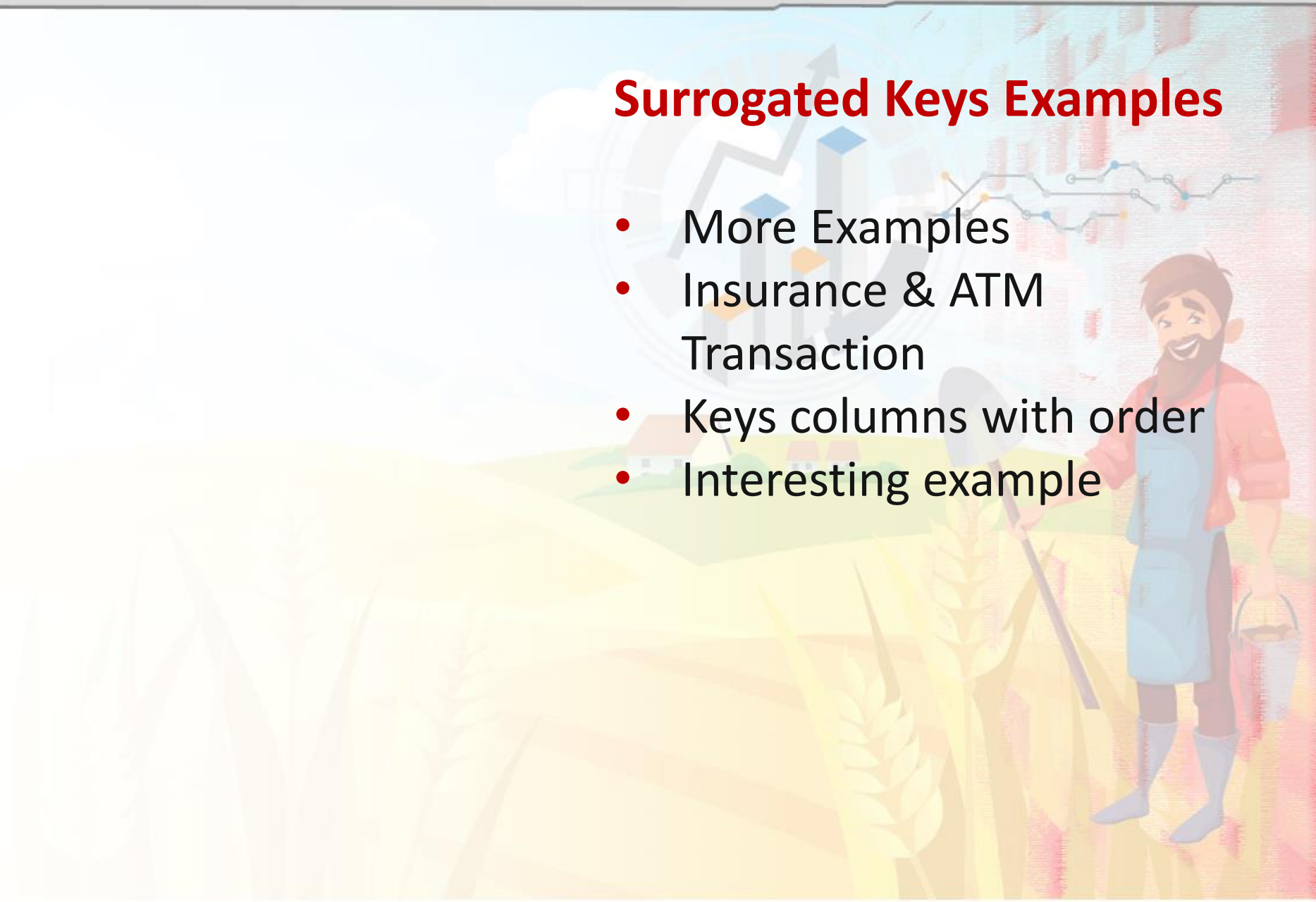
Surrogated Keys Examples



Surrogated Keys Examples

Surrogated Keys Examples

- More Examples
- Insurance & ATM Transaction
- Keys columns with order
- Interesting example



Surrogated Keys Examples

More Examples

ATMTransaction
(Card#, Serial#, Amount, DrawDate)

Need more attributes

Machine#, Type, Status, LimitBalance



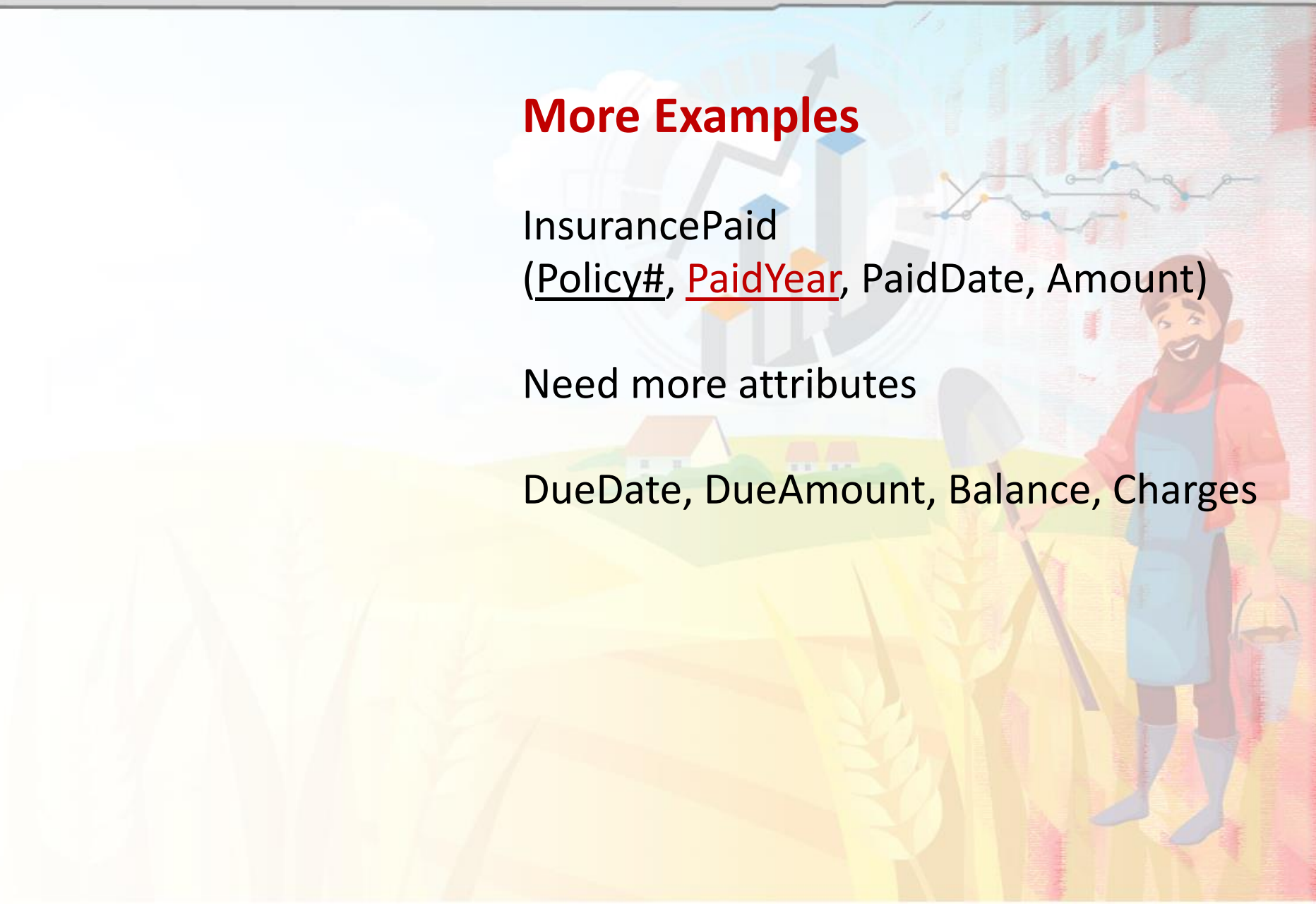
Surrogated Keys Examples

More Examples

InsurancePaid
(Policy#, PaidYear, PaidDate, Amount)

Need more attributes

DueDate, DueAmount, Balance, Charges



Surrogated Keys Examples

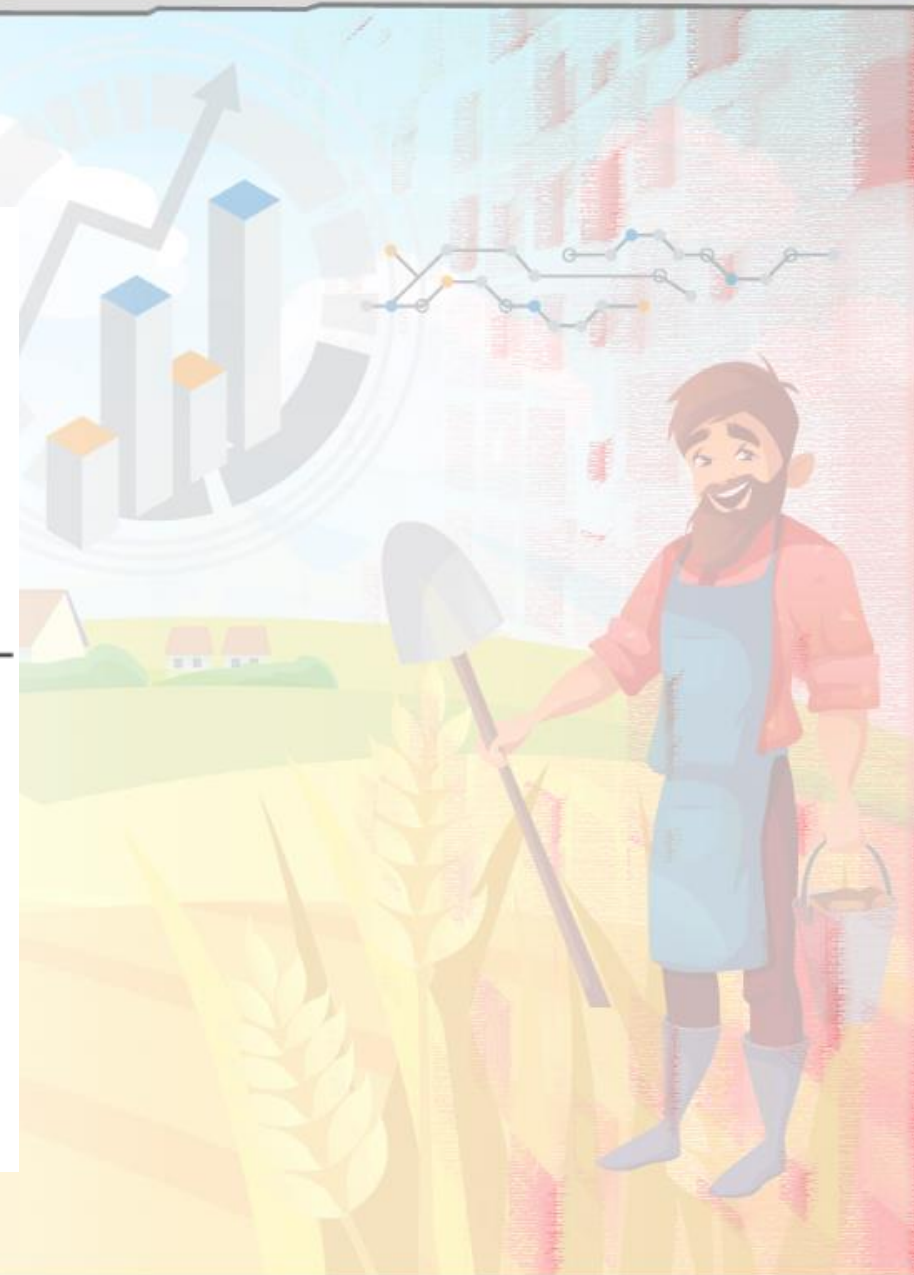
Keys with order in columns

<u>PatientID</u>	<u>Visit#</u>	Visit Date
558741	1	...
558741	2	
558741	3	
...		
589210	1	
589210	2	
599211	1	
599211	2	
...		

Better Solution
using composite keys

<u>PatientID</u>	<u>Visit#</u>	Visit Date
558741	1	...
558741	2	
558741	3	
589210	4	
589210	5	
599211	6	
599211	7	
...	...	

830000 and more



Surrogated Keys Examples

Interesting example

Lab Requisitions - drjaved

REQId	REQ Desc	Status
1	HEMATOLOGY	

Add Update Remove

Test Description	Ref. Range
1: HEMATOLOGY:Y	
2: BIOCHEMISTRY:Y	
3: SEROLOGY:Y	
4: HORMONES:Y	
5: ANTENATAL:Y	
6: URINE:Y	
7: STOOL:Y	
8: MICROBIOLOGY:Y	
9: SEMEN:Y	
10: MEDICAL CHECK-UPS:Y	
11: OTHERS:Y	

SubReqId 22 ReqId

Add Update Remove

SubLab req. id	Description
1: CBC	
2: Coagulation Studies	
3: Others Hematology	

ReqId	REQ Desc	REQ Range	REF
22	Differential		1

Add Update Remove

ReqId	Description	Ref. Range
20	Hb : M 15.0 ± 2.0; F 13.5 ± 2.0 gm/dl	
21	WBC : 4.0 – 10.0 x 10 ⁹ /L	
22	Differential :	
23	Platelets : 140 – 400 x 10 ⁹ /L	
24	RBC : M 4.6 – 6.5; F 3.9 – 5.6 10 ¹² /L	
25	Hct : M 0.39 – 0.50; F 0.34 – 0.44	
26	MCV : M 81 -94; F 79 -91 fl	
27	MCH : M 27 -32; F 25 -31 pg	
28	MCHC : M 31 -35; F 30 – 34%	
29	Reticulocytes : < 2%	

REQId	SubReqId Desc
1	CBC

Add Update Remove

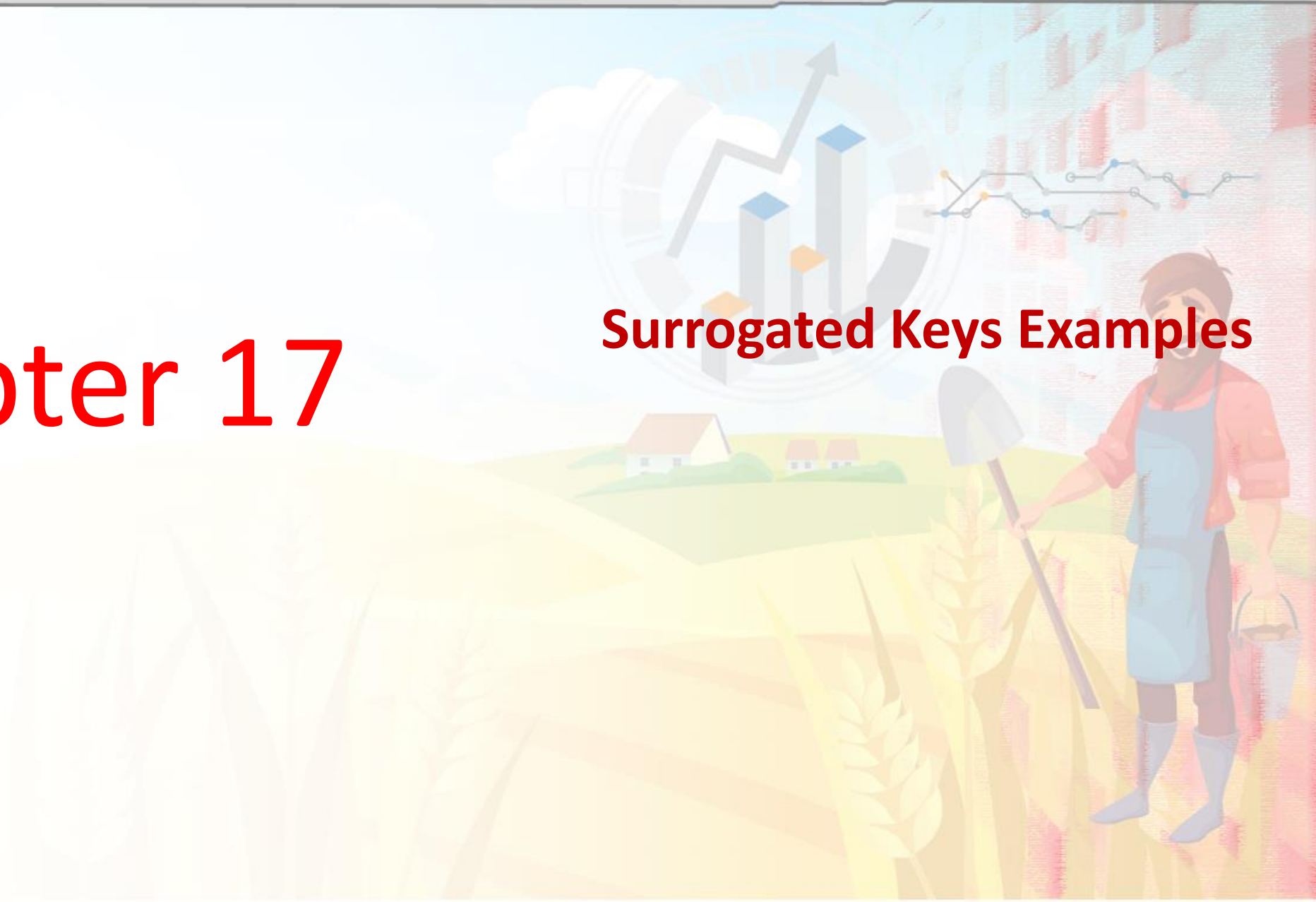
LABREQ	LABREQ_SNODESC	LABRE
38	Band Cell	22
39	Neutrophil	22
40	Lymphocyte	22
41	Monocyte	22
42	Eosinophil	22

03/11/2013 11:46:29 AM

Total Records: 11

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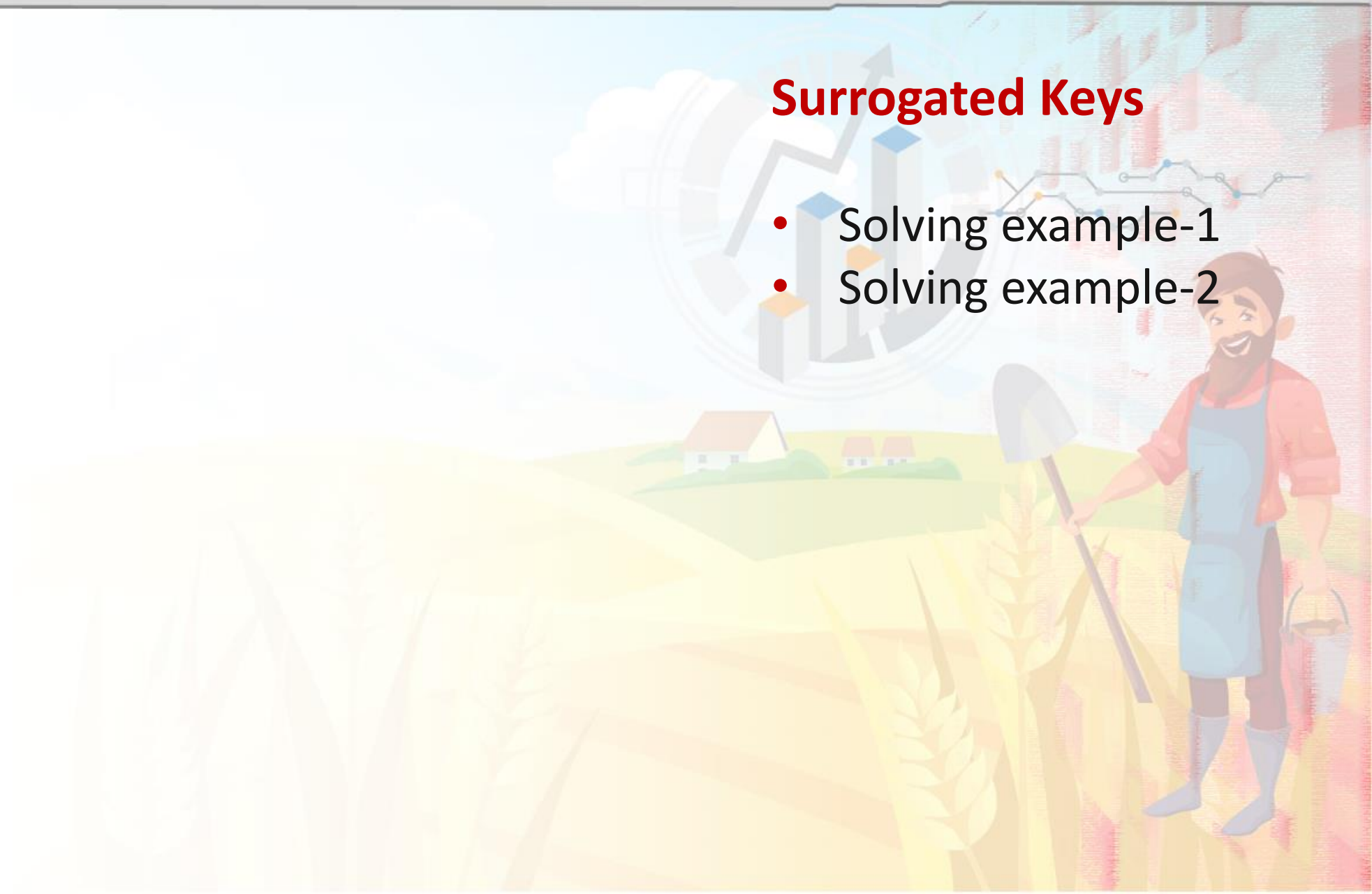
Surrogated Keys Examples



Surrogated Keys Examples

Surrogated Keys

- Solving example-1
- Solving example-2

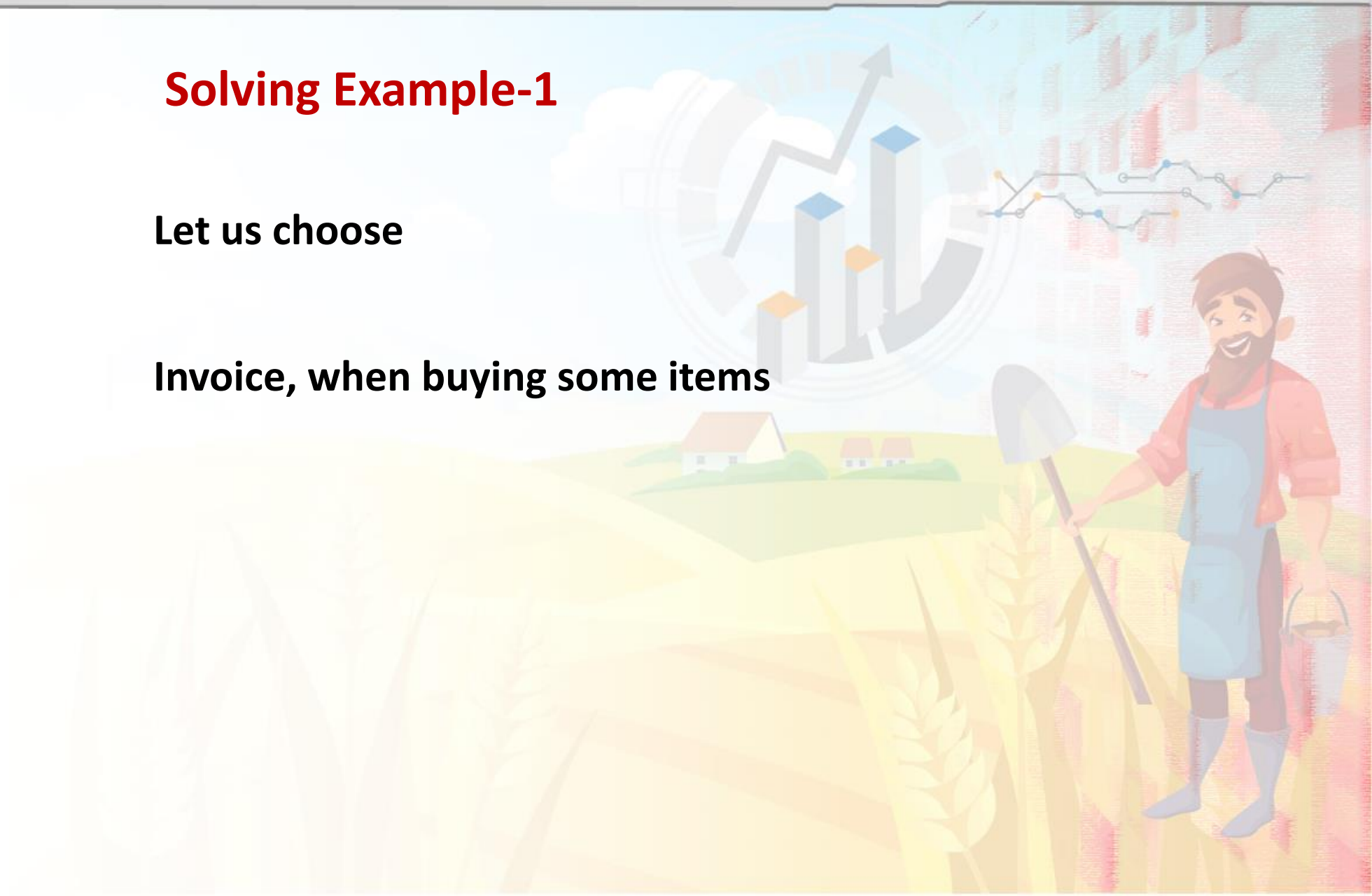


Surrogated Keys Examples

Solving Example-1

Let us choose

Invoice, when buying some items



Surrogated Keys Examples

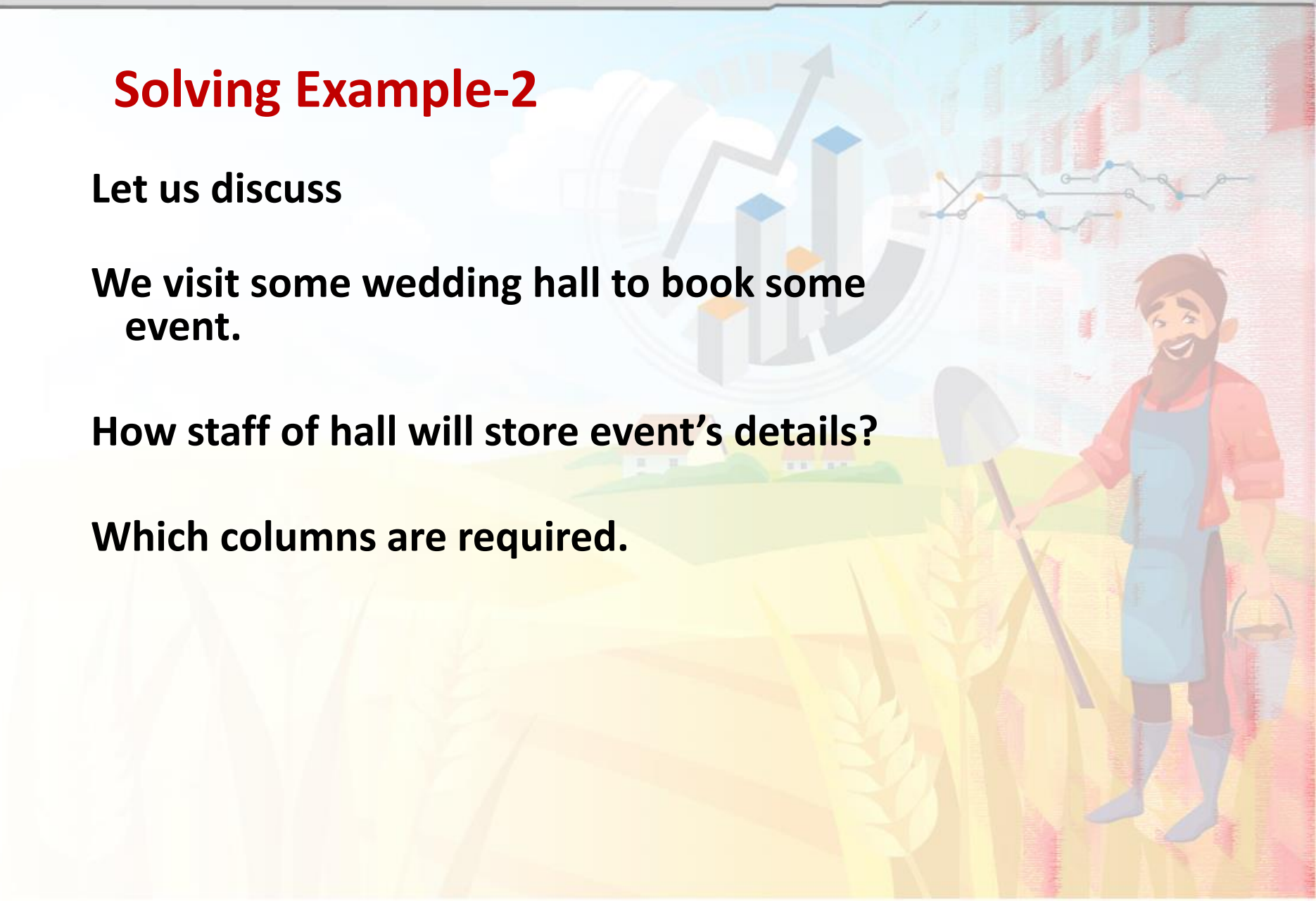
Solving Example-2

Let us discuss

We visit some wedding hall to book some event.

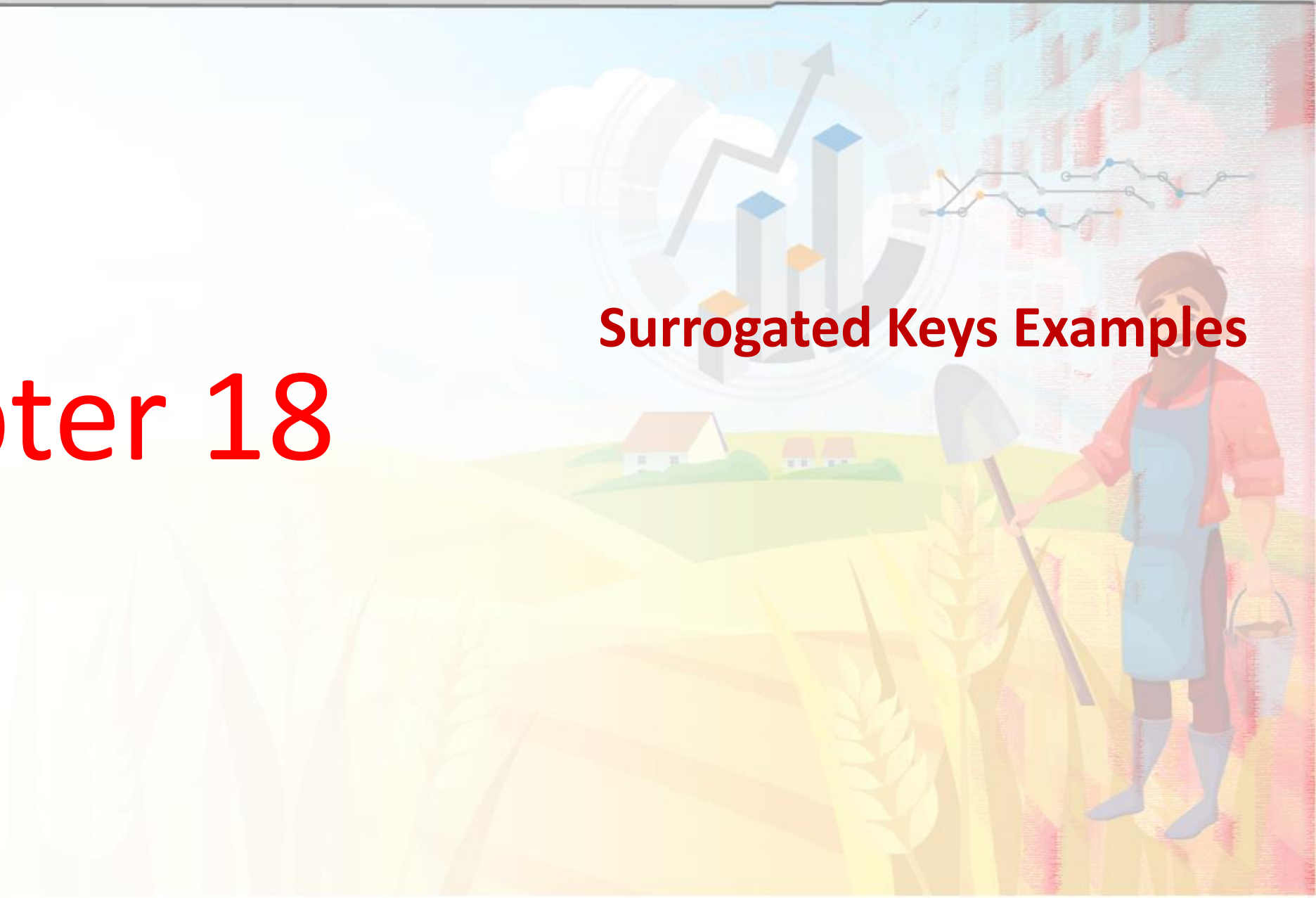
How staff of hall will store event's details?

Which columns are required.



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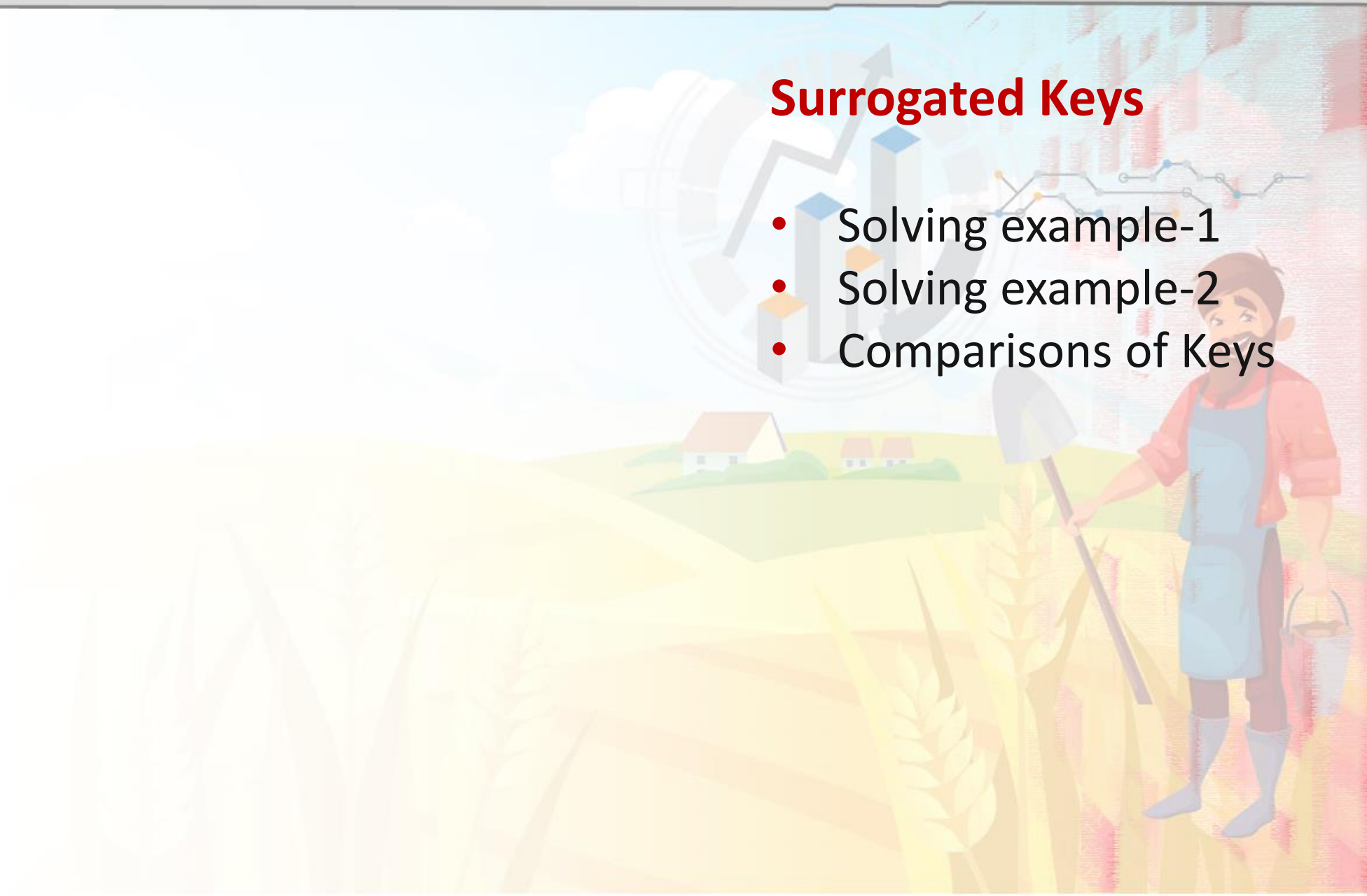
Surrogated Keys Examples



Surrogated Keys Examples

Surrogated Keys

- Solving example-1
- Solving example-2
- Comparisons of Keys



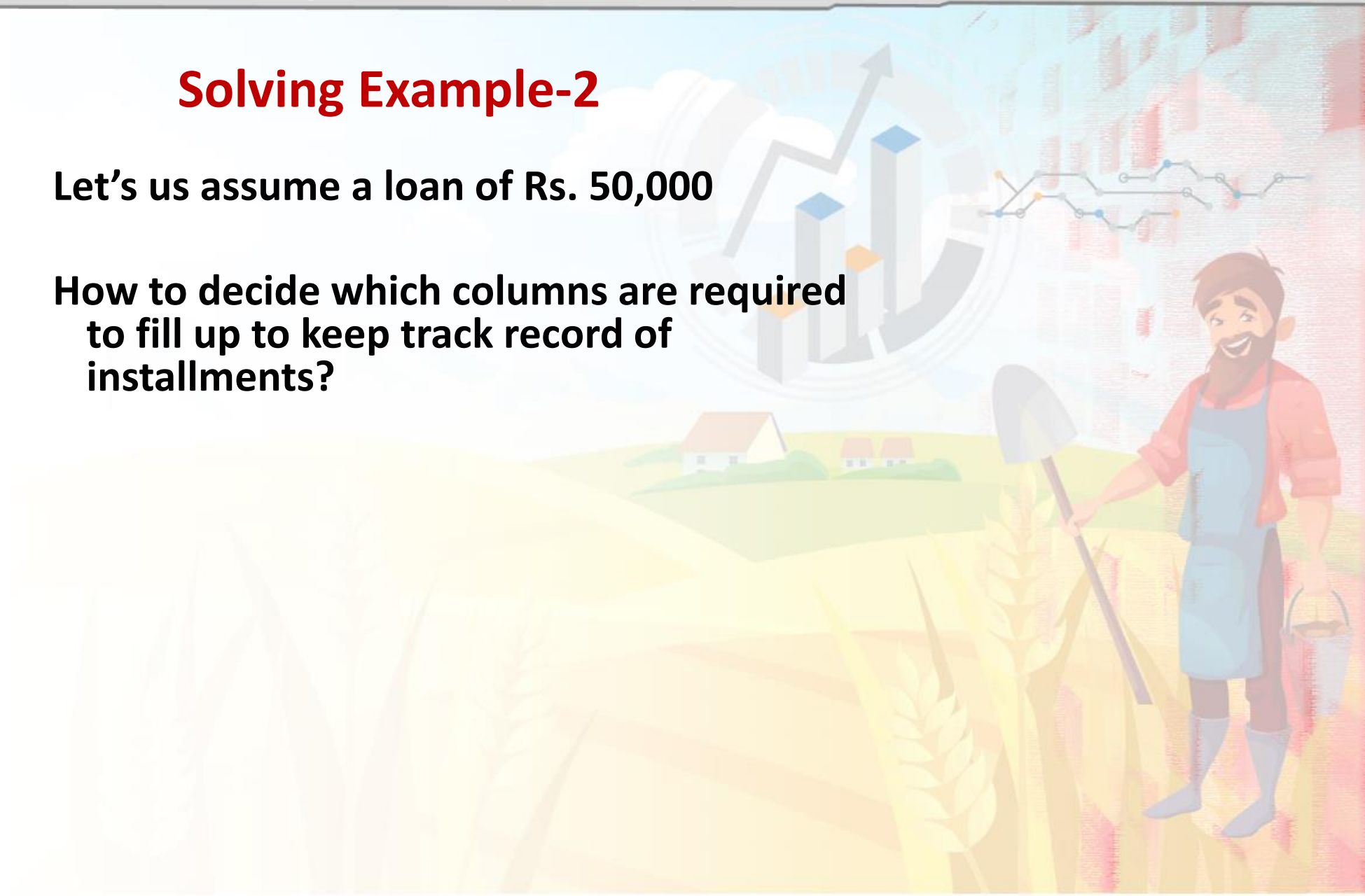
Solving Example-1 (Saving blogs)

Surrogated Keys Examples

Solving Example-2

Let's us assume a loan of Rs. 50,000

How to decide which columns are required to fill up to keep track record of installments?



Surrogated Keys Examples

Comparisons of keys

Let us discuss

What is difference between PK & surrogated key

What is difference between PK & unique key

Give three more examples of surrogated keys from real-life.

End

Foreign Keys

Chapter 19



Foreign Keys

Foreign Keys

- Definition
- Example-1
- Example-2



Foreign Keys

Definition

A **foreign key** is an attribute that refers to a primary key of same or different relation to form a link (constraint) between the relations:

- A foreign key can be a single column or a composite key
- The term refers to the fact that key values are *foreign* to the relation in which they appear as foreign key values
- Ideally Data type, Length/ Size of FK and referring PK must be same

Foreign Keys

Example-1

DEPARTMENT (DeptID, DepartmentName,
BudgetCode, ManagerName)

EMPLOYEE (EmpNumber, EmpName, *DeptID*)

Name of FK column may be different from the
name of referencing PK



Foreign Keys

Example-2

CUSTOMER(CustId, Name, Address, Telno, Email)

ORDERS(Invoice#, InvType, InvDate, CustomerId)

Does Invoice# always contain serial numbers?

A screenshot of a software window titled "Invoice". It displays an invoice from "Lloyds Chemist" for the date 11/29/2013. The invoice lists items like PANADOL and ALINAL DROPS with their quantities and prices. It also shows a total amount and a net total after discount. At the bottom, there is a signature line and page navigation controls.

Code	Med Description	Qty	Price	Amount
3906020	PANADOL	1	12.5	12.5
3900001	ALINAL DROPS	2	20	40
3906020	PANADOL	2	12.5	25
3900001	ALINAL DROPS	3	20	60
Total:				137.5
Disc.:				6
Net Total:				129.25

End

Chapter 20

Foreign Keys Examples



Foreign Keys Examples

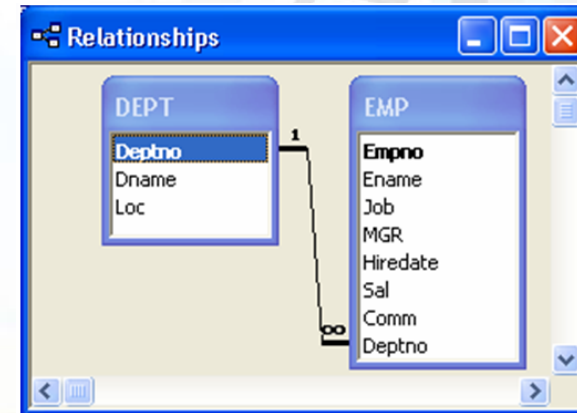
Foreign Key Examples

- Relationship details
- FK & Recursive Relationship
- Logical link between tables



Foreign Keys Examples

Relationship details



Deptno is a PK in DEPT table*

Empno is a PK in EMP table

Deptno is a FK in EMP table*

(it references to an attribute Deptno of DEPT table)

NOTE: An attribute within one relation that matches the key (Primary/same) in the same relation, is called recursive relation

Foreign Keys Examples

FK & Recursive Relationships

FK or Recursive Relationship

DEPT : Table

Deptno	Dname	Loc
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON
50	R & D	TORONTO

Record: 6 of 6

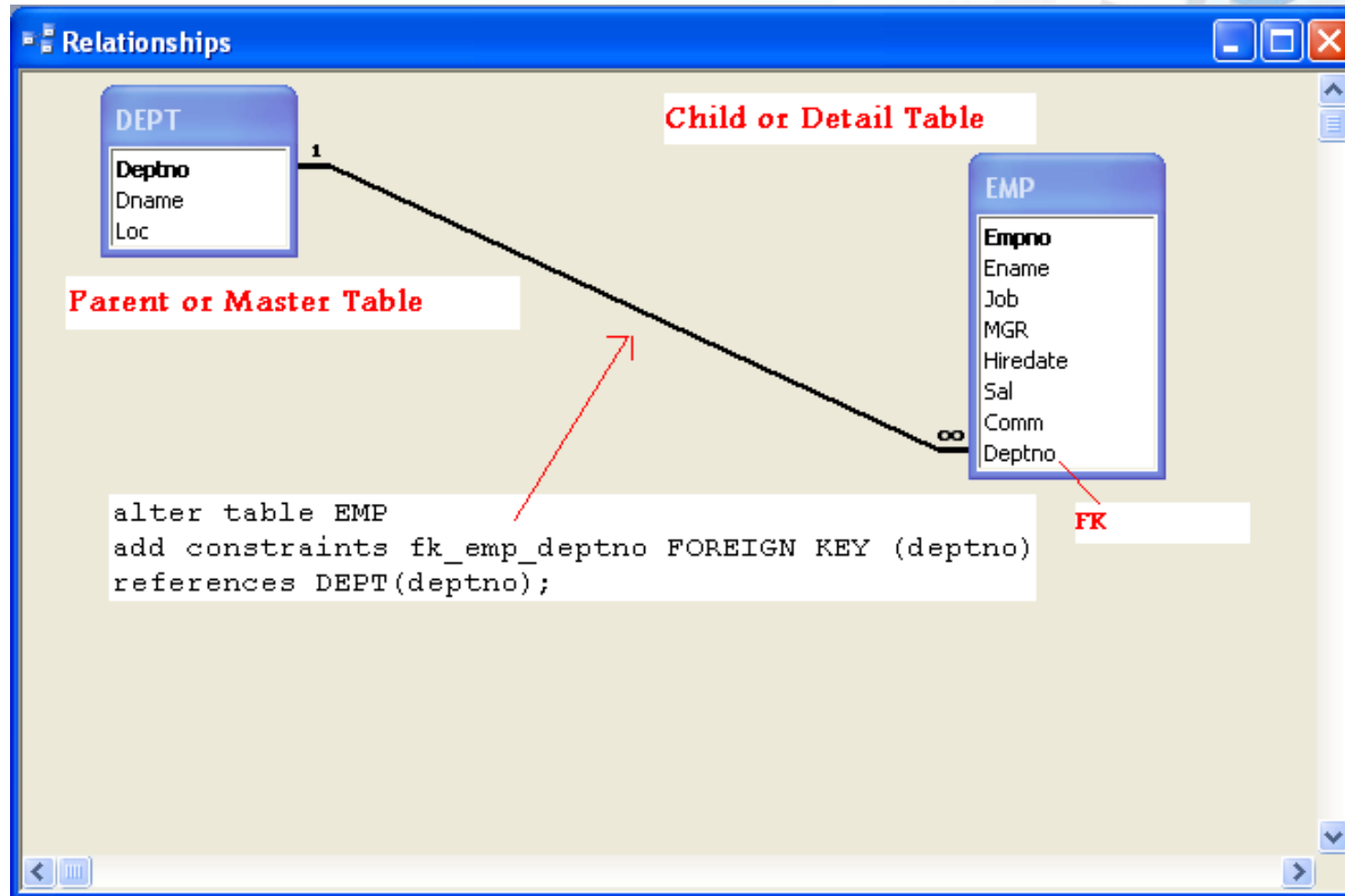
EMP : Table

Empno	Ename	Job	MGR	Hiredate	Sal	Comm	Deptno
7369	SMITH	CLERK	7902	17-Dec-80	800		20
7499	ALLEN	SALESMAN	7698	20-Feb-81	1600	300	30
7521	WARD	SALESMAN	7698	22-Feb-81	1250	500	30
7566	JONES	MANAGER	7839	02-Apr-81	2975		20
7654	MARTIN	SALESMAN	7698	28-Sep-81	1250	1400	30
7698	BLAKE	MANAGER	7839	01-May-81	2850		30
7782	CLARK	MANAGER	7839	09-Jun-81	2450		10
7788	SCOTT	ANALYST	7566	19-Apr-87	3000		20
7839	KING	PRESIDENT		17-Nov-81	5000		10
7844	TURNER	SALESMAN	7698	08-Sep-81	1500	0	30
7876	ADAMS	CLERK	7788	23-May-87	1100		20
7900	JAMES	CLERK	7698	03-Dec-81	950		30
7902	FORD	ANALYST	7566	03-Dec-81	3000		20
7934	MILLER	CLERK	7782	23-Jan-82	1300		10
7935	MILLER1	CLERK	7782	23-Jan-82	1300		

The diagram illustrates foreign key relationships between the DEPT and EMP tables. In the DEPT table, Deptno is the primary key. In the EMP table, Empno is the primary key, and MGR and Deptno are foreign keys. The MGR column represents a recursive relationship, where an employee can be the manager of another employee. The Deptno column represents a foreign key relationship to the DEPT table. Red circles and arrows highlight these relationships: a red circle around DEPT.Deptno, a red circle around EMP.Deptno, and a red circle around EMP.MGR. A red arrow points from EMP.MGR to EMP.Deptno, indicating a recursive relationship. A blue arrow points from EMP.Deptno to DEPT.Deptno, indicating a foreign key relationship.

Foreign Keys Examples

Logical link between tables



Foreign Keys Examples

Chapter 21



Foreign Keys Examples

Foreign Keys Examples

- FK is a part of composite key
- Logical link between tables
- Recursive Relationship



Foreign Keys Examples

FK is a part of Composite keys

NOTE: PK column is underlined

ITEM (Item#, Item_Name, Department, Buyer)

ORDER_ITEM (OrderNumber, Item#, Quantity, Price, ExtendedPrice)

Where ORDER_ITEM.ITEM# must exist in ITEM.ITEM#

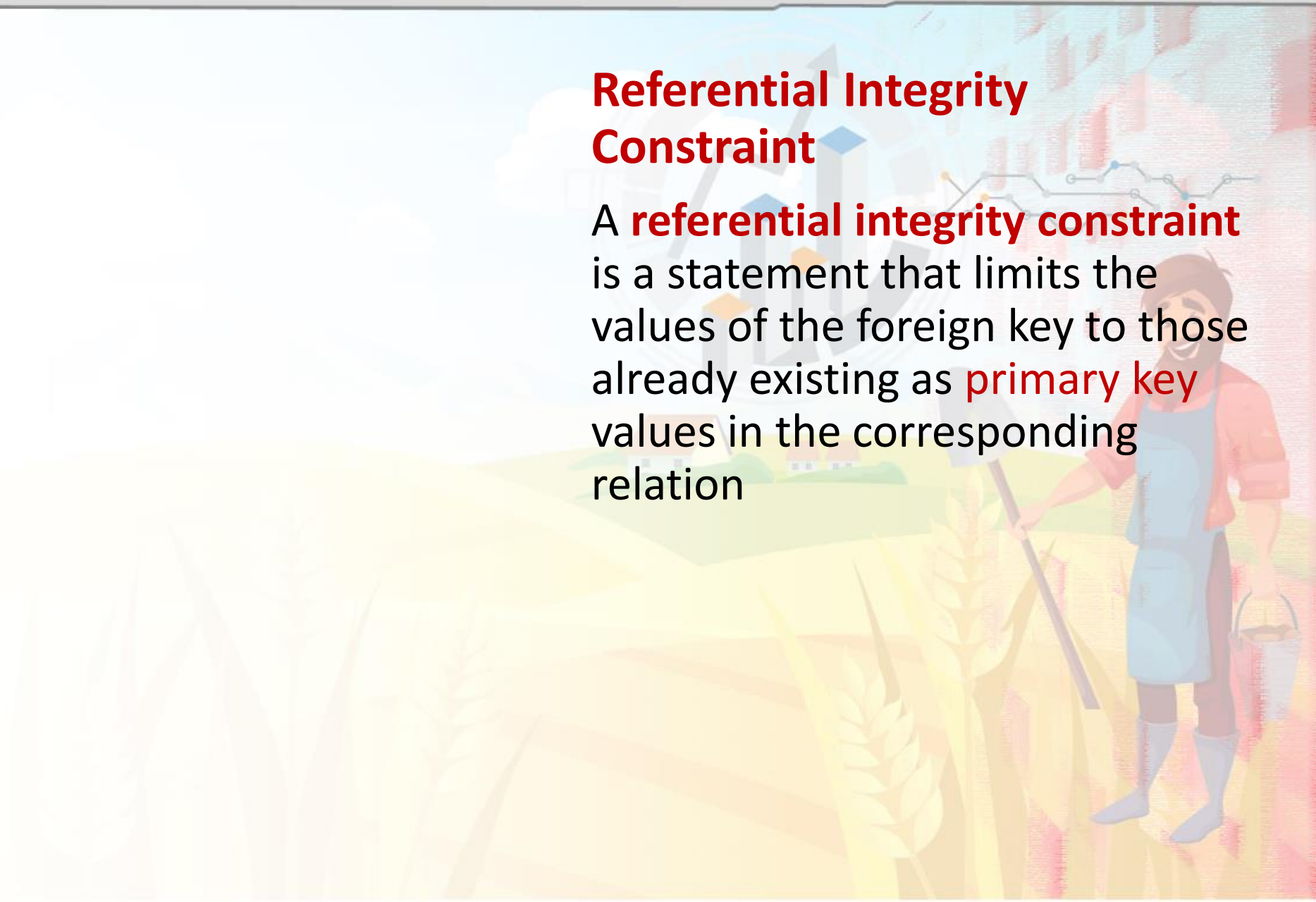
OrderNumber is an alternate name of InvoiceNumber



Foreign Keys Examples

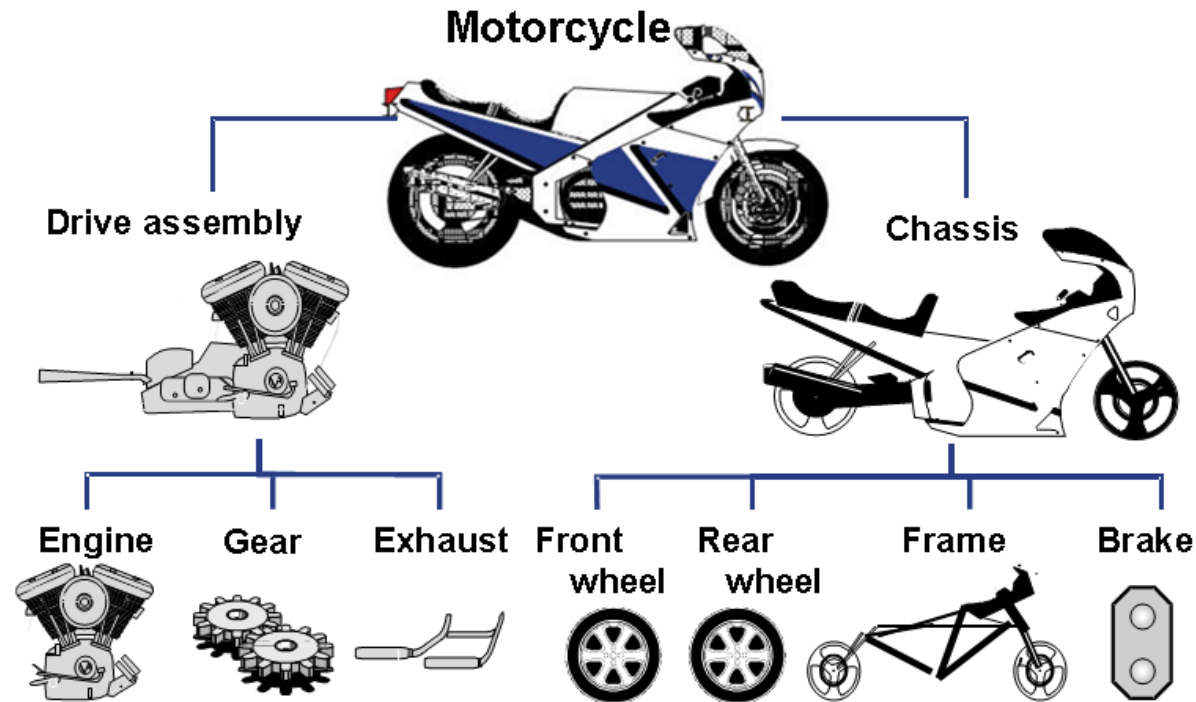
Referential Integrity Constraint

A **referential integrity constraint** is a statement that limits the values of the foreign key to those already existing as **primary key** values in the corresponding relation



Foreign Keys Examples

Recursive Relationship



Part (Part#, PName, Weight, MakeYear, Price, PType, *RefPart#*)

PType: Assembled product/ parts (e.g., Motorcycle) or single part

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Foreign Keys Examples



Foreign Keys Examples

Foreign Keys Examples

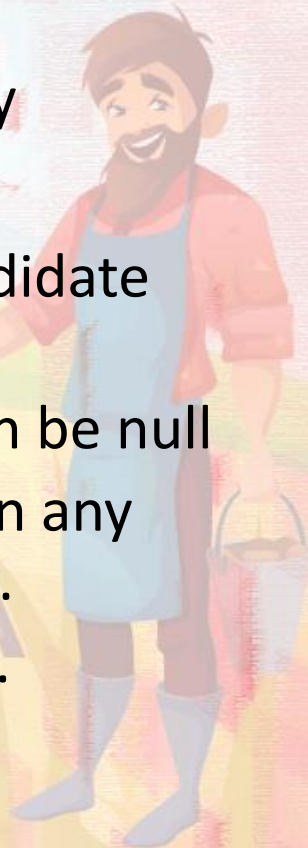
- Integrity rules
- Referential with cascade
- Integrity Example



Foreign Keys Examples

Integrity Rules

- **Entity integrity**
 - No two rows with the same primary key value
 - No null values in a primary key
- **Referential integrity**
 - Foreign keys must match candidate key of source table
 - Foreign keys in some cases can be null
 - The database must not contain any unmatched foreign key values.
 - If B references A, A must exist.



Foreign Keys Examples

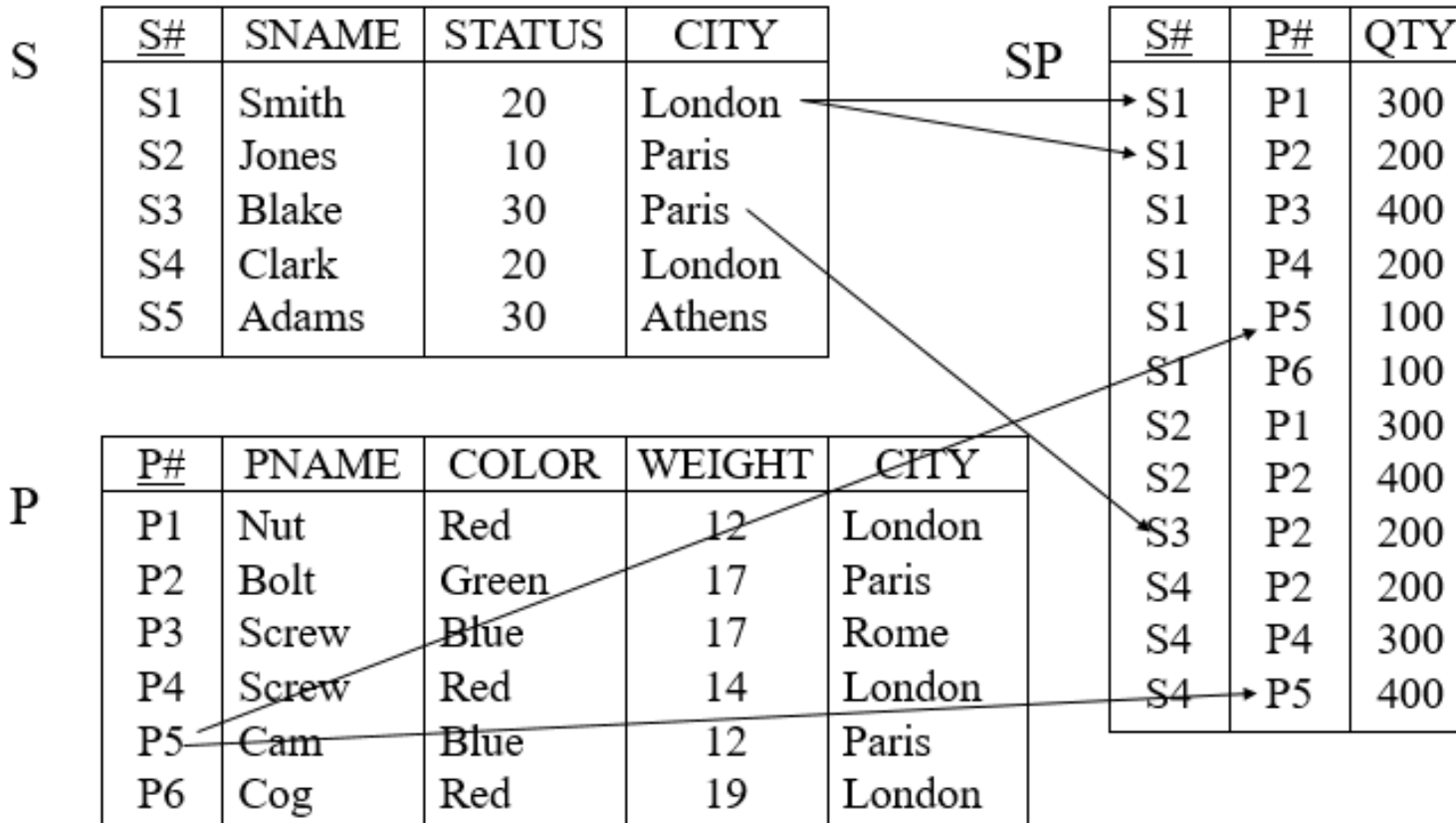
Referential with cascade

- When we update a tuple from a table, say S, that is *referenced* by another table, say SP,
- There are similar choices of *referential actions for update and delete*:
 - ON UPDATE CASCADE
 - ON UPDATE RESTRICT
- There could be other choices besides these three. , e.g.,
 - ON UPDATE SET DEFAULT.
 - ON UPDATE SET NULL.



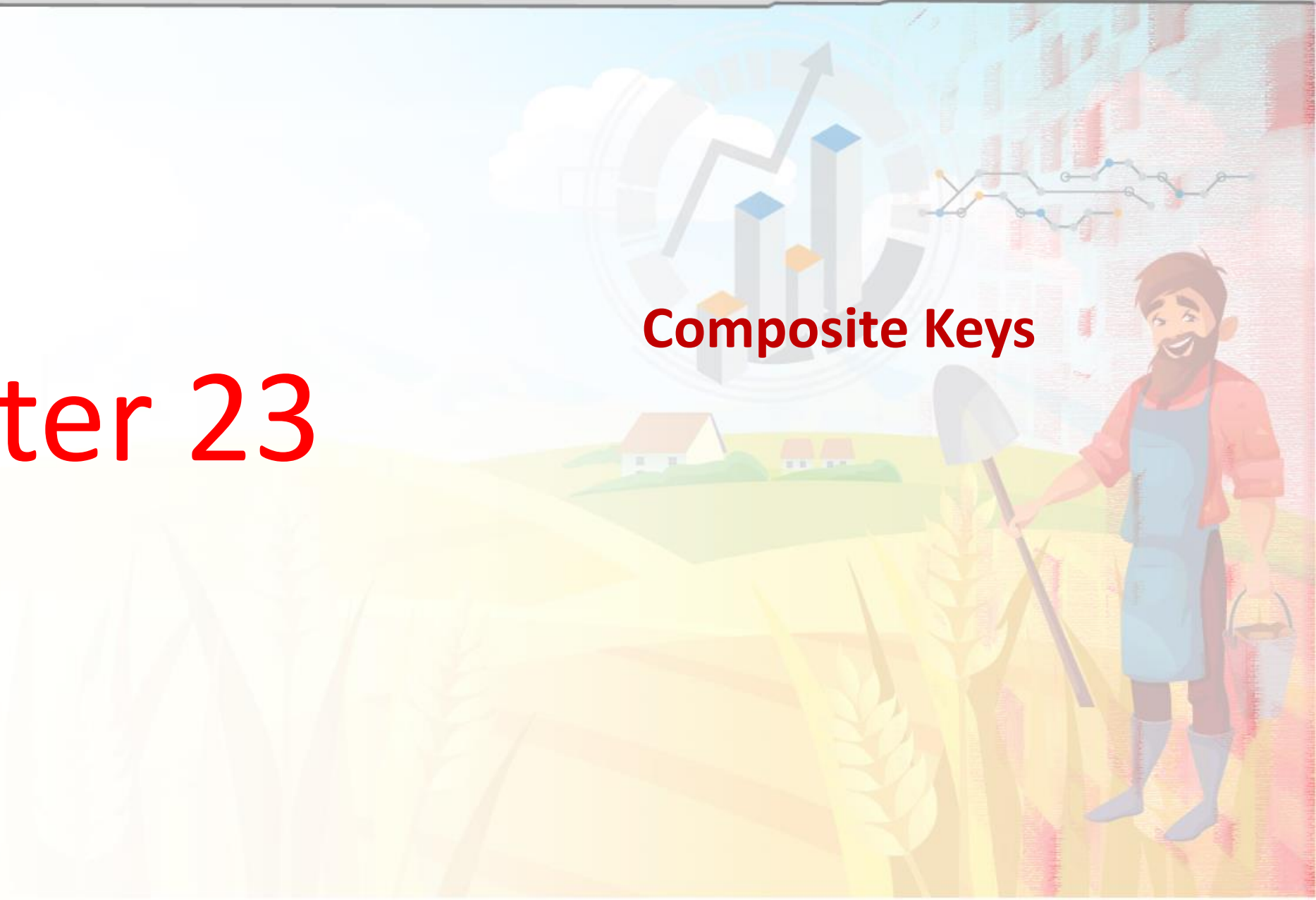
Foreign Keys Examples

Integrity Example



Chapter 23

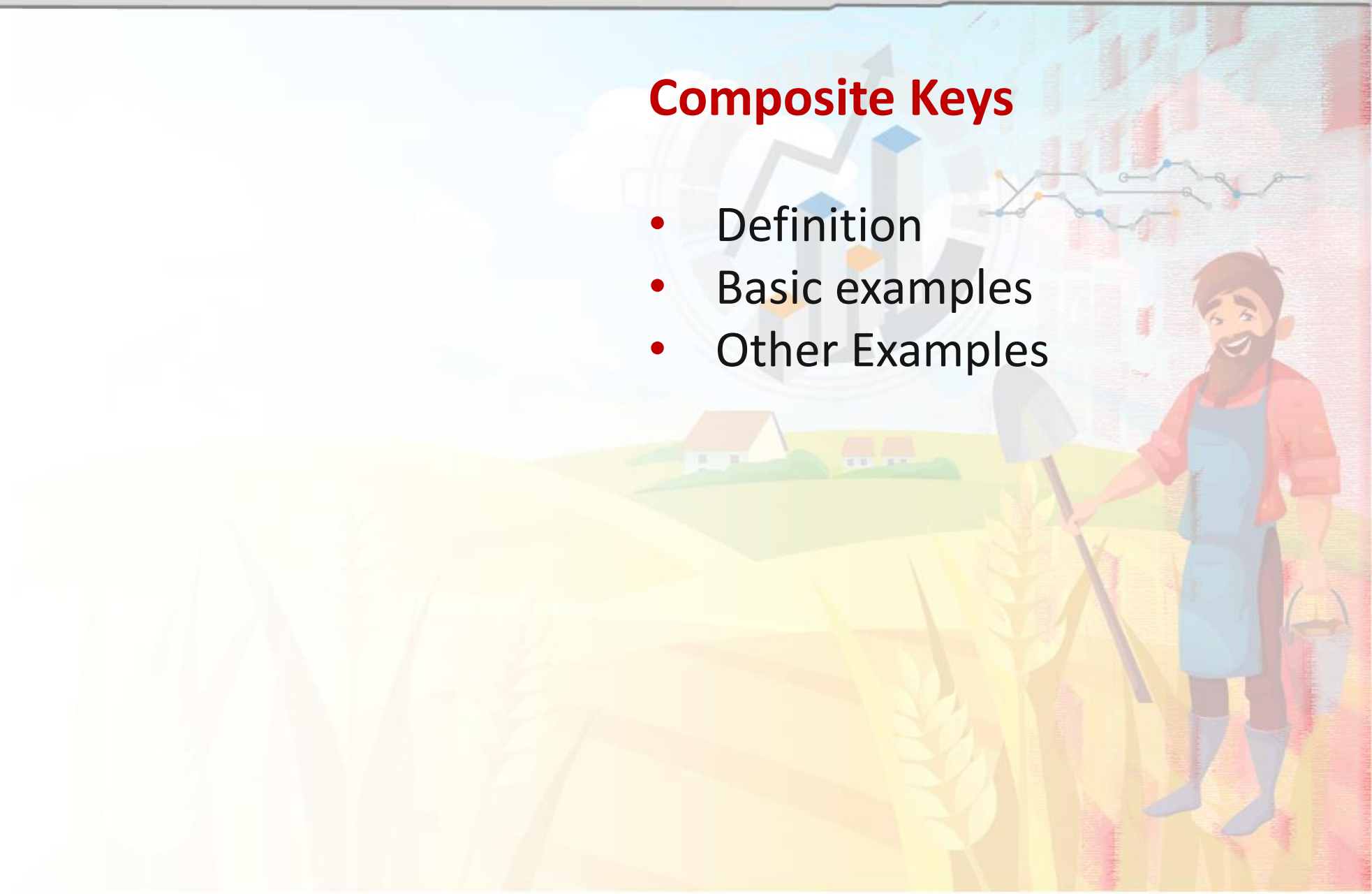
Composite Keys



Composite Keys

Composite Keys

- Definition
- Basic examples
- Other Examples

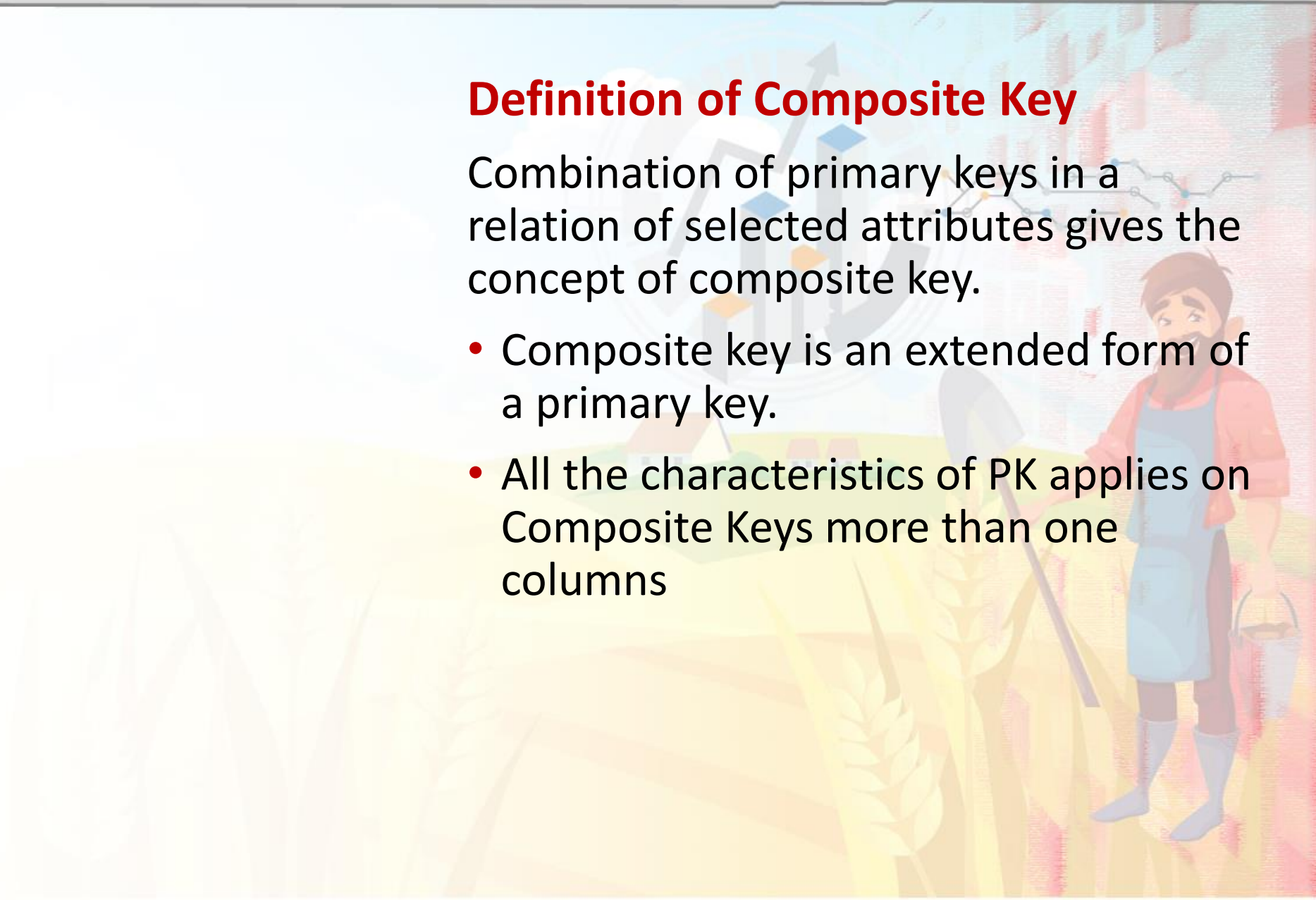


Composite Keys

Definition of Composite Key

Combination of primary keys in a relation of selected attributes gives the concept of composite key.

- Composite key is an extended form of a primary key.
- All the characteristics of PK applies on Composite Keys more than one columns



Composite Keys

Basic Example of Composite Keys

During ATM Transaction, amounts can be drawn several times on one ATM Card.

We need follow the issues

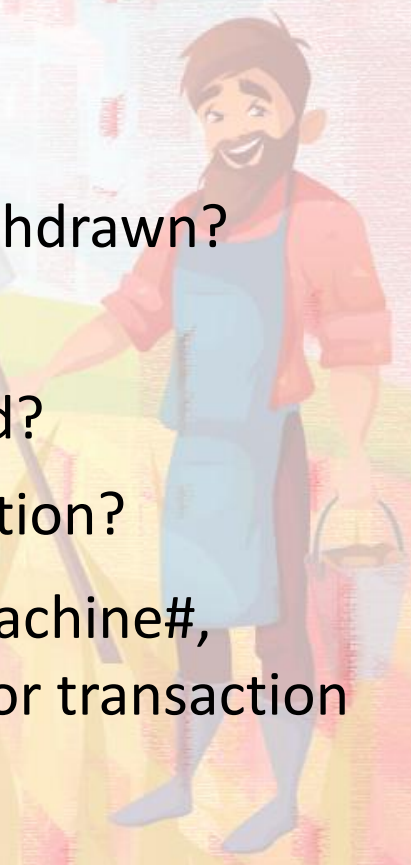
How much amount will be withdrawn?

When amount was drawn?

Which machine has been used?

What are the types of transaction?

Card#, Amount, DrawDate, Machine#, TransType are the attributes for transaction



Composite Keys

Other Examples of Composite Keys

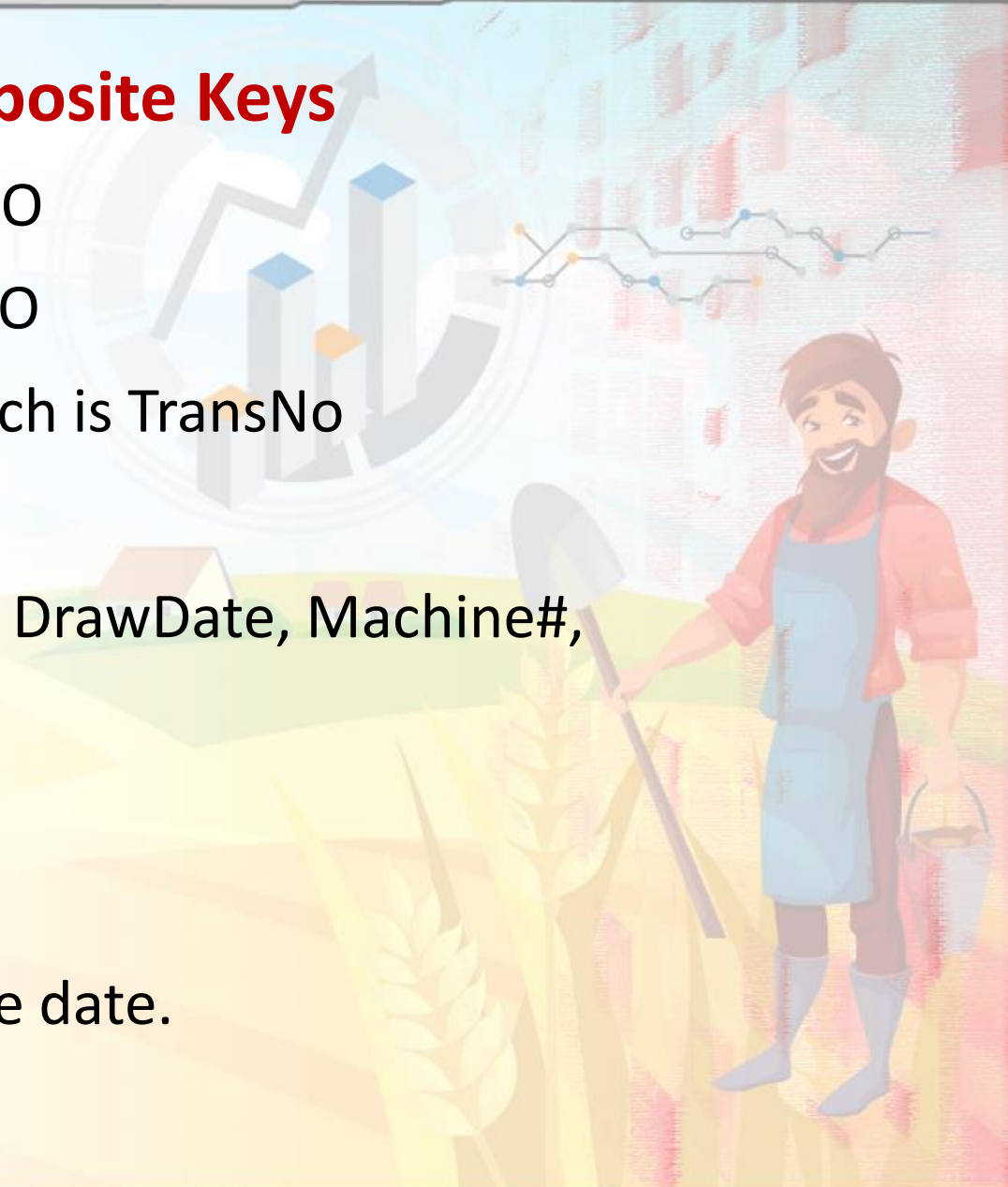
Can we say Card# will be a primary key? NO

Can DrawDate will be a key with Card#? NO

Then what to do? Add Surrogated key which is TransNo

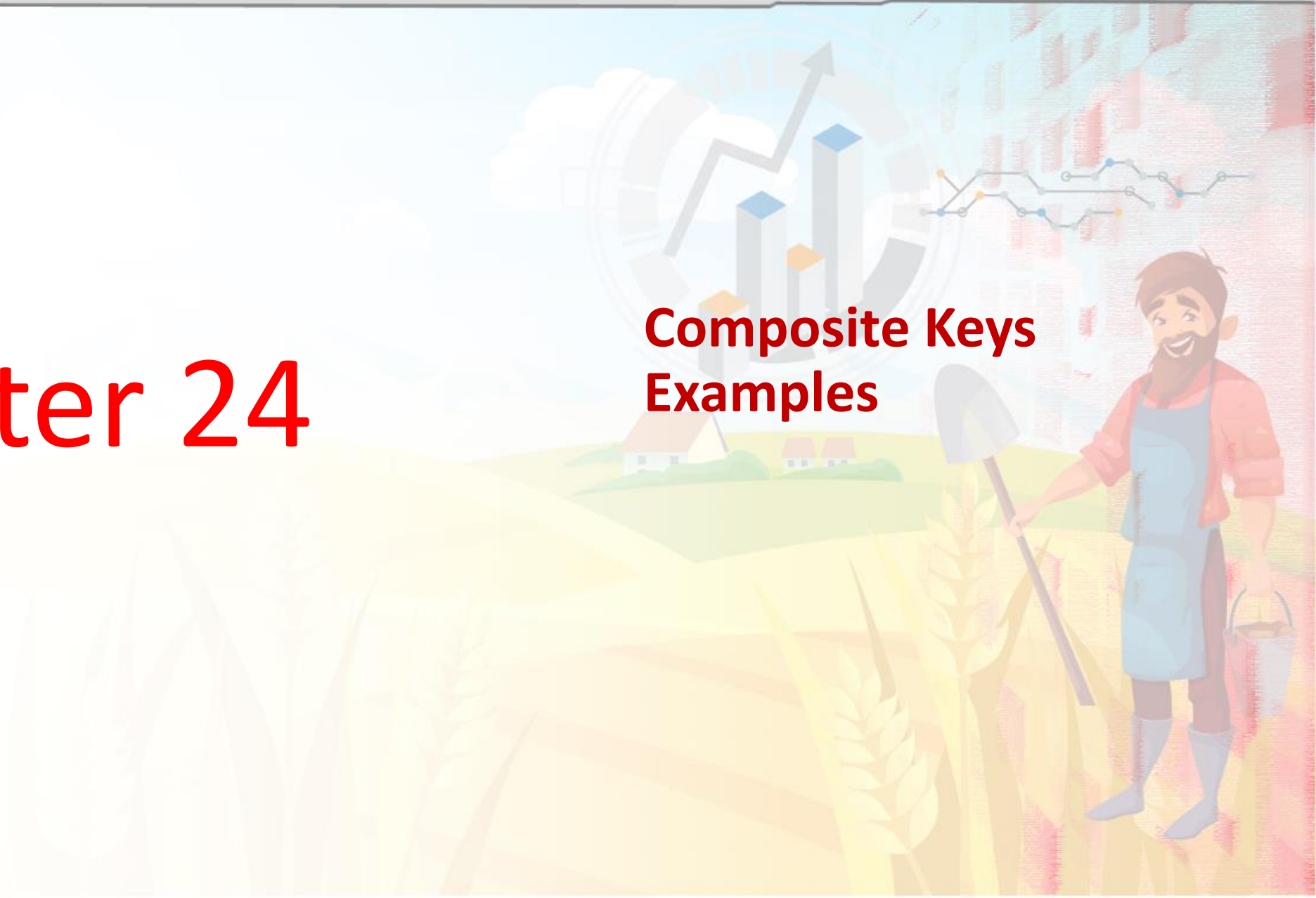
ATMTransaction(Card#, TransNo, Amount, DrawDate, Machine#, TransType)

- Date includes time in seconds as well.
- Many transactions are managed on same date.



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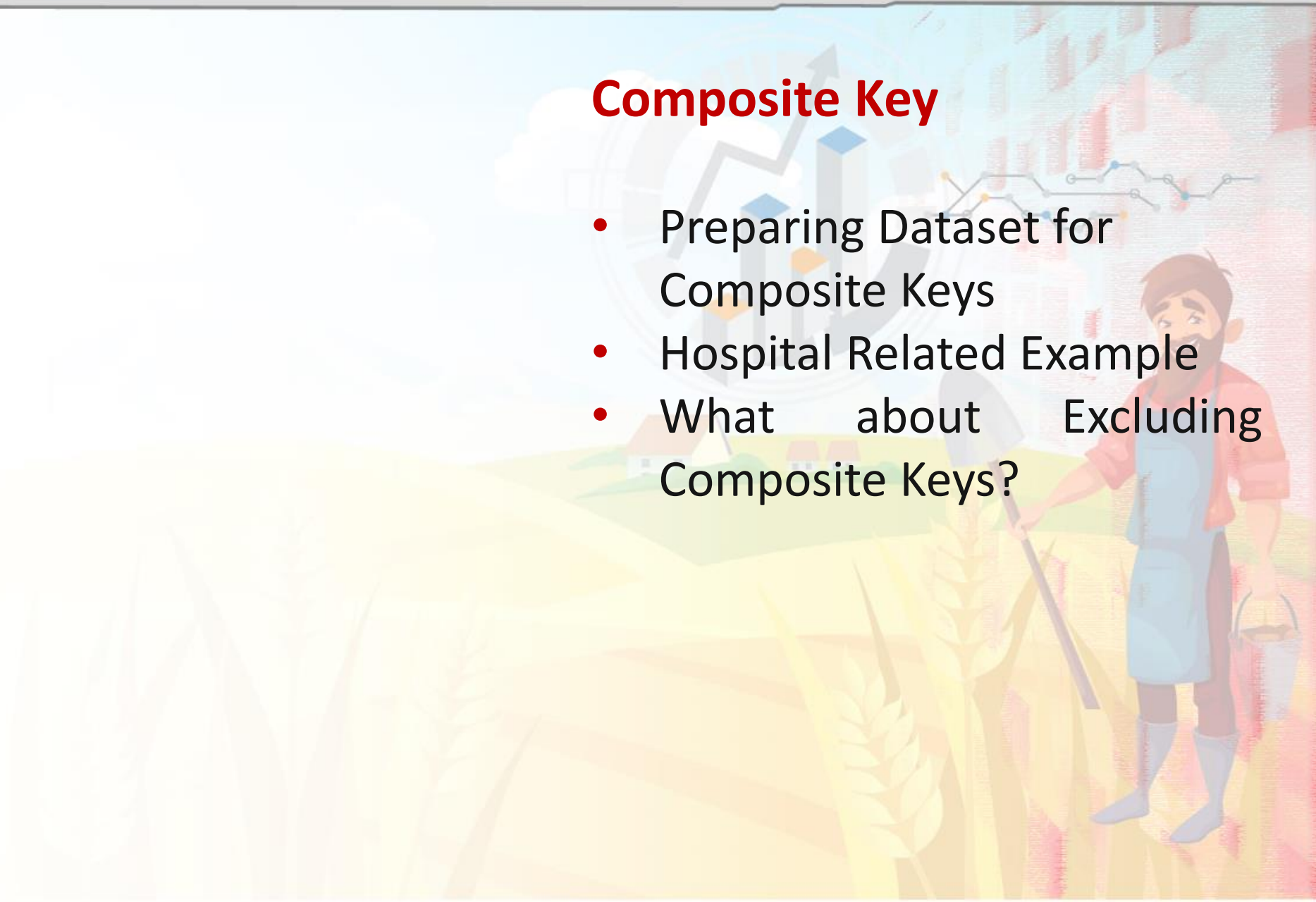
**Composite Keys
Examples**



Composite Keys Examples

Composite Key

- Preparing Dataset for Composite Keys
- Hospital Related Example
- What about Excluding Composite Keys?



Composite Keys Examples

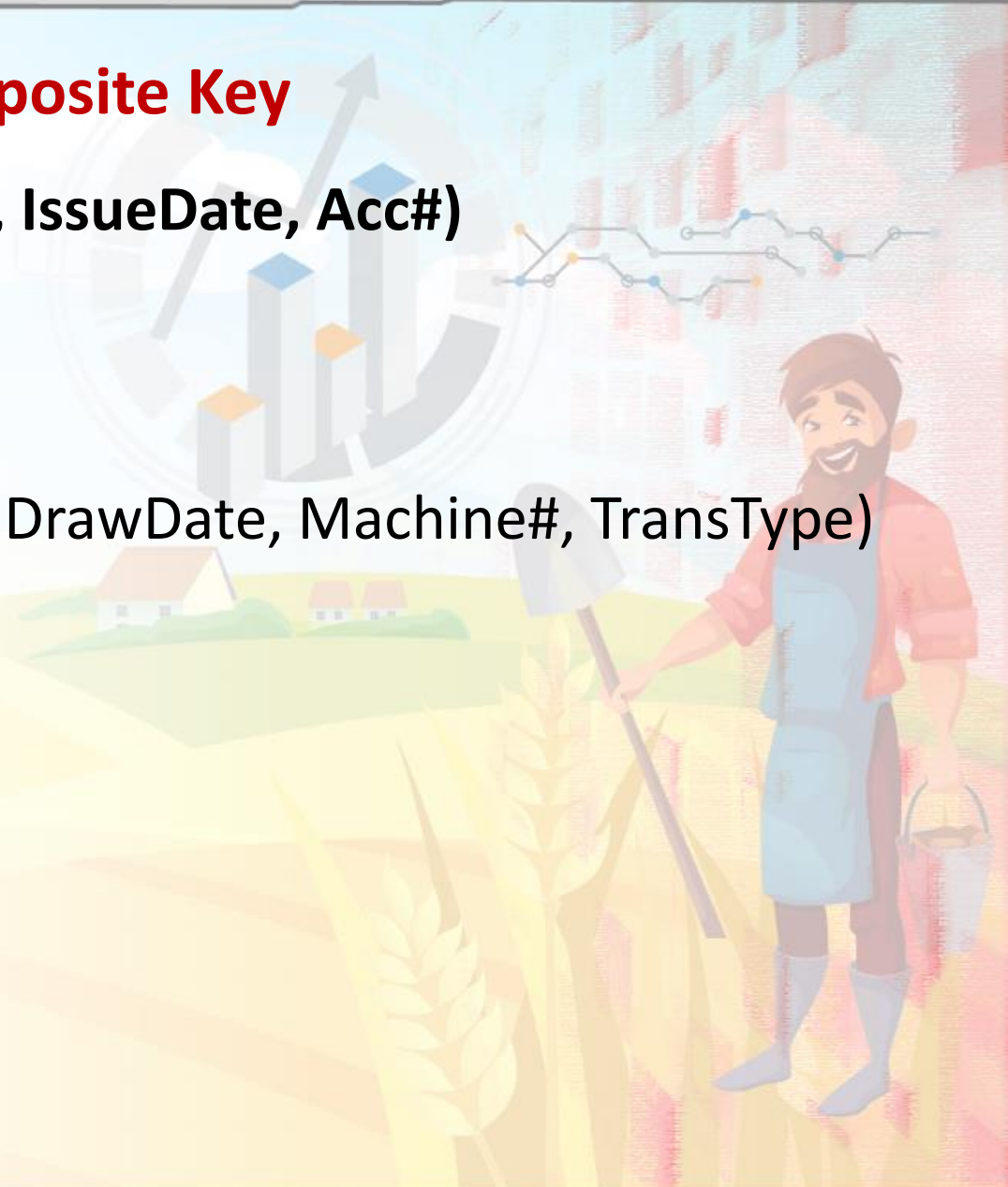
Preparing Dataset for Composite Key

Card(Card#, CardHolderName, ExpiryDate, IssueDate, Acc#)

ATMTransaction(Card#, TransNo, Amount, DrawDate, Machine#, TransType)

Do we need most closed related table?

What are PK and FK?



Composite Keys Examples

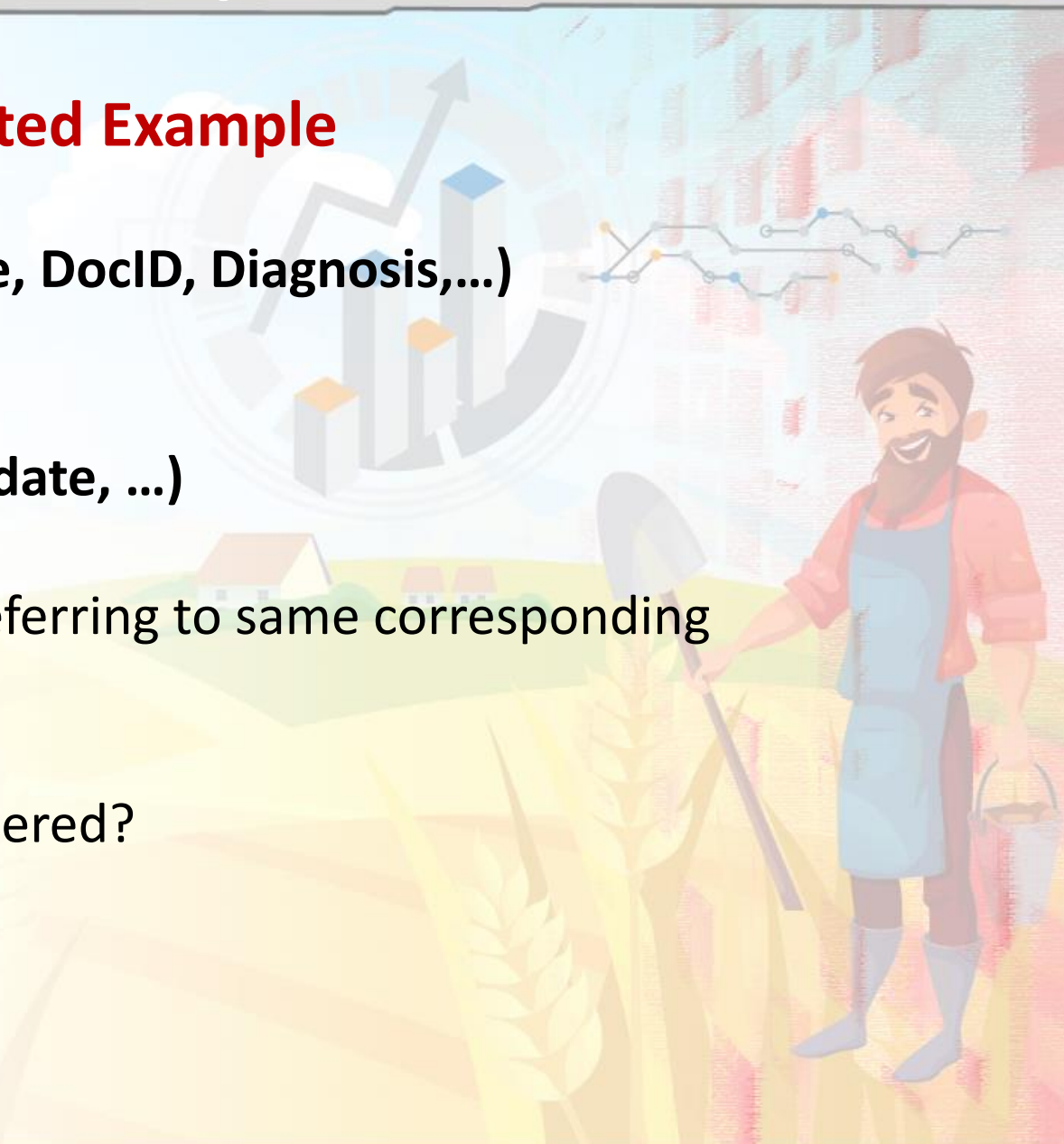
Hospital Related Example

PatientVisit(PatID, VisitSNO, Vdate, DocID, Diagnosis,...)

LabTest(PatID, VisitSNO, TestID, Tdate, ...)

LabTest has FKs *PatID, VisitSNO*, referring to same corresponding composite keys in PatientVisit

Which lab table needs to be considered?



Composite Keys Examples

What about Excluding Composite Key?

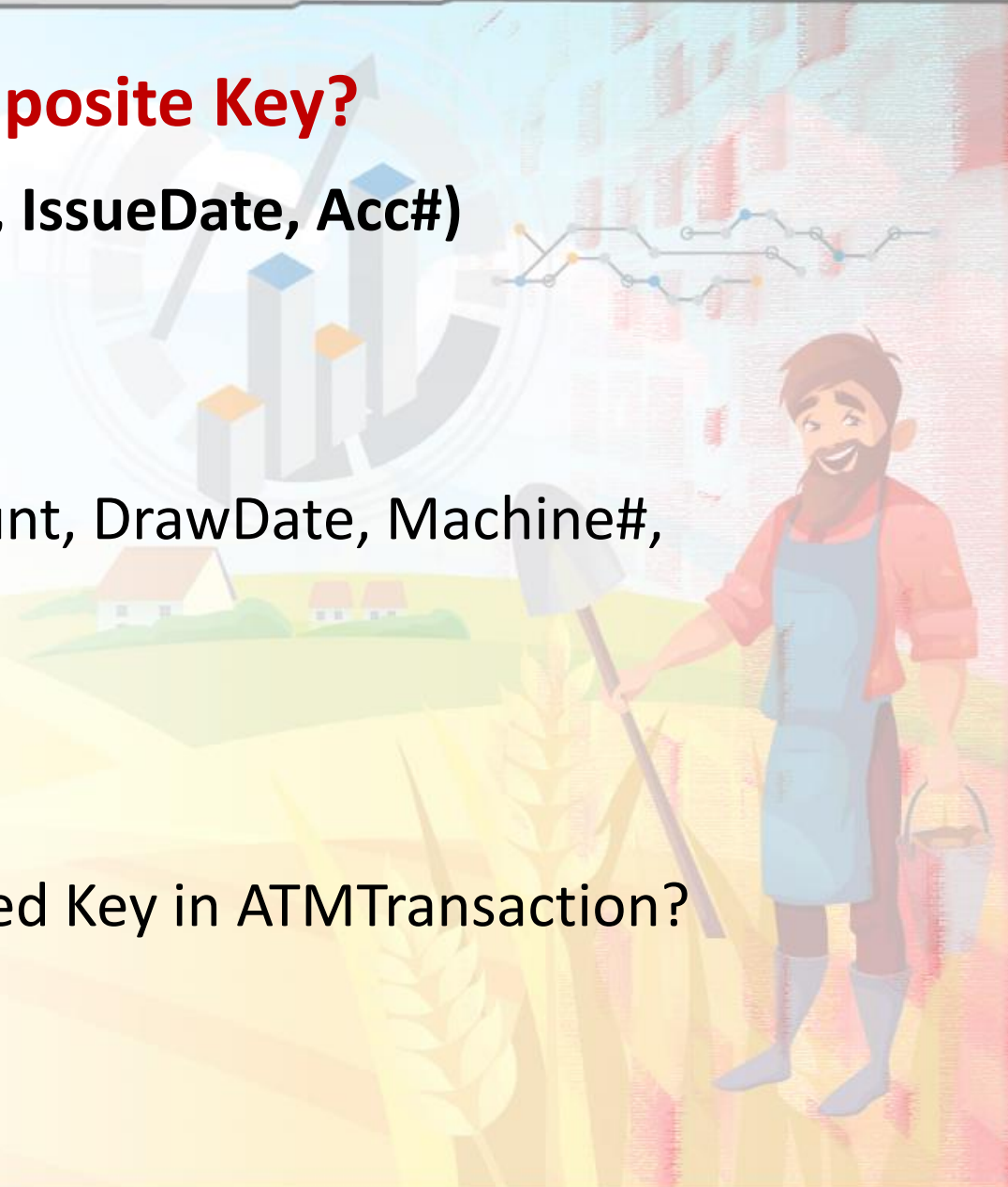
Card(Card#, CardHolderName, ExpiryDate, IssueDate, Acc#)

ATMTransaction(ID, Card#, TransNo, Amount, DrawDate, Machine#, TransType)

Do we need most closed related table?

What is a drawback of using ID as Surrogated Key in ATMTransaction?

What are PK and FK?



Chapter 25

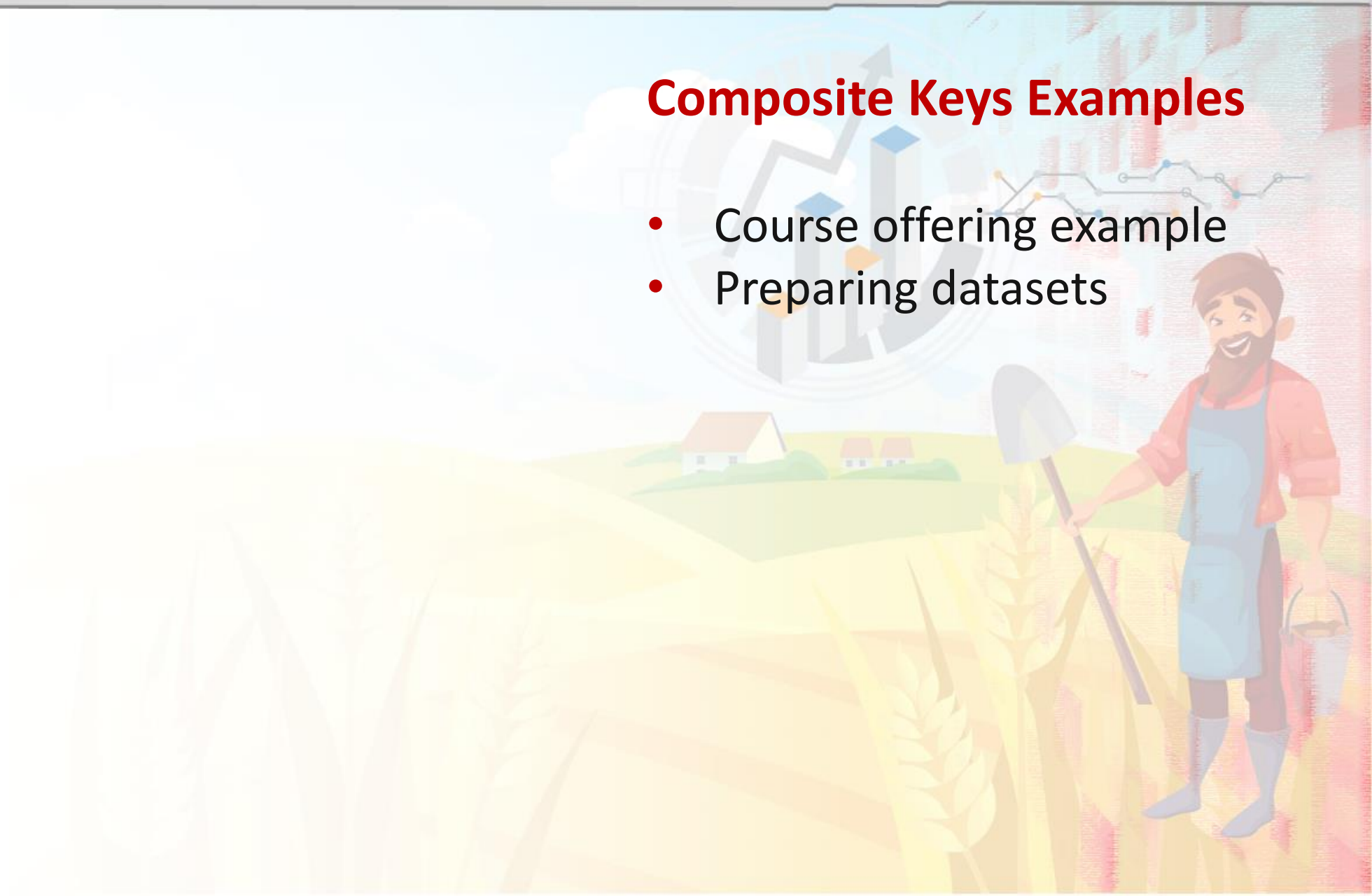
**Composite Keys
Examples**



Composite Keys Examples

Composite Keys Examples

- Course offering example
- Preparing datasets



Composite Keys Examples

Course Offering Example

Let's us assume, there are 1000 number of courses in a table or list.

Students want to register in courses.

Can they register by looking at 1000 courses? NO

What to do?

Offer selected courses?



Composite Keys Examples

Course Offering Example

Offering course table

CrsOffer(SemID, CrsID, Sec, InstrName, B#, R#)

What to do for keys, PK or Composite Key?

CrsOffer(SemID, CrsID, Sec, InstrName, B#, R#)

CrsReg(SemID, Roll#, CrsID, Sec, TotMarks, Grade)



Composite Keys Examples

Preparing Dataset for Composite Keys

CrsOffer(SemID, CrsID, Sec, InstrName, Building#, Room#)

CrsReg(SemID, Roll#, CrsID, Sec, TotMarks, Grade)

Do we need most closed related tables?

What are PK and FK?

